MSCE AGRICULTURE

INTRODUCTION

Agriculture Model Questions and Answers is a book that is designed to assist students who are preparing to sit for Malawi School Certificate of Education examinations. It captures questions from both MSCE examinations in recent years as well as questions developed from MSCE Agriculture syllabus objectives. The MSCE agriculture syllabus covers eight broad topics, which are:

1. Agriculture and the Environment
2. Crop Production
3. Animal Production
4. Agricultural Marketing
5. Farm Business Management
6. Agricultural Technology
7. Agricultural Experimentation
8. Challenges in Agricultural Development

Each of the above broader topics is narrowed down into sub topics. As such, the questions in this book have been carefully arranged so as to reflect the subtopic being examined.

At MSCE, students are supposed to sit for two papers in Agriculture. Of these papers, paper 1 is theory while paper 2 is practical. The theory paper contains sections A and B. In section A there are structured questions in which students are required to provide short answer questions. Section B usually contains three essay questions. Under this section, students are supposed to write different types of essays such as descriptive, explanatory and problem solving essays. In the practical paper, there are usually four questions in which two questions are based on specimens that are provided while the other two questions are theoretical in nature.

When answering questions in agriculture, students need to demonstrate their knowledge on a number of topics, concepts and issues that are covered in Agriculture. Apart from Agriculture, the students are also required to use and apply knowledge from other circles such as Biology, Physical Science, Mathematics, Geography, Social Studies and many other disciplines. Of greater significance is the students' communication skill. This is demonstrated by the student's ability to understand and answer the questions correctly. Students need to be very careful in the way they present their responses in the examination. The requirement is therefore for a student to have a reasonable command of the English language, because failure to communicate properly may be costly.

It is hoped that students will find this textbook very useful as they prepare for their examinations. Additionally, Agriculture subject teachers are encouraged to use this book when planning and delivering the course content in respective topics.

Chapter One

AGRICULTURE AND THE ENVIRONMENT

Questions

1.1 Physical Properties of soil

1. a. List down any five physical properties of soil.

   b. Define the term soil texture.

   c. Name any two textural classes of soil.

   d. Explain five ways in which soil texture is important in crop production.

2. A teacher brought a sample of soil to a form four class. Using "sieve method", describe how the percentage of gravel, sand, silt and clay could be determined.

3. a. Define the term soil structure.

   b. List down any three ways in which soil structure is important in crop production.

   c. Describe how the following activities help in maintaining soil structure.

      i. Planting of vegetative cover.

      ii. Addition of organic manure.

      iii. Crop rotation.

4. a. What does each of the following soil colours indicate in terms of composition of parent material?

      i. Grayish colour.
      ii. Reddish colour.

   b. Explain how soil colour affects soil temperature.

   c. State any two ways in which soil temperature can be modified.

Essay

5. *Describe any five types of soil structure.*

6. *Explain any five ways in which depth of soil affects plant growth and development.*

7. *Explain any five characteristics of soil that make it suitable for crop production.*

Practical

8. The table below shows results from Soil Science laboratory and data from meteorological observations of an Extension Planning Area. Use it to answer the questions that follow.

Table 1.1: Results from Soil Science Laboratory and Data from Meteorological Station

CHARACTERISTICS	SOIL A	SOIL B	SOIL C
Percentage sand	40	42	20
Percentage silt	8	6	10
Percentage clay	50	47	60
Soil colour	Red	Brown	Black
Effective depth (metres)	4	1	1.5
Percentage organic matter	2	5	10
Rainfall (mm per year)	2, 000	1, 500	1000
Annual rainfall distribution	More even distribution	Uneven distribution	Uneven distribution
Temperature	Medium	Medium	high
Soil pH	5.1	6.2	7.4

a. Which soil is likely to have the highest water holding capacity? (1)
b. Explain two reasons for the answer to question a (4)
c. Which soil would be good for tree crops? (1)
d. Explain two reasons for the answer to question c. (4)
e. Explain any two possible reasons for the acidity in Soil A. (4)
f. From which soil would the results show the highest population count of microorganisms? (1)
g. Explain one reason for the answer to question f. (2)
h. State any three ways of improving soil sample A. (3)

9. You are provided with the following materials:

- One tablespoonful of dry clay soil and one tablespoonful of dry sand.
- Two clear containers .
- Two containers with 50 ml of water.

Procedure

Pour the 50 ml of water into each container at the same time.

a. Observe and draw how the two containers will appear after some minutes.

b. Which soil would be good for growing of rice?

c. Give a reason for the answer above.

d. State the effect of applying organic manure to each soil sample:

  i. Clay soil.
  ii. Sandy soil.

J & K
Sand Clay

e. In which of the two soil samples would the flow of water in the irrigation channels be a problem?

f. Give a reason for the answer 9e above.

1.2 Chemical properties of soil

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the

1. Describe the term soil pH.

2. Explain how the following factors affect soil pH:

cal

   a. Soil texture.
   b. Crop removal.
   c. Weathering of parent materials.
   d. Use of chemical fertilizers.
   e. Microbial activity.

3. Explain any two effects of high soil acidity on nutrient availability to crops.

4. Explain any three ways in which soil pH affects crop production.

5. a. State any two ways of modifying soil pH.
   b. State two ways in which a farmer can raise soil pH in a maize garden.

6. a. Explain the meaning of Cation Exchange Capacity (CEC).
   b Describe how the Cation Exchange Capacity influences crop production.

7. a. Define soil salinity.
   b Explain two ways in which leaching of cations can affect the nutrient status of the soil.
   c List down any five factors that affect the nutrient status of the soil.

1.3 Soil degradation

of

1. a. Define the term soil degradation.
   b Name the three forms of soil degradation.

2. Describe the following forms of soil degradation.

   a. Chemical degradation.
   b. Biological degradation.
   c. Physical degradation.

3. Explain one way in which the following activities accelerate soil degradation.

   a. Bush fallowing.
   b. Excessive chemical use in the soil.

4. Explain one way in which each of the following measures can help to conserve soil:

   a. Construction of storm drains.
   b. Strip cropping.

5. State any five ways in which people contribute to soil degradation.

6. Explain any one biological way in which a farmer can control soil degradation in a vegetable garden.

7. Explain any five characteristics of soil that make it suitable for crop production.

Chapter One

Answers

1.1 Physical Properties of soil

1a. i. Soil texture.
ii. Soil structure.
iii. Soil colour.
iv. Soil temperature.
v. Soil consistency.
vi. Soil porosity.
vii. Soil depth.

b Soil texture is the proportion of sand, silt and clay particles in a given soil sample.

c i. Sand.
ii. Clay.
iii. Silt.
iv. Sandy clay.
(v) Loam.
(vi) Silty clay.
(vii) Silty Loam.
(viii) Sandy Loam.
ix Silty clay loam.
x Loamy sand.

d i. It affects the balance between water and air in the soil. For crops to grow they require both moisture for the dissolution of plant nutrients and air for root respiration. Soils that are well drained are also well aerated.

ii. It affects the water holding capacity of the soil. Soils that have a considerable amount of clay particles are able to hold enough moisture necessary for plant growth because they have very small pore spaces that allow less water to pass through. This is quite different with sandy soils that drain out so easily due to their large particle sizes.

iii. It affects the availability of nutrients in the soil. Soils that contain a lot of sandy particles experience a lot of leaching which depletes the nutrient levels of the soils. In contrast, clayey soils are able to hold a lot of nutrients because of low levels of leaching.

iv. Soil texture affects root growth and development. Plant roots growing in soils with a considerable amount of sand develop better as they are able to easily penetrate the soil and reach out for nutrients. Root development in clay soils is much restricted as the soils are more compact making root penetration to be very difficult.

v. Soil texture also determines how easily a given soil could be worked upon. Sandy soils are loosely packed and so much easier to work in than clay soils which are tightly packed.

2. The first step is to weigh and record the total mass of the collected soil sample. The second step is to separate the inorganic components of the soil which include gravel, sand, silt and clay. This can be done using sieves of different mesh diameter. The soil sample is placed in the sieves starting with the one with bigger holes. When in the sieve, the soil sample is shaken vigorously. This action causes small sized soil particles to pass through the holes, while trapping the larger particles. The trapped soil particles are put in separate places. Once separated, the weight of these inorganic components of the soil is computed. The mass of each component is then expressed into a percentage of the total soil sample.

3 a. Soil structure is the arrangement of individual soil particles to form soil aggregates.

b i. It affects the movement of water and air in the soil.
ii. It affects the amount of water and air in the soil.
iii. It affects the transfer of heat in the soil.

c i. This provides protection to the soil particles against the impact of heavy raindrops that cause the detachment of soil particles.

ii. The manure increases the organic matter content of the soil that has a binding effect on the soil particles.

iii. Alternation of plants with tap and fibrous root systems help to provide adequate binding which reduce cases of soil detachment.

4 a. i. Grayish colour- indicates that the parent material is rich in Silica or Quartz.
ii. Reddish colour: indicates that the parent material is rich in Iron oxides.

b Darker soils are able to absorb more heat energy from the sun while soils with a lighter complexion reflect more sun heat back into the sky. Due to this reason, darker soils are usually warmer than lighter soils.

c i. By applying mulches.
i. Irrigation of soils in order to cool them.
ii. Drainage of waterlogged soils.
iii. Addition of organic matter.

Essay

5 *The first type of soil structure is called platy structure. This is a type of structure in which the soil particles are arranged into thin flat plates that lie horizontally on top of each other. This type of structure is commonly found in areas with compacted soils.*

The second type is called a blocky or cuboidal structure. In this type, the soil particles aggregate to form irregular blocks of soil.

Another type of soil structure is columnar structure. It is also called Prismatic structure because the soil particles are vertically arranged to form prism like structures.

Granular structure exists where the individual soil particles are loosely packed in small round groups. This structure is characterized by the presence of small pore spaces and organic matter.

A crumby structure forms where there is a large presence of organic matter. The soil aggregates resemble the crumbs of bread. It is an ideal structure for most crops because it contains a large proportion of organic matter and air spaces.

6 *Firstly, soil depth affects root penetration and development because is associated with soil fertility. For example, soils that are well developed and mature contain a lot of plant nutrients. These plant nutrients enable the roots to be well developed. On the other hand, shallow soils are usually deficient in plant nutrients. As such, plants growing in such soils have poorly developed roots.*

Soil depth also affects the drainage of the soil. While deep soils enable easy drainage of water, shallow and poorly developed soils are usually waterlogged. Following this, plants that grow in deep soils are healthier than those in shallow soils.

Water retention is another characteristic of soil that is affected by soil depth. Soils that are well developed are able to retain sufficient amounts of water than those that are poorly developed. This is because deep soils contain a lot of organic matter that enables it to absorb and keep moisture.

The other effect of soil depth is aeration of the soil. A well developed mature soil contains a large amount of organic matter which makes the soils to be well aerated. This aeration helps to supply oxygen for root respiration. For this reason, plants growing in deep soils grow better than those in shallow soils.

The depth of soil also determines the extent to which plants are physically supported. It can therefore be noticed that in shallow soils, plant lodging is more common than deep soils because deep soils provide maximum anchorage, especially to deep rooted plants.

7 *The first characteristic is the availability of nutrients. Soils need to have a good proportion of essential plant nutrients as these are necessary for the support of various plant processes like photosynthesis and respiration.*

Aeration of the soil is also an important characteristic of soil. This is the ease with which air is able to circulate in the soil. Soils that are well aerated enable plant roots to access oxygen for root respiration.

Crop production is also affected by soil pH. If the pH is too high or too low, most plants fail to fully develop. Most plants do well within the optimum pH range of 5.5 to 7.7. This is because within this pH range most plant nutrients are available in forms that plants can absorb. Soil pH is also very important in the work of microorganisms that help in decomposition of organic matter.

Soil moisture is another important soil characteristic. The moisture is important to plants because it helps in dissolving plant nutrients and transportation of food to all parts of the plant.

Soils that are waterlogged result into poor plant development. Well drained soils are ideal for plant growth because they are more aerated and contain a lot of plant nutrients. As such, drainage of the soils is a crucial factor in plant growth and development.

Practical

8 a. Soil sample C.

b. i. Has a high percentage of clay particles (60%). Clay particles have very tiny particles which do not allow water to pass through.

ii. Soil C has a high amount of organic matter which helps to hold more water.

c. Soil A

d. i. Has a higher soil depth which can effectively support root development.

ii. Receives high amount of rainfall that is evenly distributed thereby enabling tree roots to easily access moisture.

ii. Has a considerable amount of sand that enables easy root penetration.

e. i. Soil sample A has a high percentage of sandy particles which are large and loosely packed thereby creating very large spaces within the soil. These characteristics of sandy particles make them prone to more leaching. The process of leaching leads to the removal of cations, which are replaced by hydrogen ions that cause soils to become more acidic.

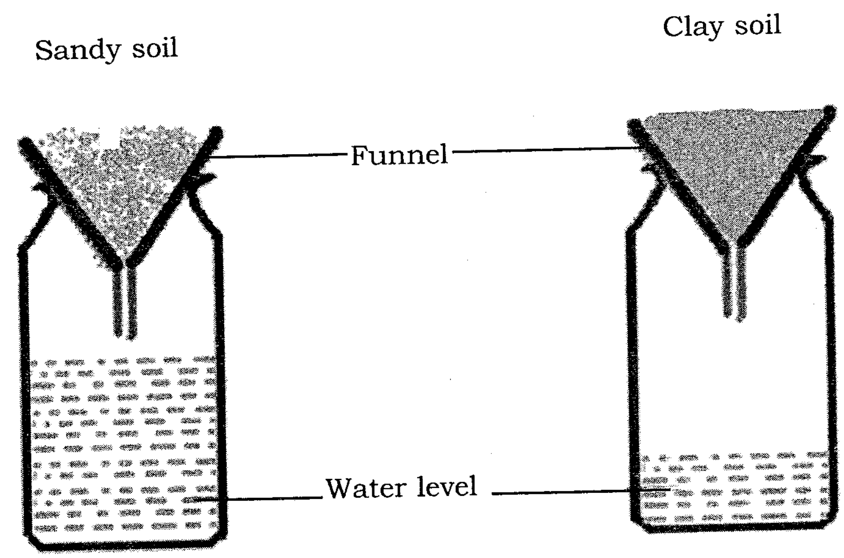
ii. Soil sample A receives a high amount of rainfall which helps to accelerate the loss of soil nutrients through leaching and soil erosion.

f. Soil sample C.

g. The soil is black in colour which is caused by an abundance of humus in the soil. This humus is a result of decomposition that is done by micro organisms.

h. i. Addition of organic manure so as to increase organic matter content.
ii. Liming to increase soil pH.
iii. Cultivating the soils in order to loosen the soil particles.

9.a. Drawing of results



b Clay soil.

c It is able to retain a lot of water which is good for the growing of rice.

d i. The organic matter will help in loosening the clay particles, a thing that would encourage aeration and water drainage.

ii. The organic manure will help to bind the loosely packed sandy soils together thereby improving its water retention.

e Sandy soil.

f It has very big pore spaces which encourage a lot of infiltration. This is a problem in channel irrigation because enough water may fail to reach crops that are at the end of the channels.

1.2 Chemical properties of soil

1. Soil pH refers to the acidity or alkalinity of soil. It is determined by the presence of negative and positive electrical charges called ions. The positively charged ions are called hydrogen ions, and when they are in higher concentration in the soil, the soil is said to be acidic. The negatively charged ions are called hydroxyl ions. When there is a higher concentration of hydroxyl ions, the soil is said to be alkaline. However, an equal concentration of hydrogen and hydroxyl ions leaves the soil in a neutral situation.

2. a. The size of the soil particles determines the rate at which cations are going to be leached from the soil. For example, clay soils due to their small particles and very tiny pore spaces experiences less leaching. For this reason, clay soils are less acidic. On the other hand, sandy soils are more acidic because the rate of leaching is very high due to their large particles and large pore spaces that allow easy passage of water.

   b As plants are growing in the soil, they make use of plant nutrients. The removal of these nutrients from the soil greatly alters the pH of the soil.

   c Rocks that form soils contain minerals that are either acidic or alkaline. Depending on the mineral composition of the parent material, the process of weathering can result into modification of the soil ph. For example, rocks that contain sulphur result into the formation of acidic soils through the formation of sulphuric acid. In contrast, weathering of calcium rich rocks such as limestone give way to alkaline soils.

   d Depending on the chemical composition of the fertilizers, soil pH can either be reduced or increased. For example, CAN fertilizers increase soil pH because they contain calcium. On the other hand, use of sulphate of ammonia results into the soils becoming more acidic.

   e Decomposition of organic matter by microorganisms results into the release of carbonic acids which lower the pH of the soil.

3. a. It causes some plant nutrients like phosphorous to become unavailable for plant absorption through the formation of insoluble compounds.

   b Some plant elements like zinc, cobalt, aluminium and iron become excessively available for crop absorption in acidic soils. This results into the nutrients building up toxic levels in both the plants and the soils.

4. a. It determines the type of crops to be grown in a given area. There are some crops that do well in acidic soils while others in alkaline soils. Still more, some crops tolerate a wide range of soil pH. For example, tea is a crop that grows better in acidic soils while bananas does well in alkaline soils. Crops like coffee, groundnuts, maize and sorghum tolerate both slightly acid and slightly alkaline soil conditions.

   b Soil pH determines the rate at which beneficial soil microorganisms work in the soil. For example, bacteria work best within the optimum pH range of 5.5 to 7.8.

   c Soil pH also affects the availability and release of plant nutrients. For instance, phosphorous is available for plant absorption as phosphates when the pH is within the range of 6.5 to 7.5.

d Some soil pH ranges also cause an increase in plant diseases mainly due to deficiencies in certain plant nutrients. For example cases of potato scab are high in alkaline soils while acidic soils accelerate the cases of club root in vegetables like cabbages.

5. a. i. Addition of chemical fertilizers.
   ii. Liming.

   b i. Applying agricultural lime into the soils.
   ii. Addition of Calcium Ammonium Nitrate Fertilizer.

6. a. It is the ability of soils to replace lost cations on the exchange complex per unit weight of soil at a particular pH.

   b i. The Cation·Exchange Capacity affects the total amount of nutrients available to plants as exchangeable cations. If the soil has a low Cation Exchange Capacity it implies that the soil is low in exchangeable plant nutrients and so plants may fail to grow healthily.
   ii. CEC also affects the availability of basic ions as opposed to hydrogen ions. If the soil has a low concentration of the basic ions the soil pH becomes low. This low soil pH affects the availability of certain nutrients, apart from making the soils to become toxic.

7. a. This is when the soil is having an abundance of soluble salts.

   b It results into increased concentration of hydrogen ions which cause the soil to be acidic. This increased soil acidity makes some nutrients to be unavailable for plant absorption.

   c i. Soil pH.
   ii. Leaching of basic nutrients.
   iii. Soil erosion.
   iv. Nutrient uptake by plants.
   v. Farming methods.

1.3 Soil degradation

1. a. This is the loss of the quality and value of soil.

   b i. Physical degradation.
   ii. Biological degradation.
   iii. Chemical degradation.

2. a. This results when the chemical properties of soil have been modified to extremes. The most important chemical properties concerned include soil pH and Cation Exchange Capacity. The altering of these chemical properties affects the availability of plant nutrients as well as the work of beneficial microorganisms in the soil leading to low crop productivity. Chemical degradation is accelerated by such activities like use of chemical substances and poor farming practices like deforestation.

   b This is a form of soil degradation in which the organic components of the soil have been disturbed or destroyed. These organic components include vegetation, humus, micro and macro soil organisms. The destruction of these organic components is caused by the release of organic compounds on the land. Furthermore, human activities like setting out of bush fires and deforestation result into biological soil degradation.

c This form of soil degradation is closely associated with the destruction of soil structure and texture. Basically the destruction of these physical soil properties results into compaction of soils, poor aeration, reduced infiltration and increased run off that lead to accelerated soil erosion. A number of factors contribute to physical degradation such as poor soil management practices like continuous cultivation and deforestation among, many others.

3. a. This practice involves using a piece of land continuously until it looses fertility and then abandoning it for some other pieces of land. Continuous cultivation on the same piece of land results into rapid depletion of soil nutrients.

   b Some of the chemicals used in agriculture can result into altering of soil pH which can limit the availability of certain plant nutrients.

4. a.Storm drains help to divert overland flow into natural waterways such as streams and rivers. This reduces soil erosion that results from surface run off.

   b This helps to provide adequate ground cover to the soil thereby reducing soil erosion through running water and raindrops.

5. a. Wanton cutting down of trees.
   b Construction of ridges along the slope.
   c Cultivating along river banks.
   d Over use of agricultural chemicals.
   e Continuous cultivation.
   f Monocropping.
   g Use of heavy machineries.
   h Setting out of bush fires.

6. a. Addition of organic manure in the soil helps to improve soil structure as well as to increase the nutrient content of the soil.

   b Inclusion of leguminous crops like beans in crop rotations to facilitate the fixation of nitrates in the soil.

   c Use of mulches to protect the soil from the impact of heavy rain drops which cause splash erosion.

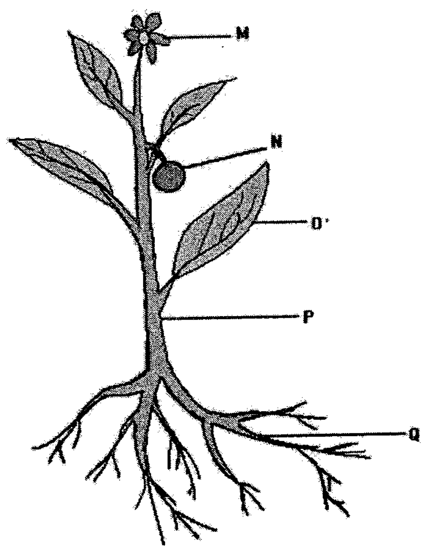
Chapter Two

CROP PRODUCTION

Questions

2.1 Plant parts and functions

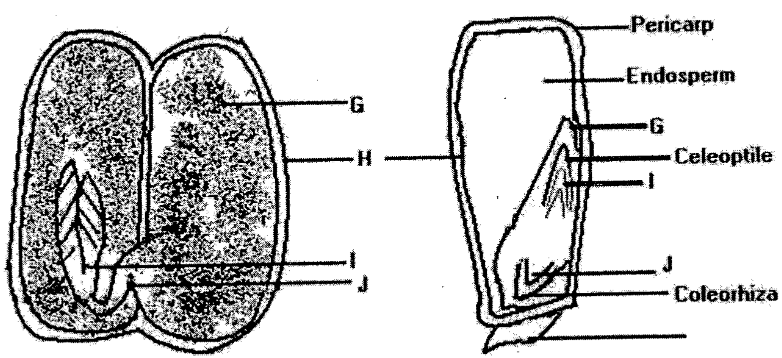
1.Study the figure below and answer the questions that follow.



a. Identify parts M, N, O, P and Q.

b. State the functions of parts M, N, O, P and Q.

2. Study the figures below of plant seed and answer the questions that follow.



Seed E

Seed F

a. Identify seed types E and F.

b. Give examples of plants that produce the seeds.

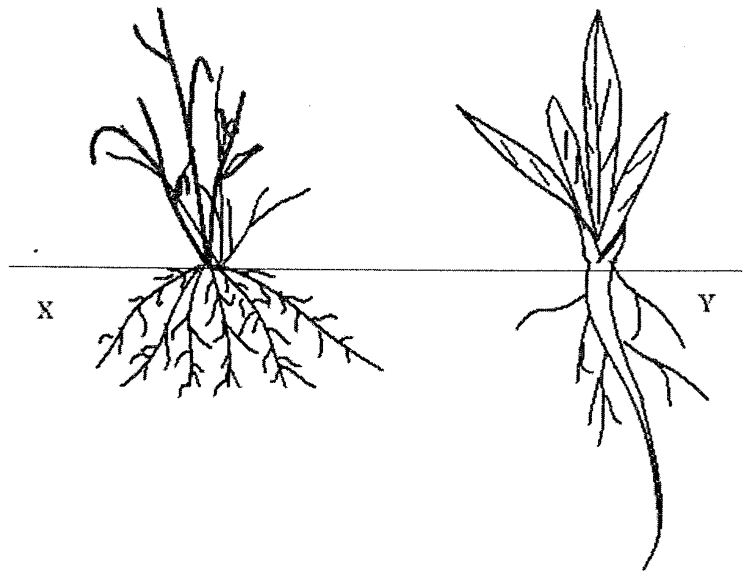
c. Name the parts labeled G, H, I and J.

d. What are the functions of the labeled parts in seed E?

e. What is the function of the celeoptile and coleorhiza in seed type F

f. Explain the function of the endosperm in seed type F.

3. Study the figures below of two root systems and use them to answer questions that follow.



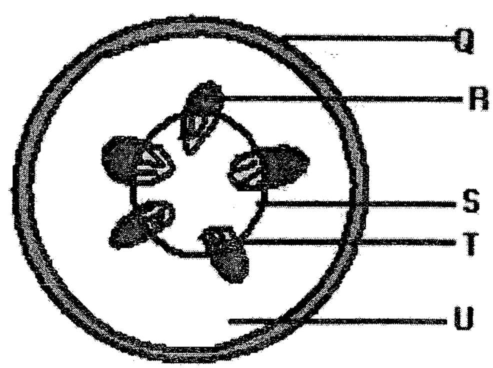
a. Identify root systems X and Y.

b. What advice can be given to a farmer on fertilizer placement if he/she grows crops that have the root X and Y?

c. Explain how the rooting depth of X and Y affects the amount and frequency of irrigation.

d. Explain the advantage of rotating crops with root systems X and Y.

4. The figure below is a diagram of the internal structure of a root. Use it to answer questions that follow.



a. Name the parts labeled Q, R, S, T and U.

b. Explain one function of each of the parts labeled Q, R, S, T and U.

2.2 Seeds and planting materials

1. a. List down any five advantages of using shelled groundnut seeds as planting materials.

   b What are some of the limitations of using shelled groundnut seeds for plant propagation?

2. a. List down any five examples of vegetative planting materials.

   b State four disadvantages of vegetative propagation.

3. Explain each of the following advantages of vegetative propagation:

   a. Ensuring genetic uniformity.

   b. Ensuring early maturity of plants.

4. Explain any five advantages of asexual propagation in crop production.

5. Explain the difference between runners and rhizomes.

6. Describe how budding is done in crop propagation.

7. With the aid of well labeled diagrams, describe the procedure in:

   a. Grafting.
   b. Layering.

2.3 Essential plant nutrients

1. a. List down examples of macro and micro plant elements.

   b List down the sources of plant nutrients.

2. a. State any five functions of nitrogen in plants

   b What are some of the problems of excessive application of nitrogenous fertilizers to crops?

3. If a farmer applies Urea fertilizer in a maize field, describe how the plant nutrient in the fertilizer could be depleted from the soil.

4. a. State any two functions of Sulphur in crops.

   b State any two deficiency symptoms of Sulphur in crops.

   c Name any one inorganic fertilizer that can be applied to soil to correct Sulphur deficiency in crops.

5. Briefly explain how the following factors cause depletion of essential plant nutrients.

   a. Crop removal.
   b. Soil erosion.
   c. Leaching.
   d. Volatilization.
   e. Fixation.

6.Describe the yellowing of maize caused by the deficiency of the following plant elements:

a. Nitrogen.

b. Potassium.

c. Magnesium.

7. A crop showed signs of interveinal chlorosis. Use this information to answer the questions that follow.

a. Name the plant nutrient that was deficient.

b. State two ways of correcting the deficiency.

Practical

8. You are provided with the following materials:
- Half teaspoonful of (Specimen Y) - Urea
- Half teaspoonful of (Specimen Z) - Calcium Ammonium Nitrate (CAN)
- Two small clear containers with 20 ml of water in each

Put each of the fertilizer samples in the separate containers of water at the same time. Shake the contents.

a. Observe and record what happens.

i. Specimen Y (Urea).
ii.Specimen Z (CAN).

b. Which fertilizer sample would be recommended for the following areas?

i. High rainfall areas: Y (Urea).
ii.Low rainfall areas. Z (CAN).

c. Explain each of the answers in 6 (b) above.

d. If a farmer delayed applying fertilizer to a crop, which fertilizer sample would be suitable?

e. Give a reason for the answer in 6 (d) above.

9. During an experiment that was conducted by students on essential plant nutrients, the following signs were noticed in four different plots

Plot 1	**Plot 2**
◆Increased lodging of crops ◆ Short nodes and little growth ◆Yellowing of leaves ◆Reduced resistance to diseases	◆Dying tips ◆Light green bands along leaf margins ◆Young leaves remaining closed

Plot 3	Plot 4
• Slow growth • Delayed maturity • Thin and tall plants • Purplish leaves	• Withering of stems • Death of terminal buds • Wilting of leaves

a. Identify the plant elements that were lacking in the four plots that were being experimented on.
b. From the table above, what conclusions can be made on the functions of the deficient plant nutrients in the four plots?
c. Give any four ways in which the plant element in plot 1 can be lost.
d. Name the sources that can help to add the deficient nutrient in plot 1.

2.4 Crop protection: Weeds and weeding

1. State any one advantage and any one disadvantage of biological weed control.

2. Describe any three methods that can be used to control weeds in a vegetable garden.

3. Tabulate any three advantages and disadvantages for mechanical and chemical weed control measures.

Essay

4. *Describe any five effects of weeds on crop production.*

5. *Describe any five cultural methods of controlling weeds.*

2.5 Crop protection: plant pests and diseases

1. a. Explain the meaning of the term crop protection.

   b. What is the difference between a pest and diseases?

2. Explain the reason why crop pests and diseases are said to have an economic significance.

3. Describe the following pest and disease control methods.

   a. Cultural control.
   b. Biological control.
   c. Chemical control.

4. Explain how integrated pest management works.

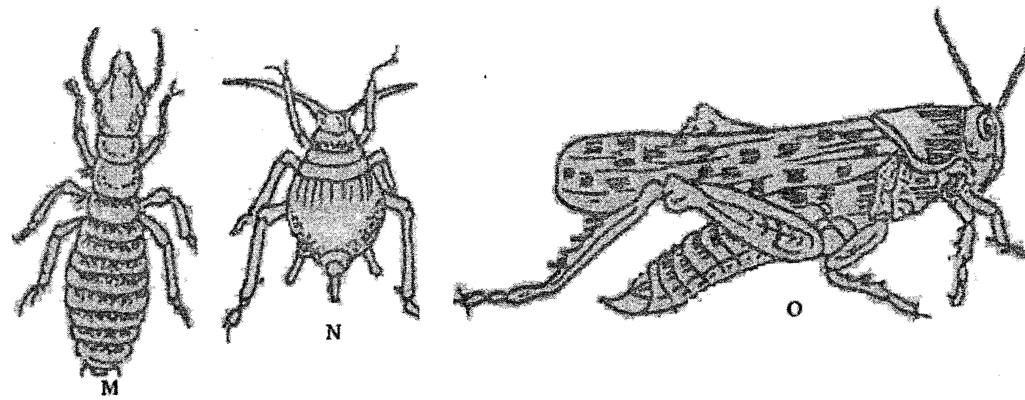
5. Describe one way in which each of the following legislative measures can help to control pest and diseases.

   a. Prohibition.
   b. Notification order.

6. a. Describe any two ways in which a weevil damages stored maize.

   b. State any two ways of controlling weevils in maize.

7. Figure below shows crop pests that reduce crop production. Study it and answer the questions that follow:



a. Identify M, N and O.

b. Which of these pests is both a field and a storage pest?

c. Name any two cultural methods of controlling the pest mentioned above.

d. Describe any two ways in which a farmer may break the life cycle of O.

e. Describe any two ways in which N contributes to low crop production.

f. Describe any one way of controlling N.

8. Give any two pests of mangoes.

b. Explain any one way in which each of the following practices on a farm can reduce pest attack.

i. Storing produce in air tight facilities.

ii. Closed season.

9. Write an essay explaining any five ways in which aphids can affect vegetable production.

2.6 Cropping systems

1. Explain two ways in which mixed cropping can help to conserve soil.

2. Describe the activities that are involved in the following cropping systems:

a. Bush fallowing (land rotation).

b. Shifting cultivation.

c. Agroforestry.

d. Crop rotation.

3. Explain any five ways in which organic farming would ensure food security.

Practical

4. The two tables below show designs of two cropping systems labeled A and B. use them to answer questions that follow.

	Year 1	Year 2	Year 3
Plot 1	Beans	Maize	Tobacco
Plot 2	Maize	Tobacco	Groundnuts
plot 3	Tobacco	Groundnuts	Beans
Plot 4	Groundnuts	Beans	Maize

A

	Year 1	Year 2	Year 3
Plot 1	Maize	Maize	maize
Plot 2	Maize	Maize	maize
Plot 3	Maize	Maize	maize
Plot 4	Maize	Maize	maize

B

a. Name the cropping systems represented by A and B.

b. Which cropping system would help to control witch weed infestation.

c. Explain the answer in b.

d. Which cropping system would allow a farmer to attain specialization of crop production?

e. Explain the answer in d.

5. Explain the difference between monocropping and mixed cropping in terms of:

a. Capital requirement.

b. Use of machinery.

c. Risk bearing.

6. Explain the difference between crop rotation and shifting cultivation in terms of:

a. Food security.

b. Facilitation of deforestation.

c. Conservation of soil.

7. a. State any three principles to consider when designing a crop rotation system.

b Complete the crop sequence for the years 2, 3 and 4 in the following crop rotation design:

	Year 1	Year 2	Year 3	Year 4
Plot 1	Maize			
Plot 2	Cassava			
Plot 3	Cotton			
Plot 4	Groundnuts			

8. You are provided with specimen labeled Q. (Maize grain)
   A farmer planted the specimen labeled Q in a field and a week later planted groundnuts in the same field.

   a. Name the farming system.

   b. Explain any three advantages of the farming system named in a.

   c. Give any two agro based industries that can use the specimen labeled Q as a raw material.

   d. Name the main nutrient that an animal can get from feed made from the specimen labeled Q.

2.7 Fruit production

1. a. Give any two varieties of mangoes.

   b. Name one sucking pest of mangoes.

   c. Explain any one way in which the insect pest named in 1b can damage mangoes.

   d. Explain any one way of controlling sucking insect pests in mangoes.

2. Explain any four factors to consider when selecting a site for a mango orchard.

3. a. List down any three diseases of mangoes.

   b. State the damage that is caused by the following pests of mangoes.

      i. Mango stone weevil.
      ii. Mango scales.

   c. Give two cultural measures for controlling mango stone weevil.

Essay

4. *Describe any five husbandry practices which should be followed when establishing a mango orchard.*

5. *Explain any five ways in which fruits are important.*

6. *Explain any five ways in which diseases can affect mango production.*

2.8 Crop improvement

1. a. State the two aims of crop improvement.

   b Give any five objectives of crop improvement.

2. Describe the following methods of crop improvement:

   a. Introduction.

   b. Selection.

3. Describe the following stages in hybridization.

   a. Choosing parents.

   b. Self pollinating parental lines.

   c. Cross pollinating pure lines.

4. A farmer was provided with four inbreeds of maize with the following characteristics:
   - Inbred W: early maturing.
   - Inbred X: high resistance against maize streak virus.
   - Inbred Y: short height that resists plant lodging.
   - Inbred Z: high yielding per unit area.

   With the aid of a well labeled diagram, describe how the farmer would develop a new variety that would retain all the characteristics from the four inbreeds.

2.9 Crop storage and processing

1. State any two ways in which processing of mangoes is important.

2. Explain any two ways in which groundnut seeds can be processed into livestock feed.

3. a. Name any two products that can be obtained from processing of tomatoes.

   b. Explain any two ways in which processing of tomatoes is important.

2.10 Pasture production

1. a. Name the two types of pastures.

   b. Give any three examples of exotic grass pastures in Malawi.

   c. Explain any two advantages of improved pasture over natural pasture.

2. Explain any two ways in which pastures help to prevent soil erosion.

3. Explain any two disadvantages of broadcasting as a method of sowing pasture.

4. Describe the following criteria when selecting improved grasses and legumes.

   a. Compatibility with desired species.

   b. Proposed method of utilization.

5. Explain any two ways in which preservation of pastures in form of hay can benefit a farmer.

6. A farmer would like to establish pastures. Briefly describe how the farmer can establish the pastures under the following headings:

   a. Land preparation.

   b. Selecting appropriate type of pastures.

c. Determining the seed rate.

d. Treating the pasture seeds.

e. Sowing (planting) of pastures.

7. Describe the following methods of pasture conservation.

   a. Hay.

   b. Silage.

   c. Foggage.

8. List down any five factors that affect the quality of :

   a. Hay.

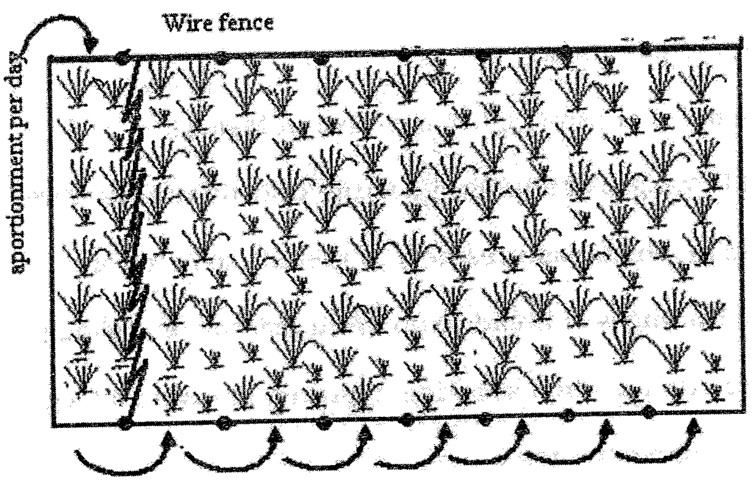
   b. Silage.

9. Explain the importance of the following processes in pasture establishment

   a. Inoculation.

   b. Pelleting.

10. The figure below is a diagram of the systems of grazing animals. Use it to answer the questions that follow.



   a. Name the system of grazing animals.

   b. Explain any two advantages of the system of grazing animals.

   c. Why is the system of grazing animals not commonly used in Malawi?

11. Write an essay explaining any five ways in which pastures are important.

Practical

12. Table 2 shows average monthly rainfall for a certain area in Malawi in the 2007/2008 farming year. Use it to answer the questions that follow.

Month	S	O	N	D	J	F	M	A	M	J	J	A
Rainfall	0	0	80	220	180	50	10	0	0	0	0	0

a. Explain one effect of the distribution of rainfall on a variety of maize which matures in 120 days if it was planted:

i. Towards the end of December.

ii. Early November.

b. If a farmer allowed cattle to mate at the beginning of September, explain one effect of such rainfall distribution on cattle production.

c. Explain how such rainfall distribution affected digestibility of pasture during the following periods.

i. May to August.
ii. November to January.

d. Explain the importance of each of the following steps in silage making:

the

i. Cutting grass before flowering.

ii. Partial drying.

iii. Compressing grass.

13. Table 3 shows quantity and quality of two grass pastures. Each grass pasture was grown on three hectares of land. Use it to answer the questions that follow.

Grass pasture	Dry matter (kg/ha)	Crude protein (%)
X	100	4.0
Y	120	3.6

$X = \frac{4}{100} \times \frac{100}{ha} \frac{3h}{1}$

$Y = \frac{3.6}{100} \times \frac{120}{ha} \times$

a. Calculate the crude protein yield for each of the grass pasture.

b. Explain two ways in which calculation of crude protein yield for the two grass pastures is important to the livestock farmer.

c. Give three ways in which burning is important in pasture management.

d. Explain any two ways in which grass pastures sown together with legume pastures would give high livestock production.

14. A farmer established a grass pasture using the following information:

Seed size = 200, 000 seeds/ kg
Purity = 75%
Germination = 50%

If the expected plant population was 60,000 plants per hectare, calculate seed rate of the grass pasture. Show your working.

Chapter Two

CROP PRODUCTION

Answers

2.1 Plant parts and functions

1. a. i. M is a flower.
   ii. N is a fruit.
   iii. O is a leaf.
   iv. P is a stem.
   v. Q is a root.

   b M: For reproduction so that new plants are formed.

   N: Contains seed.
   Acts as a food reserve.

   O: For photosynthesis, respiration and transpiration.

   P: Provide support to buds, leaves, flowers and fruits.
   For translocation of sugars.
   For storage of manufactured food.

   Q: for absorption of mineral salts and water.
   For storage of manufactured food.
   For supporting the whole plant.

2. a. i. Seed E: Dicotyledonous seed.
   ii. Seed F: Monocotyledonous seed.

   b i. Seed E: Legumes like groundnuts and beans.
   ii. Seed F: Cereals like maize and rice.

   c i. G: Cotyledon.
   ii. H: testa (seed coat).
   iii. I: plumule.
   iv. J: radicle.

   d i. G: Stores food that is used during germination by the embryo.
   ii. H: Provides protection to the embryo against mechanical injury and entry of pathogens.
   iii. I: It develops into a shoot system.
   iv. J: Develops into a root system.

   e They provide protection to the apical meristems. For example, the celeoptile protects the shoot while the coleorhizae protects the root as it pushes through the soil.

   f It stores food that is absorbed by the scutellum (cotyledon) and used by the embryo during germination.

3. a. i. X is a fibrous root system.
   ii. Y is a tap root system.

   b i. Root system X: In this root system, fertilizer should not be applied too deep into the soil because the root system is shallow and cannot access the nutrients at such a depth.

   ii. Root system Y: Fertilizer should be applied deeper into the soil because the plant has a tap root system which is able to get nutrients from deeper layers of the soil.

   c i. Root system X requires fewer amounts of water for irrigation because the rooting depth is very shallow. However, irrigation has to be done frequently because the water can easily dry out.

   ii. Root system Y requires more amounts of water to be applied to make sure that plant roots are able to access the moisture since the root system is deep. Irrigation of the plants with root system Y needs not to be frequent as the huge amounts of water will take a lot of time to dry.

   d It helps in the effective use of nutrients by getting back leached nutrients into the top soil. Since root system Y is deep rooted, the plant will be able to absorb nutrients deep down from the soil profile. When this plant is harvested and its residues are ploughed back into the soil and decompose, the nutrients can be made available for plants with root system X.

4. a. i. Q is epidermis.
   ii. R is Phloem.
   iii. S is a cambium layer.
   iv. T is a Xylem.
   v. U is cortex.

   b Part Q: provides protection to the internal parts of the stem.

   Part R: It is responsible for the translocation of sugars manufactured during photosynthesis to all parts of the plant.

   Part S: Has actively dividing cells that form the phloem and xylem vessels.

   Part T : It is responsible for transporting water and mineral salts to all parts of the plant. : The woody nature of part T (the xylem vessel) helps to offer physical support to the plant. Part U: stores food.

2.2 Seeds and planting materials

1. a. i. Seeds are cheap.
   ii. Seeds can be stored for a very long time.
   iii. They reduce the cases of disease transmission from parents to offspring.
   iv. Seeds can be sown mechanically, therefore making them easy to sow.
   v. Cross pollination of plants enable the development of new plant varieties.

   b. i. Most seeds contain limited food reserves to support the germinating plant
   ii. Does not ensure uniformity in the plants.
   iii. In some plants a very long juvenile period is required before fruits are produced.
   iv. Some plants have a very hard seed coat that cause them to have a long dormancy period.

2. a. Stem cuttings, rhizomes, runners, stolons, leaves, suckers, tubers.

   b. i. If the parent plant is affected by diseases, there is a high chance of transmitting the disease to the offspring.
      ii. Some methods of vegetative propagation like layering and grafting require specialist skills to perform.
      iii. Most materials used in vegetative propagation are bulky and difficult to transport and store.
      iv. More labour is required because the materials are planted manually.
      v. The materials can easily dry out if not planted promptly.

3. a. The planting materials have the same genetic make up like those of the parent plants. As such all the plants that are propagated have similar characteristics as those of the parent.

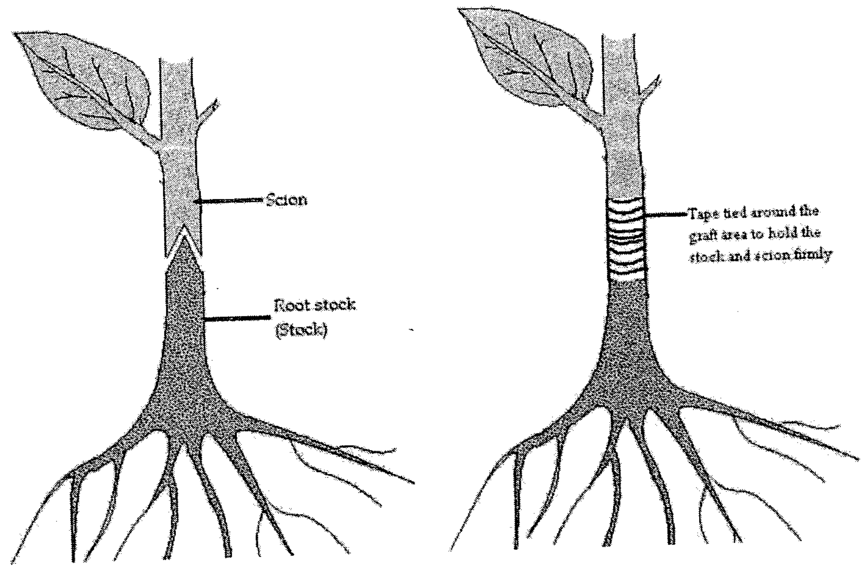
   b. The planting materials assume the age of the parent plant from which they are obtained. As such, they mature and produce earlier.

4. a. Asexual propagation helps to produce new individual plants without using seeds.
   b. Most of the planting materials can be obtained locally, making the method to be cheap.
   c. Vegetative planting materials have large food reserves that support the young plants in the early stages of development.
   d. It encourages early maturity of the plants because the planting materials assume the age of the parent plant.
   e. It ensures uniformity in the plants which enables easy mechanization of farm operations.

5. A runner is a stem that grows horizontally on the surface while a rhizome is a thick, horizontal underground stem.

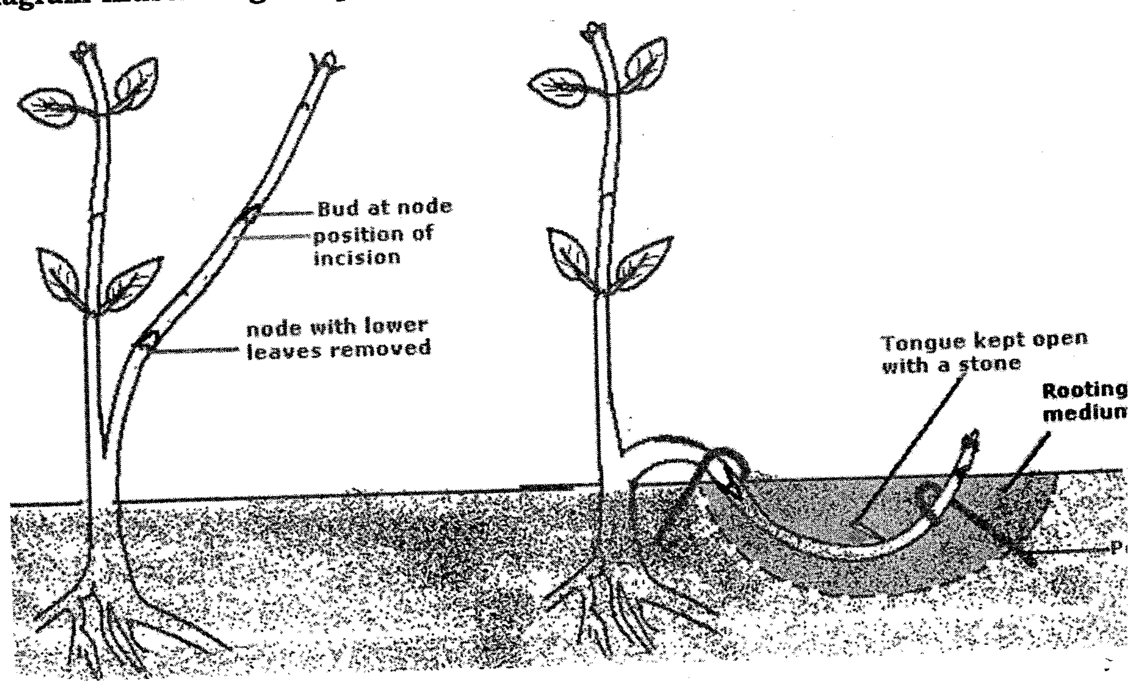
6. Budding involves removing a bud and the bark around it with a sharp knife from one tree and placing it on another tree of the same species or a closely related plant which is called a stock. Budding is basically done to ensure that the plant that is developed has got a strong root base. As such, when obtaining a bud, it should be from a plant that has good qualities other than a vigorous stem. On the stock, a T- shaped incision or cut is made into which the bud is inserted. To ensure that the bud develops, the incision in the stock must reach to the cambium layer. The bud is then inserted and held in position with the aid of a tape and wax is applied to prevent rotting.

7. a. **Diagram illustrating the procedure in grafting.**



Grafting is a method of plant propagation in which a shoot system of one plant is joined into the root stock of another plant of the same species. Initially, grafting involves obtaining an upper stem of one plant called a scion and fixing it into the base and root stock of another plant called a stock. For grafting to be successful the following should be undertaken. First, the cambium layers of the stock and scion should be joined together. To achieve this, the two plants that are used should be of the same diameter. Secondly, a V shaped cut should be made on both ends of the scion and stock. This ensures that the two plants are firmly fixed into position. A water proof tape is then wound around the point of union to avoid rotting of the wound.

b. **Diagram illustrating the procedure in layering**



This method aims at inducing root development on a stem that is still attached to t parent plant. To achieve this, an incision (tongue) is made on a stem just below a node order to expose the cambium where cell division would eventually give rise to roots. Th cut is kept open by placing a small stone or moss plants. This part is then buried into the soil in which there is a rooting medium like sand and leaf mould mixture to induce further rooting. The stem is kept in position using pegs. After some time, roots develop on the node that is buried into the ground. The stem is then cut and used for further propagation.

2.3 Essential plant nutrients

1. a.

Macro plant nutrients	Micro plant nutrients
◆ Nitrogen	◆ Boron
◆ Phosphorus	◆ Zinc
◆ Potassium	◆ Copper
◆ Sulphur	◆ Iron
◆ Magnesium	◆ Molybdenum
◆ Calcium	◆ Manganese
◆ Carbon	
◆ Hydrogen	
◆ Oxygen	

b. i. Artificial fertilizers such as CAN, Urea, superphosphate, Muriate of Potash e.t.c
ii. Organic matter such as farm yard manure, green manure and compost manure, crop residues.
iii. Weathering of mineral bearing parent material.
iv. Nitrogen is also fixed in the soil by Rhizobium bacteria as well as through lightening.

2. a. i. Used in the formation of chlorophyll.
ii. It is an essential component of the protein molecule.
iii. It encourages vegetative growth.
iv. It increases the size of grains in cereals.
v. Promotes succulency in plants such as cabbages and carrots.

b. i. It encourages excessive succulency which causes grain crops to fall down.
ii. It weakens stems and fruits.
iii. The leaves are scorched.
iv. Delayed maturity.

3. a. Plant absorption: as plants are growing in the soil, they make use of dissolved nutrients from the soil for use in various plant processes.

b. Leaching is whereby plant nutrients are washed downward of the soil profile where plants cannot access them. Once the fertilizer is dissolved, water that infiltrates into the soil will carry along with it the dissolved nutrients.

c. Some of fertilizer may also be removed from the soil through soil erosion. This occurs in areas where running water is carrying away soil particles from the garden.

4. a. i. It helps in the formation of root nodules on leguminous plants
ii. It forms an essential component of amino acid molecules such as cystein and cystine which are responsible for protein synthesis.
iii. Helps in the formation of oils in onions and garlic that gives them their sweet smells.

b. i. Retarded plant growth.
ii. Young leaves turn light green to yellowish colour.
iii. Plants are short and small.

c. 23:21:0 + 4S, CAN, Super phosphate and Ammonium Sulphate.

5. a. As plants are growing in the soil, they utilize nutrients from the soil causing them to be depleted.

b. Removal of the top fertile soil by wind and water result into washing away of nutrients from the soil.

c. This is very common in high rainfall areas and coarse textured soils. The rain water dissolves nutrients from the top soil and washes them down the soil profile where plant roots can not access them.

d. This is the loss of nitrogen in a gaseous form through the works of microorganisms.

e. Plant nutrients are made available to plants in certain simplified forms which the plants can easily absorb. However, due to certain reasons, the nutrients are converted into forms that plants can not absorb them.

6. a. Nitrogen deficiency is characterized by a gradual change in colour in the older mature leaves from their normal green colour to uniformly yellow colour.

b. Potassium deficiency is characterized by a yellowing of leaf margins in mature leaves which is later followed by interveinal scorching towards the midrib.

c. The leaves deficient in magnesium are characterized by spotted/mottled chlorotic areas in areas between the veins (interveinal chlorosis) while the veins remain green.

7. a. Magnesium

b. i. Applying inorganic fertilizers.
ii. Applying organic fertilizers.
iii. Applying dolomite lime.

Practical

8. a. i. Specimen Y (Urea): dissolves slowly.
ii. Specimen Z (CAN): dissolves very fast.

b. i. High rainfall areas: Y (Urea).

ii. Low rainfall areas. Z (CAN).

c. In high rainfall areas specimen Y (Urea) cannot leach very easily because it dissolves slowly while Specimen Z (CAN) can easily be dissolved with the less moisture in low rainfall areas.

d. Specimen Z (CAN).

e. Specimen Z (CAN) is able to dissolve very fast and therefore can easily be absorbed by the plant.

9. a. i. Plot 1: Potassium.
ii. Plot 2: Calcium.
iii. Plot 3: Phosphorous.
iv. Plot 4: Boron.

b. Plot 1(potassium)
Potassium is essential in:
i. Strengthening of cellulose in stems and roots to reduce lodging.
ii. Formation of chlorophyll than brings about the green colour in plants.
iii. Increases disease resistance.

Plot 2 (Calcium)
Calcium is essential for:
Cell division that encourages growth in apical meristems.

Plot 3 (Phosphorous) essential for:
i. Promoting root and shoot development.
ii. Promotes maturity of plants by encouraging seed formation and fruit ripening.

Plot 4(Boron)
Boron is essential for:
i. Promoting water absorption.
ii. Promoting translocation of sugars.

c. i. Crop removal.
ii. Leaching.
iii. Soil erosion.
iv. Fixation.

d. i. Crop residues.
ii. Inorganic fertilizers like muriate of Potash.
iii. Weathering of Potassium bearing rocks like biotites and feldspars.

2. 4 Crop protection: Weeds and weeding

1. Advantage(s)

i. Biological weed control uses natural enemies that are usually found in the local environment.
ii. It requires less labour for introducing the natural enemies into the field
iii. Use living organisms that are less hazardous to the environment.

Disadvantage(s)

i. It is difficult to breed organisms that can only attack the weeds alone.
ii. It requires careful attention in maintaining a balance between the pests and the weeds.
iii. It requires thorough knowledge on the behaviour of the pest and the weeds.
iv. The pest may attack other crops in near by plots.

2. a. Chemical control

This is a method of weed control in which chemical substances called herbicides are used to poison and eventually kill the weeds. The herbicides achieve this in a number of ways such as through translocation and contact. Translocated or systemic herbicides are absorbed through the leaves or roots and translocated to all parts of the plant through the xylem and phloem vessels. This translocation ensures that the whole plant is poisoned and killed. Contact herbicides on the other hand require that the herbicides be in contact with the weeds. Upon contact, the herbicides diffuse into the weed and kill them. Other types of herbicides help in preventing the growth of weeds in the garden. These are applied to the soil. Chemical weed control is an effective method of controlling weeds in the garden because it has immediate results.

b. Biological weed control

This is achieved through the introduction of living organism into the garden. These may be pests that feed on the weeds or pathogens that can transmit diseases to the weeds. Additionally, certain parasitic plants can also be used to control the weeds. The use of these living organisms makes biological weed control an effective method in ensuring environmental sustainability as no chemicals are used. However, farmers must exercise biological weed control with caution for a number of reasons. For example some of the pests that are introduced may also attack cultivated plants. In other cases, the introduction of biological control agents may also lead to the multiplication of other harmful organisms that can damage crops. This may happen if the biological agents are preys of other organisms.

c. Mechanical weed control

This method involves the use of machineries to remove weeds in the garden. The machineries are usually operated by man. Use of machineries helps to complete weeding operations on the farm in good time. However, it is more economical only where the land holding is very big. Another disadvantage of the method is that the machineries can cause physical damage to the crops.

3. Table showing advantages and disadvantages of mechanical and chemical weed control methods

Method	**Advantages**	**Disadvantages**
Mechanical control	◆Very fast	◆Expensive
	◆Less tiring	◆Requires skill to operate
	◆Enables cultivation of large areas	◆Can cause damage to crops
Chemical control	◆Very quick	◆High concentration can damage crops
	◆Reduced labour demand	◆Lead to environmental pollution
	◆Ensures timely weed control	◆Require repeated application

Essay

4. *The first effect is that weeds compete heavily with cultivated plants for sunlight, moisture, nutrients and space. Since the weeds are better adapted to competition for these resources, they are able to get established earlier. As a result, the weeds end up choking up the crops leading to low productivity.*

Weed infestation in crop fields requires the use of control measures to overcome them so that high yields are maintained. The use of most of the weed control measures involves expenditure. Any expenditure on weed control has an implication on the total cost of production. It can therefore be stated that any level of weed infestation increases the cost of producing crops.

During harvesting, weeds become mixed up with cultivated crops. This increases the impurities in the crops, thereby reducing the quality of the plants. Low quality crops end up being sold at lower prices on the market.

Weeds act as volunteer plants to certain pests and pathogens, offering them food and shelter. This enables the pests to multiply to serious levels that greatly reduce plant yields.

Some weeds like witch weed are parasitic. Such types of weeds absorb food from the host plant. This end up in starving the host plants, thereby reducing productivity.

5. *Mulching involves covering the soil with vegetative materials like leaves and straws. This method helps to prevent weeds growing beneath the soil from receiving sunlight. If the weeds are unable to get sunlight, they fail to conduct photosynthesis which is the food making process in plants. The result is that the weeds eventually die.*

   Correct plant spacing ensures that the soil is adequately covered. In so doing, weeds that grow below the plants are unable to receive sunlight necessary for photosynthesis.

   Crop rotation involves alternating the growing of crops on a piece of land in a rotating sequence. This is an effective method of controlling host specific weeds like witch weed that attack maize.

   Flooding is a practice that works effectively in controlling weeds in rice fields. If the soil is fully submerged in water, most terrestrial weeds are suffocated as these are not adapted to living in aquatic environments.

   Burning of the fields is practiced in order to destroy weeds and their seeds. Complete destruction of the weeds help to eliminate the weeds in the garden. However, burning should be practiced with caution as it causes destruction of organic matter and beneficial soil microorganisms.

2.5 Crop protection: plant pests and diseases

1. a. Crop protection refers to keeping crops safe from weeds, pests and diseases that reduce crop yield.

   b. A pest is a living organism that causes damage to crops while a disease is a disorder in a living organism that has a specific cause and recognizable signs and symptoms.

2. This stems from the fact that pests and diseases cause a lot of damage to crops leading to reduced quality and quantity of the crop yield. The reduced yield quantity and quality means that the profitability of the farmer is also drastically reduced.

3. a. This is a method of pest and disease control that makes use of the normal crop husbandry practices in crop production. The method aims at reducing the multiplication and spread of pests and diseases.

   b. This is a method of pest control in which natural enemies of pests are introduced in the field in order to attack or cause diseases on the pests. The natural agents that are used include predators, parasites and pathogens. Due to its emphasis on use of biological agents to control pests, this method is regarded as one of the best methods in pest control as it does not cause pollution in the environment. Despite this, the method is very slow in bringing about the results. In addition, careful monitoring of the biological agents is required once they are introduced as they may as well cause damage to crops and other beneficial organisms in the environment.

c. In this method, farmers make use of pesticides which are inorganic substances to kill the pests at different stages of their lifecycle. Chemical control is regarded as a quickest way of controlling pests as more pests can be killed within a short period of time over a very large area. The major set back of this method is that apart from killing the pests, it may also affect other beneficial insects.

4. This is a pest control method that makes use of several control techniques simultaneously in order to provide the best control for pests. When used effectively, the integrated pest control method has long term impact. It emphasizes more on use of cultural methods, biological control and physical control measures whenever resources permit. Where ever applicable, legislative measures that help to reduce multiplication of pests are enforced. The use of chemical control is not encouraged as it has short term impact.

5. a. This restricts the importation of any material that is suspected to contain pests and pathogens. In cases where such materials have been imported, the laws provide for the inspection of the materials.

   b. This requires reporting certain serious pests like red locusts and army worms to authorities. It has an advantage of controlling the pests before they multiply and cause serious damage to crops.

6. a. i. The excreta from the weevils taint the colour of the maize grain thereby lowering its quality apart from making it unpalatable.

   ii. The larva of a weevil bores into the grain and feed on the endosperm. The adult weevil has biting mouth parts with which it damages the grain. The damaged grain is then exposed to further attacks by other insects like the red flour beetle that feed on grain dust. The result is a very big reduction in the yield of the grain.

   iii. Respiration of the weevils can raise temperature and humidity of the granary, thereby causing the stored maize to start germinating and developing moulds.

   b. i. Storing clean maize grains with very little moisture.
   ii. Fumigation of the granaries/silos with right chemicals.
   iii. Using air tight storage facilities.
   iv. Disposing of infested grains in air tight containers

7. a. i. M is a termite.
   ii. N is an aphid.
   iii. is a grasshopper.

   b. Pest M.

   c. i. Avoiding banking in dry weather.
   ii. Harvesting the maize cobs when they are fully dry.

   d. i. Collecting and destroying the eggs so that they fail to hatch into larva.
   ii. Destroying the larva or nymphs so that they fail to mate and reproduce.

e. i. They transmit diseases to crops that lower crop yield.
   ii. They suck cell sap and other juices from plants causing them to die.

f. i. Using chemicals (insecticides) that help to poison and kill the insect.
   ii. Using predators such as the lady bird beetle whose adult and larvae are known to prey on aphids.

8. a. Mango scales, Mango stone weevil.

   b. i. Help to keep away pests and disease causing organisms from entering the storage facility and causing damage to the stored produce.

   ii. It helps to starve and eventually kill pests by destroying crop residues that can provide food and shelter and breeding ground to the pests

9. The piercing mouth parts of aphids suck cell sap and cell juices from plants. The sap contains food nutrients for the plant which help to support plant growth. If a plant is heavily infested by aphids, more of this sap is taken up and cause the plants to droop, wilt and eventually die.

   As the aphids are feeding on the plants, they tend to produce sticky waxes on which sooty moulds can develop. The sooty moulds interfere with photosynthesis as they block sunlight from reaching the leaves. The effect of this is failure of the plants to manufacture their foods resulting into loss of plant vigour.

   Aphids also transmit viruses to vegetables that cause diseases. These diseases lower the quantity of vegetables harvested.

   The saliva of some aphids is toxic to vegetables. When these toxins are injected into the plant, they cause distortion and discoloration of the leaves. This lowers the quality of the leaves.

   The control of aphids also has an economic implication. Some methods of controlling aphids like chemical and physical require some money to be invested. The implication of this is increased costs for producing vegetables.

2.6 Cropping systems

1. a. Mixed cropping ensures adequate ground cover which offers maximum protection to the soil against soil erosion that is caused by rain drops and running water.

   b. If legumes are included in a garden, they help to replenish nitrates that are used up by other plants like cereals.

2. a. This system of farming involves continuous cultivation on a piece of land until the soil is exhausted and temporarily abandoning it for another land. The land is abandoned temporarily to enable it to regain fertility. Once this fertility is regained, the farmer returns to it and start cultivating on the land. This is why it is also called land rotation. This system of farming is restricted to areas with very low population densities where land is plenty. The major set back of bush fallowing is that it encourages deforestation apart from accelerating land degradation.

b. The system involves cultivating a piece of land continuously and abandoning it once the soil has become exhausted. When opening up a new piece of land, the farmer is involved in clearing the land from trees and any other vegetation. The cleared vegetation is then burnt out on the land. For this reason, shifting cultivation is also known as slash and burn method of cultivation. The ash produced from the burning vegetation acts as a source of potash which helps in increasing yields in the early years. However in the later years it is characterized by a tremendous reduction in crop yields due to continue cropping. Just like with bush fallowing, shifting cultivation is suited to areas where land is abundant.

c. This is a system of growing crops together with trees. The advantage of including the trees is that they help in improving fertility of the soil. In addition, the trees provide maximum protection to the soil against agents of erosion due to the deep penetrating roots. Furthermore, the cultivated crops receive protection from the trees against strong winds that cause plant lodging and other extreme weather conditions.

d. This system of cropping involves dividing a piece of land into different plots on which crops are planted in a definite sequence. It has an advantage of maintaining and improving soil fertility.

3. The first way is that organic farming uses cheap inputs such as compost manure and farm yard manure that can be made locally. These inputs are useful in adding fertility of the soil that help in increasing crop yield.

The second way is that organic farming help to effectively protect crops from damage that can be caused by weeds, pests and diseases through a combination of control methods such as cultural, physical and biological methods. These methods help in timely control of weeds, pests and diseases leading to increased crop yields.

Farming the organic way helps in the effective management of the soil. Practices such as crop rotation, mulching and addition of organic manures helps to improve soil structure as well as reducing soil erosion. These practices help in keeping the soil productive for long periods which lead to increased crop yields.

The other role is that organic farming practices such as mulching and addition of organic matter help in the conservation of soil moisture. This moisture helps in dissolving plant nutrients that are absorbed and used up by the plants.

Organic farming uses practices that do not pollute the soil and the whole environment because of less emphasis on chemical usage. This helps to keep the environment in a state of balance that can adequately support plant growth.

Practical

4. a. i. A is crop rotation system.
   ii. B is monocropping.

b. Cropping system A.

c. Witch weed is a parasitic weed whose specific host plant is maize. By rotating maize with other plants the witch weed can fail to get food for its survival from the non host plants. The weed may eventually die due to lack of food.

d. Cropping system B.

e. In cropping system B most of the farmer's time and resources are concentrated on one crop. In this regard, the farmer is able to gain more skills on how best the crop can be managed.

5. a. Monocropping requires less capital investment in buying inputs because one crop is grown. This is true where monocropping is practiced on a very large farm where inputs are bought in large quantities at whole prices which are reasonably lower. In contrast, mixed cropping involves large capital investment for buying different inputs in smaller quantities at retail prices that are normally very high.

   b. Under monocropping, it is very easy to use different machinery simply because the farms are usually very big. Since only one crop is grown, the machine can be permanently adjusted to suit the crops under cultivation. On the other hand, machine use under mixed cropping is made difficult because different crops with different requirements for given types of machinery are grown. This makes it necessary to adjust the machine each time when working with a different crop.

   c. Monocropping is highly risky in the instances of droughts, heavy rainfall or lower prices for produce. Under these types of circumstances the farmer is likely to lose since only one crop is under cultivation. In case of mixed cropping, where different crops are grown, the farmer may have a chance to harvest and sell some of the crops whose production has not been affected by climatic or economic factors. This is because different crops are affected differently by these factors.

6. a. Crop rotation is a system of farming that can ensure food security because it efficiently utilizes the soil by among other things improving soil fertility. Due to this increased fertility, a farmer is able to harvest more yields. On the other hand, shifting cultivation is a system that encourages soil exhaustion as the land is continuously cultivated without any effort to improve fertility. As a result, shifting cultivation is characterized by progressively declining yields that can not ensure food security.

   b. Shifting cultivation is a system that encourages deforestation because farmers are always on the move searching for new pieces of land to cultivate. When on these new pieces of land, a lot of trees are cut down leading to more areas becoming bare. On the other hand, farmers practicing crop rotation utilize smaller pieces of land where fewer trees are cut. For this reason, crop rotation does not encourage deforestation

   c. Shifting cultivation is characterized by increased loss in soil fertility because the land is continuously cultivated until fertility is lost. It also encourages loss of beneficial organic matter content of the soil, especially where the cleared vegetation is burnt out by fire. In contrast, crop rotation is a practice that has a lot of measures that encourage soil conservation. For instance, inclusion of leguminous plants in rotations helps in fixing nitrates in the soil, while as the alternation of deep and shallow rooted plants help in protecting soils from erosion.

7. a. i. Deep rooted plants should be followed by shallow rooted plants.
   ii. Crops with adequate ground cover should be followed with those with less ground cover.
   iii. Crops that use a lot of plant nutrients should be followed by those that absorb less nutrients from the soil.
   iv. Crops that belong to the same family should not succeed each other.

b. Completed crop rotation sequence for years 2, 3 and 4

	Year 1	Year 2	Year 3	Year 4
Plot 1	Maize	Cassava	Cotton	Groundnuts
Plot 2	Cassava	Cotton	Groundnuts	Maize
Plot 3	Cotton	Groundnuts	Maize	Cassava
Plot 4	Groundnuts	Maize	Cassava	Cotton

8. a. Mixed cropping.

b.i. It makes effective use of soil nutrients in that when deep rooted plants are included, they absorb nutrients that are leached down.

ii. If fibrous rooted plants are included, their roots hold soil particles together thereby preventing soil erosion.

iii. Inclusion of leguminous plants in mixed cropping help to raise the nitrogen content of the soil through the activities of rhizobium bacteria that are contained in the root nodules of the leguminous plants.

iv. It slows down the spread of certain pests and diseases as some of the plants act as physical barriers to pests and disease causing organisms.

v. Some crops are resistant to drought, pests and disease attack. As such, including them in mixed cropping help to lower the risk of total crop failure that may arise in such incidences when only one crop is grown.

vi. Early maturing crops help to safeguard against food shortage as farmers are able to harvest earlier.

vii. Farmers obtain higher total yields per unit area.

viii. It makes efficient use of labour as most farm operations are done all at once.

c. i. Rab processors limited.
ii. Bakhresa Grain and milling company.
iii. Chibuku Products limited.

e. Carbohydrates.

2.7 Fruit production

1. a. Domasi, Dodo, Kent, Anderson, Davis-Haden, Palmer, Zill.

b. Fruit fly.

c. It lays eggs on the fruit which upon hatching give way to shiny white maggots that enter the fruit causing them to change colour before ripening as well as causing flesh parts of the fruit to become liquid.

d. Collecting and burying the fruits to prevent the maggots from developing into adult fruit flies.

2. **a. Altitude**
   Mangoes do well in low altitude areas below 1500m above sea level. These areas are important because they provide hot conditions that are necessary for the ripening of the fruits.

   b. Soil type
   Mangoes prefer well drained fertile soils such as sandy loams. These are very important soils because they enable free root penetration and development of the mango tree roots.

   c. Soil pH
   This should be within the range of 5.5 to 7.5 as this is the optimum range in which most of the essential plant nutrients are in forms that are readily available for crop absorption.

   d. Rainfall
   Mangoes do well in moderate rainfall areas. The minimum annual rainfa requirement is 650mm. However, during flowering and fruiting, the rains are nc required as they may cause the flowers to fall down.

3. a. Anthracnose, Powdery Mildew and Flower Blight.

   b. i. - Larva enters fruit and attack seed.
   - Causes premature fruit fall.
   - Inside of fruits develop white hard areas.

   ii. - Attack leaves, stem and fruits.
   - They leave a sticky substance (honey dew) on leaves, stems and fruits.

   c. i. Practicing orchard hygiene.
   ii. Collecting and burying fallen fruits.

Essay

4. *Land preparation is one of the important husbandry practices in mango production. involves digging holes in which the mango trees are to be planted. When making tl holes, top soil should be placed separately from sub soil. The holes for plantii mangoes should measure 90cm x 90cm x 90cm. Spacing of the holes varies with varie and fertility of the soil. If the mango variety has less vegetative growth or the soil is ve fertile, mangoes can be planted at a spacing of 9m x 9m. But if the variety has a larg vegetative growth or the soil is less fertile, the holes can spaced at a distance of 12 apart. Before planting, the holes should be filled up with the top soil that is thorough mixed with manure. The sub soil can be used for the construction of a basin around t tree. This helps in the conservation of moisture.*

   Planting of mangoes can be done directly in the main field or in nurseries. What ever the case, planting of the mangoes in the main field should be done at the start of t rainy season. This ensures successful establishment of the fruit as it is able to fu utilize the rainy season.

   Mulching of the mango orchard is another recommended practice. Covering of the s with leaves, straws and any other suitable vegetation helps in conserving enou moisture for the young plants.

   Mango plants require a lot of protection from weeds, pests and diseases. Weedi should be done regularly to ensure that the plants do not face stiff competition sunlight, moisture and nutrients. Weeding can be done using hands slashers and hoes. Chemical weed killers may also be of greater use.

At the time of planting, fertilizer application to the mango seedlings is recommended. This may be in form of organic manure and inorganic ones. The most important fertilizers in mango production include CAN, super phosphate, and muriate of potash. These help to supply nutrients for the growth of the plant.

5. Firstly, fruits are important to farmers because they provide them with income though selling.

   Secondly, fruits are important because they provide raw materials to fruit processing industries. These companies process the fruits and sell them to both local and international markets.

   Fruits also provide people working in orchards and fruit processing industries with employment opportunities. Through such employment, the people earn salaries which help to promote their living standards.

   When fruits and their products are exported to other countries, they enable a country to obtain foreign currency. This foreign currency is very useful when a country would like to obtain foreign goods that are not manufactured locally.

   Lastly, when eaten, fruits provide people with valuable food nutrients, minerals and vitamins . These food nutrients are very important in promoting health.

6. To start with, diseases in mangoes greatly reduce the quantity of the fruits that is realized. This happens because some diseases like powdery mildew cause the flowers and fruits to fall before ripening.

   Secondly, the diseases also reduce the quality of the mango fruit. The lowered quality results because the fruits are made to ripe prematually apart from rotting as is the case with fruits attacked by anthracnose.

   Some mango diseases attack the leaves such that the ability of the plant to manufacture food is reduced. Serious cases of disease attack may therefore lead to drying and dying of the mango tree.

   The lowered quantity and quality that results due to disease attack affects the profitability of the mango enterprise. Basically, the amount of money that farmers fetch after selling mangoes is very little due to the seasonal nature of the mangoes. This problem is therefore compounded in situations where the farmers have failed to produce more quality fruits due to disease attack.

   Lastly, some of the measures of controlling disease are very costly. For example, control of anthracnose and powdery mildew requires the use of chemicals that are expensive. This may make the production of mangoes to be a less attractive fruit enterprise.

2.8 Crop improvement

1. a. i. To develop plants that are able to produce high yields.
   ii. To develop plants with improved quality .

   b. i. To increase the biomass.
   ii. To increase partition.

iii. To increase ability to resist pest and disease attack.
iv. To promote uniformity.
v. To increase nutritive value of crops.

2. a. This involves importation of plants from their native lands into new areas. For the introduced plants to perform better, the environmental conditions in the new land must be similar to those of the original land where the plants were grown.

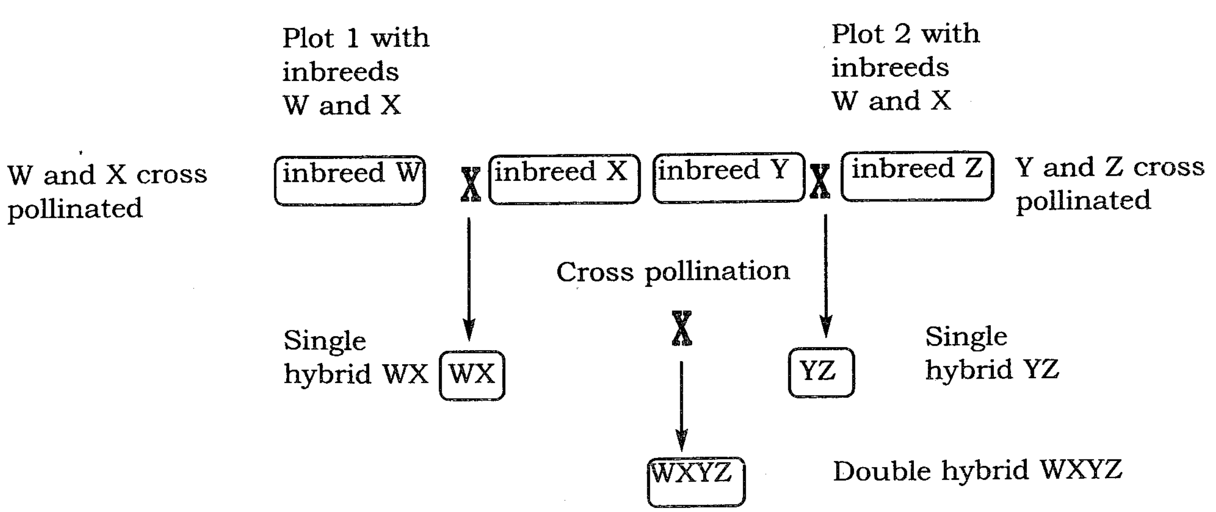
   b. This involves identifying and choosing plants that have superior characteristics for breeding. It is based on the notion that within any given population of plants, there are bound to be individual differences. The farmer therefore, observes and identifies those plants that are outstanding in the population and use them for breeding. Examples of characteristics that farmers look for in plants include high yielding, early maturity and high disease resistance. Although plant selection does not add any new characteristics in the plants, it is very helpful in promoting the desirable traits that are already present in the plant population.

3. a. This is concerned with identifying plants with desirable traits to be used for breeding. The parent plants should be those with complementary characteristics. For example, a high yielding plant that is susceptible to disease attack can be crossed with a low yielding but disease resistant plant. This can result into the production of a high yielding disease resistant plant variety.

   b. Once identified, the parent plants are successively self-pollinated in separate fields for up to six generations. The aim for this is to completely get rid of undesirable traits that are in the parents. This however, results into production of pure breeds that are characterized by reduced plant vigour.

   c. This stage allows the superior genes from the two parents to combine and form a new plant. It is achieved through crossing the pure lines from the two parent plants. The newly produced plant is rather characterized by increased performance than the parent plants. The success of producing the hybrids relies very much on preventing self pollination of the pure lines by killing the male part of the plant. In addition, there is need to prevent cross pollination of the pure lines by foreign pollen grains. This can be achieved by covering the female flowers with a plastic paper.

4. **Illustration of hybridization**



From the diagram above, the farmer should first of all cross breed inbreed W with inbreed X in one field while another field should be planted with inbreed Y and Z. The result of this cross breeding will be the production of maize varieties that have desirable characteristics. For instance, the crossing of inbreeds W and X will produce a variety of maize WX that is characterized by early maturity and high resistance to maize streak virus. In the same way crossbreeding inbreeds Y and Z results into maize variety YZ that is short and high yielding. If the farmer further cross pollinate in breed WX with YZ, a new variety of maize WXYZ is produced. This variety will be characterized by early maturity, high resistance to maize streak virus, short height and high yielding.

2.9 Crop storage and processing

1. a. i. It helps to add value to the products.
   ii. It provides employment to people working in the processing industry.
   iii. It helps to increase foreign currency earnings through exports of products from processed mangoes.

2. a. i. By extracting oil from the groundnut seed and using the remains as groundnut seed cake.

   ii. By mixing it with other feed stuffs to produce balanced feed rations.

3. a. Tomato sauce and Jam.

   b.i. It adds value to the tomatoes thereby increasing the profits made by the tomato processing industries.

   ii. It helps in preserving the tomatoes so that they are kept for long period of time without getting bad.

   iii. It provides job opportunities to people working in the processing companies thereby enabling them to earn incomes for uplifting their lives.

2.10 Pasture production

1. a. Improved pastures and natural pastures.

   b. i. Rhodes grass.
   ii. Guinea grass.
   iii. Buffel grass.
   iv. Bushman Mine Panic.

   c. i. Improved pastures have a higher percentage of proteins than natural pastures.

   ii. Most improved pastures have a long juvenile period. This ensures that these pastures are able to maintain their digestibility.

2. a. Pastures provide vegetative cover to the soil which helps to protect the soil from the impact of heavy rain drops.

   b. The roots of pastures help to bind the soil particles together thereby preventing the soils from getting eroded.

   c. They also help to reduce the flow of running water so that the rate at which soils are eroded is reduced.

3. a. The method is very wasteful in that a lot of seeds are required to cover a unit area of land.

   b. It results into uneven distribution of the plants leading into overcrowding of the plants in one place. Overcrowding of the plants result into poor pasture development as the plants struggle to get enough sunlight, moisture and nutrients that are vital for proper plant development.

   c. Some farming operations like weeding and fertilizer application are made more difficult, especially in places where the plants are overcrowded.

4. a. The selected pasture species should be those that can successfully grow together. For example, legume species with a climbing growth habit like silver leaf grow successfully with taller grass species like thatching grass. In this type of pasture stand, the legumes benefit from the grasses as they are supported to reach out for the sunlight. On the other hand, the grasses are able to utilize the nitrates fixed by the legumes.

   b. Pastures are utilized in a number of ways such as hay, silage, cut and carry. Some pasture species are best utilized in one form than the other. For example, Rhodes grasses are suitable for use as temporary pastures than permanent ones because their productivity declines after four years of utilization.

5. a. It provides a cheap source of livestock feed especially during the dry season when there is less pastures.

   b. Surplus hay can be sold out to other farmers, hence providing incomes to farmers.

   c. Properly stored hay is able to maintain its nutritive value, which boost livestock production.

6. a. The land for pasture establishment should be prepared early. This is important because it helps the cleared bush to have more time to decompose so that soil fertility is improved. Additionally, the farmer is able to sow the pasture with the first soaking rains, thereby enabling the pastures to make full use of the rainy season. When preparing seed beds for sowing pastures, the farmer should also ensure that the surface is made to a fine tilth. This is important because it enables the seedling to emerge without difficulties and so reduce the seed rate.

   b. When selecting pasture, the farmer should go for species that are of higher quality and able to give high yields. Such types of pastures are important in that they help to increase productivity in the livestock. In addition, farmers should select pastures that can withstand the local climatic conditions as well as those that can resist parasites and diseases. This ensures that the pastures are able to give high yields that can meet the needs of the livestock through out the year.

   c. This is concerned with deciding on the quantity of seeds or planting materials to be required over a given piece of land. The amount of seeds and planting materials would vary depending on a number of factors like method of sowing. Normally, seed rate is very high when seeds are sown using broadcasting because a lot of seeds fail to germinate. Secondly, where germination percentage is very high, less amount of seeds are required because there is less need for replanting, than where the germination percentage is low. Another important factor is the type of pasture stand. Where a pure stand is established the seed rate for the established stand is high because a large area is involved. But when a mixed stand is established, the seed rate for each pasture species is reduced. Other crucial factors include growth habit of the pastures, proposed method of utilization and seed size.

d. Pasture seeds are treated in special ways for a number of reasons. The first method is called scarification and involves the softening of the seed coat in order to encourage germination. Some leguminous seeds like silver leaf and stylo have hard pods that delay germination. To encourage germination, the pods of these plants are removed through a method called hulling. This has another advantage of ensuring even distribution, especially where broadcasting has been used to sow seed. Inoculation is another method for treating leguminous seeds. It involves coating leguminous seeds with a right specie of rhizobium bacteria. The advantage of this is that it promotes nodulation and nitrogen fixation. Lastly, there is pelleting in which the seeds are mixed with substance like lime, gypsum and rock phosphate in order to improve soil pH and nutrients deficiencies.

e. Pastures are usually sown at the start of the rainy season to enable them to fully utilize the short rainy season. Pastures can be planted using either seeds or vegetative planting materials. There are a number of methods that can be used when sowing pastures. The first method is broadcasting. It involves scattering pasture seeds on thoroughly prepared seed beds using hands or fertilizer spreaders. Effective broadcasting requires that the seeds be firstly mixed with materials like sand or fertilizer. This ensures even distribution of the seeds. Raking of the soil after broadcasting ensures even distribution of the seeds. Broadcasting as a method of sowing pastures is very wasteful as more seeds may fail to germinate due to washing away by water and wind.

Drilling is another method in which pasture seeds are sown in drills (small tunnels) that are made on the seed beds. These can be made using sticks or seed drillers. It is an efficient method of sowing pastures as the seeds are placed at a depth where they can easily germinate. It is also economical because fewer amount of seeds are required over a given area than in broadcasting.

Other pasture species are planted using parts that are obtained from the parent plant other than the seeds. This is referred to as vegetative planting or propagation. The most commonly parts used include stem cuttings and other rooted cuttings. Compared to seeds, vegetative planting materials have large food storage organs that help them to get easily established. Its shortfalls, however are that large quantities of planting materials are required per given area and that they require to be planted as soon as they are cut.

7. a. This is herbage that is cut and partially dried to preserve it for use as animal feed. Materials for hay making are cut before flowering stage because this is the time when herbage is very nutritious. Once cut, the herbage is dried (preferably sun dried) for a few days until there is very little moisture left to avoid moulds from developing on the hay for this lowers the quality of the herbage. The hay is then stored by baling or in loosely stacked piles. The hay is stored in leak proof sheds to ensure longer preservation. the quality of hay is largely dependent on a number of factors such as storage, stage of cutting and weather conditions among many other factors.

b. This is forage that is cut before flowering and preserved by the process of fermentation. The process of silage making involves cutting and partially drying the forage. The dried forage is then chopped into fine pieces. This is necessary because it enables easy compaction of the silage to drive out as much oxygen as possible. The chopped materials are later on filled in a silo where they are compressed to remove oxygen. Each layer is sprinkled with carbohydrate concentrate to provide energy for the bacteria in the silo. This is followed by sealing of the silo using a plastic sheeting to prevent oxygen from getting into the silo. The absence of oxygen in the silo makes it possible for the preservation of the silage through the work of anaerobic bacteria.

c. This is also called standing hay because it is left to dry in the field and used as animal feed during the dry season. It is the cheapest mode of preserving herbage for livestock feed. In order to produce good quality for age, farmers should ensure that they cultivate those pastures that are highly digestible even when mature as well as to include more legume species. The disadvantage of this method is that the herbage is of very low nutritive value because it is left to mature, becoming fibrous in the end.

8. a. i. Weather conditions at drying.
   ii. Time of exposure on the sun.
   iii. Stage of cutting the herbage.
   iv. Storage place.
   v. Nutrient content of the grass and legume species used.

   b. i. Speed of sealing the silo.
   ii. Nutrient content of the herbage used.
   iii. Amount of moisture in the herbage.
   iv. The pH of the herbage.
   v. Stage of growth at cutting the legume and grass.

9. a. It ensures successful nodulation that encourage fixation of nitrates in the soil by rhizobium bacteria species.

   b. It corrects nutrient deficiencies and soil pH thereby improving the establishment of legumes.

10. a. Strip grazing.

    b. i. It ensures that livestock get a fresh supply of feeds every day.
    ii. Helps in breaking the lifecycle of certain parasites as the livestock do not stay on one strip for several days.
    iii. It ensures even distribution of dung and urine all over the pasture land thereby improving soil fertility.

    c. Shortage of land.

11. Pastures provide a cheap source of livestock feeds. These pastures are locally found in the environment as compared to commercially produced livestock feeds.

    The growing of pastures helps in maintaining and improving fertility of the soil. For example, when leguminous pastures are included in mixed pasture stands they help to fix nitrates into the soil. This is due to the presence of the nodules on their roots which provide a habitat for rhizobium bacteria, which is able to convert atmospheric nitrogen into nitrates. The decomposition of the pasture plants also helps in replacing lost nutrients into the soil.

    Another role played by pastures is control of soil erosion. Pastures manage this by providing vegetative cover to the soil as well as helping in binding the soil particles with their roots. In addition, the growing of pastures on the land helps to reduce the flow of water, thereby reducing the rate of soil erosion.

    Pastures are also important in that they help in maintaining and improving the structure of the soil. As the pastures are decomposing, they increase the humus and organic matter content of the soil. This organic matter is very important in the formation of soil aggregates as they help in cementing the individual soil particles together.

Lastly pastures are beneficial for they help to control crop pests and diseases. If th pastures are rotated with other crops they can help in breaking the life cycles c certain pests and disease causing organisms. For example, eel worms that cause roo knot diseases in tobacco and other crops can be effectively controlled by rotatin these crops with love grass.

Practical

12. a. i. The maize will not be able to make full use of the rainy season because there wil be very little rains in the critical stage when the maize is about to star producing ears. This will eventually result into low yields.

ii. The maize plant will be able to fully utilize the rains such that high yields ma be attained.

b. The cow would calf in late May or early June when the pastures are dry and less abundant. The effect will be production of less milk for the calf as the pastures are less nutritious during this time.

c. i. During this period the pasture will be dry and less digestible as this is a dry period.

ii. The pastures will receive sufficient rainfall which will make the pasture to be tender and easily digestible.

d. i. The grasses are highly nutritious before flowering. This enables the silage that is produced to contain more nutrients.

- The young grasses are very tender, as such the silage that is made from such grasses are highly palatable and digestible.

ii. - This stage removes some moisture from the silage for easy ensiling.

- It also helps in ensuring that the pastures retain their succulency. This makes the pasture to be easily digested.

iii. It removes as much air as possible from the silage in order to facilitate anaerobic fermentation process which is good for preserving the silage.

13. a. Grass Pasture X:

Dry matter yield = 100

Size of land = 3 hectares

Crude protein % = 4%

Crude protein yield $= \frac{4}{100} \times \frac{100kg}{ha} \times \frac{3ha}{1}$

$= \underline{12kg}$

Grass Pasture Y:

Dry matter yield = 120

Size of land = 3 hectares

Crude protein % = 3.6%

Crude protein yield $= \frac{3.6}{100} \times \frac{120kg}{ha} \times \frac{3ha}{1}$

$= \underline{12.96kg}$

b. i. The farmer is able to formulate rations that meet specific requirements of the livestock.

ii. It helps the farmer to select appropriate grass pastures that have higher crude protein content.

iii. The farmer is also able to identify grasses that can be improved.

c. i. It helps to control pests and diseases by burning off the pests and infected pastures species. The heat from the fire also help in sterilizing the soil.

ii. It burns off unpalatable pasture species that remain ungrazed on the pasture land.

iii. It prevents bush encroachment.

d. i. It helps to increase the yield of grass pastures due to the nitrates that are fixed by the legumes. This ensures that feeds are available to the livestock throughout the year.

ii. The combined yield for the pastures is high, resulting into abundant supply of feed to the livestock.

iii. Leguminous pastures have a high protein content. Proteins are an essential component in livestock because they help in repairing of worn out tissues.

14.

$$\text{Seed rate} = \frac{\text{Expected plant population}}{\text{ha}} \div \text{Seed size} \div \text{Purity \%} \div \text{Germination \%}$$

$$= \frac{60{,}000 \text{ seeds}}{\text{ha}} \div \frac{200{,}000 \text{ seeds}}{\text{Kg}} \div \frac{75}{100} \div \frac{50}{100}$$

$$= \frac{60{,}000 \text{ seeds} \times 1\text{Kg} \times 100 \times 100}{\text{ha} \times 200{,}000 \text{ seeds} \times 75 \times 50}$$

$$= 0.8\text{Kg/ha}$$

Chapter Three

ANIMAL PRODUCTION

Questions

3.1 Livestock feeds and feeding

1. a. Mention the two major classes of livestock feeds.
   b. List down feed nutrients for livestock ,and examples of feeds that provide the nutrients.

2. Briefly explain the functions of the feed nutrients listed below.

   a. Carbohydrates.
   b. Lipids (fats and oils).
   c. Proteins.
   d. Water.
   e. Vitamins.
   f. Minerals.

3. Explain two major reasons for feeding livestock.

4. Explain any three factors which should be considered when balancing a ration.

5. a. Describe any two functions of a Soya bean meal to layers.

   b. Explain the difference between a production and maintenance ration.

6. Write an essay explaining any five factors that should be considered when feeding animals.

Practical

7. Table 1 shows information on feeding of layers. Use it to answer questions that follow.

AVAILABLE FEED	PROTEIN CONTENT
Maize meal	7%
Soya bean meal	40%

Using a Pearson square method, formulate a 990kg ration for layers containing 21% protein. Show your calculation.

8. You are provided with the following materials.
   - One teaspoonful of maize flour.
   - One teaspoonful of groundnut flour.
   - Two pieces of clear white paper.
   - Iodine solution.
   - Dropper.

   a. Take half of each specimen. Add two drops of iodine solution to each of the specimens. Observe and record the results.

i. Maize flour.
ii. Groundnut flour.

b Which of the two specimens would be used as a source of carbohydrates for pigs?

c Explain your answer above.

d Rub each of the remaining specimens against the white pieces of paper provided. Observe and record the results.

e From the results, state three ways in which agro industries can increase value of groundnut flour.

9. You are provided with the following materials:

Specimen W (Groundnut seed). = Bean
Specimen X (Star grass).

a. Explain any two roles of specimen labeled X in livestock production. (2)

b. Explain two ways in which the specimen labeled W can be processed into livestock feed.

c. In which group of livestock feeds does each of the following belong?

i. Specimen labeled X.
ii. Processed product from specimen W.

d. Explain any two situations in which a farmer would provide the group of livestock feed as that of processed product of W to beef cattle.

e. If you were to sow stylo in a pasture land containing the specimen labeled X, describe how this could be done. (2)

3.2 Livestock management and production systems

3.2.1 Sheep production.

1. a. Mention any two breeds of sheep.

   b. Name any two characteristics that a farmer may look for when selecting a breed of sheep to raise.

2. Describe any three characteristics of a good sheep house.

3. Describe the feeding behavior of sheep.

4. Describe the following three methods that are used when castrating lambs.

   a. Using a Burdizzo.

   b. Using a knife (open method).

   c. Using an elastrator band.

5. a. Name the activity that is done during docking and trimming in sheep.

   b Explain any two advantages of docking in sheep.

3.2.2 Goat production

1. Give any three breeds of goats.
2. Explain one way in which each of the following management practices can improve goat production:

   a. Keeping goats in a warm house.

   b. Providing protein concentrates to goats.

   c. Selecting an appropriate breed of goats.
3. Describe the two feeding habits of goats.
4. Explain any two reasons for keeping goats.
5. Explain any two qualities of a good goat house.
6. Explain any five characteristics of a goat to be selected for breeding.

3.2.3 Beef production

1. Give any three examples of beef breeds of cattle.
2. Explain any three advantages of stall feeding.
3. State any two ways in which castration of calves is important in cattle production.
4. Explain two benefits of dehorning in livestock production.
5. Explain any two feed requirements of beef animals.

Practical

6. Table below shows live weight gain for two exotic breeds of cattle that were stall fed using same amount of silage for 40 days. Weighing was done at 5 days interval. Use it to answer the questions that follow.

Days	Breed A (live weight gained in kg)	Breed B (live weight gained in kg)
5	1	4
10	3	7
15	4	10
20	6	13
25	8	16
30	10	19
35	12	22
40	14	25

a. Draw two line graphs on the same axes to show the live weight gained in the two cattle breeds.

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b. What is the difference in live weight gained between the two cattle breeds on day 33?

c. If you were a beef cattle farmer, explain any two advantages you would get by raising the breed B.

d. Given the following additional information on the breed B.

Total cost of feed	-K8, 000
Veterinary costs	- K2, 500
Casual labour	- K6, 500
Income obtained	- K35, 000

Calculate the gross margin for the breed B. Show your working.

7. Table 1 shows results from a stall feeding project for two different breeds of cattle of the same age and under same conditions. The project was run for 130 days. Use it to answer questions that follow.

Breed	Initial Weight (kg)	Finishing Weight (kg)
M	198	340
N	190	395

a. Calculate the live weight gain per day for each breed M and N.
b. Which breed was economical to stall feed?
c. State three reasons for the answer in b.

8. Explain any two advantages of extensive system of managing beef cattle.

3.2.4 Dairy production

1. a. Explain any two reasons for steaming up cows two months before calving.
   b. State any one advantage of colostrum to calves.

2. Explain one way in which each of the following factors would affect milk yield:

   a. Age of the animal.
   b. Milking technique.

3. Describe any two management practices a farmer should provide to a cow during gestation period.

4. a. Describe any two characteristics of a good dairy cow.

   b. List down any three ways in which a dairy farmer can ensure production of clean milk on a farm.

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Practical

5. Table below shows a record of milk production in a cow.

Time (months)	Average amount of milk per day (litres)
1	6
2	7
3	8
4	7
5	6
6	5.5
7	5
8	4.5
9	4
10	3.5
11	3
12	2.5

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e it

a. Draw a graph using the information in the table above.

b. When is milk production at its peak?

c. If this cow calves every year, when should it be allowed to have a dry period?

d. Explain two ways in which the record given in table 1 would assist a farmer in livestock management.

e. Describe two other types of records that dairy farmers may keep besides the record given in table 1.

6. A farmer kept a record of four dairy cows, namely Kambani, Jolly, Dolly and Bettie. The record shows the quantity of feed eaten by each animal per day, milk obtained from each cow per day and the calving interval for each animal. All these are shown in the table below: study it and answer the questions that follow.

NAME OF COW	QUANTITY OF FEED (Kg/Day)	MILK YIELD (Kg/Day)	CALVING INTERVAL (days)
Kambani	1.0	4.5	360
Jolly	1.0	3.9	400
Dolly	1.5	5.0	380
Bettie	1.0	4.6	340

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a. From the table, select the best yielding cow for further breeding.

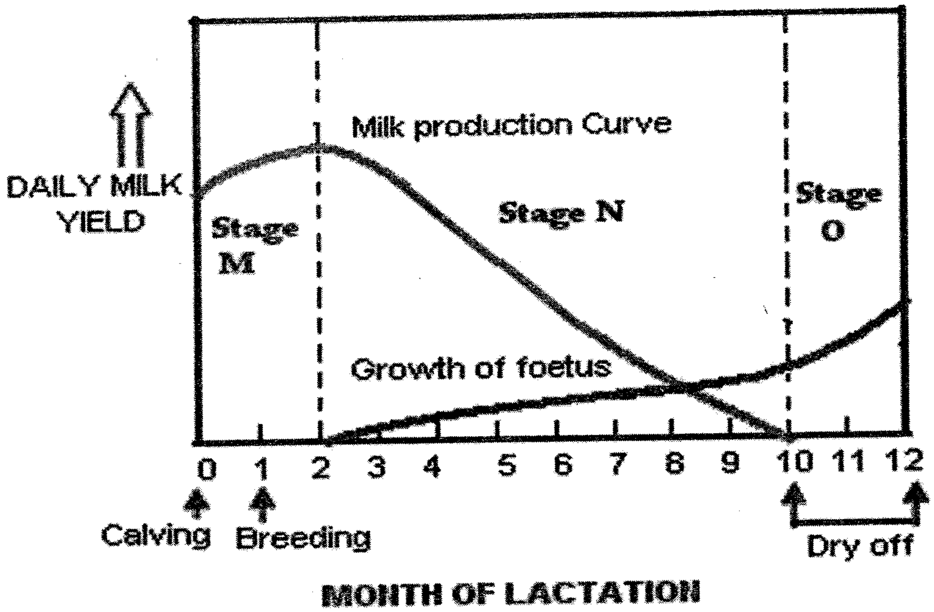
b. Explain two reasons for the answer in 3 a above.

c. State the meaning of the term calving interval.

d. Explain how a farmer may reduce the calving interval in his dairy animals.

e. Describe any two factors which affect quality of milk produced.

7. The figure below is a graph showing milk production of a dairy cow. Use it to answer questions that follow.



a. In which stage is the milk yield increasing?

b. Explain one reason for the answer in part a.

c. Explain the reason for the decline in milk yield in stage N.

d. Define the term dry off period in stage O.

e. Explain any two reasons for the two months dry period the cow requires before parturition.

3.3 Livestock parasites and diseases

1. A farmer observed the following signs of diseases attack in goats:
   Swollen feet.
   Difficulty in walking.

   a. Name the disease the goat was suffering from.

   b. State one way of controlling the disease.

2. a. State any four symptoms of mastitis in a cow.

   b Explain any two ways of controlling mastitis in a cow.

3. a. State four symptoms of pneumonia in goats.

   b Explain any two ways of controlling pneumonia in goats.

4. A farmer observed the following symptoms in sheep.
   Continuous salivation.
   Wounds and blisters in the mouth and feet.

   a. What disease was the sheep suffering from?

   b. State any three ways of controlling the disease.

5. a. List down any three symptoms of brucellosis in cows.

   b. Name any two ways of controlling brucellosis in cows.

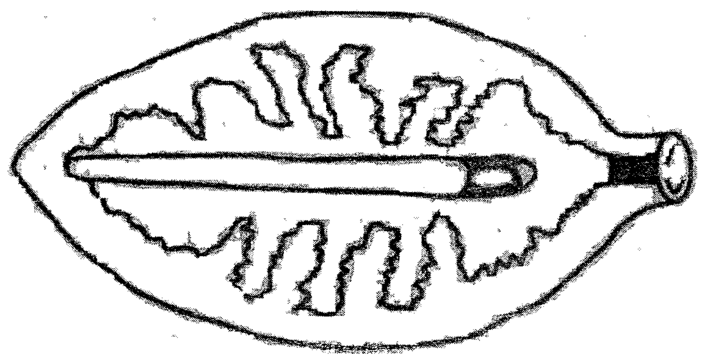
6. Explain any two ways in which trypanosomiasis affect cattle production.

7. Explain how each of the following practices helps to control livestock diseases.

   a. Vaccination.

   b. Sanitation.

8. Study the figure below of an animal parasite and answer the questions that follow.



   a. Name the parasite.

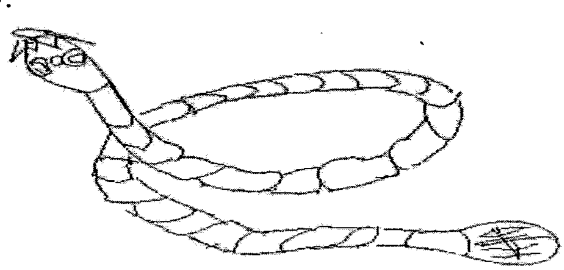
   b. Describe any four symptoms which would be observed when an animal is attacked by the parasite.

   c. Give three ways of controlling the parasite.

9. a. Explain two ways in which this parasite affects animal production.

   b. Name three animals that are attacked by this parasite.

10. The figure below is a diagram of a livestock parasite. Use it to answer the questions that follow.



   a. State any two ways in which the parasite affects livestock production.

b. Explain two ways of controlling the parasite.

11. Describe any two harmful effects of internal parasites on farm animals.

12. a. Describe the life cycle of a two host tick.

b. Explain any two ways in which knowledge of a two host tick life cycle can be us in the prevention and control of the parasite.

3.4 Anatomy and physiology of reproductive systems of livestocl

1. The table below shows reproductive cycles of four types of farm animals. Use it answer questions that follow.

Type of animal	Length of Oestrus cycle (days)	Duration of oestrus cycle (hours)	Length of gestation (days)
Cow	21	18	283
Nanny	21	36	150
Sow	21	48	115
Ewe	17	36	150

a. Which animal has the longest heat period?

b. Explain one advantage of having a longer heat period.

c. A cow was on heat on 27th May 2008 but was not served.

i. State any three ways in which a farmer would have known that the cow was heat.

ii. What was the next appropriate date for serving the cow? Show your working.

d. A farmer plans to have his ewes give birth on 31st December 2009.

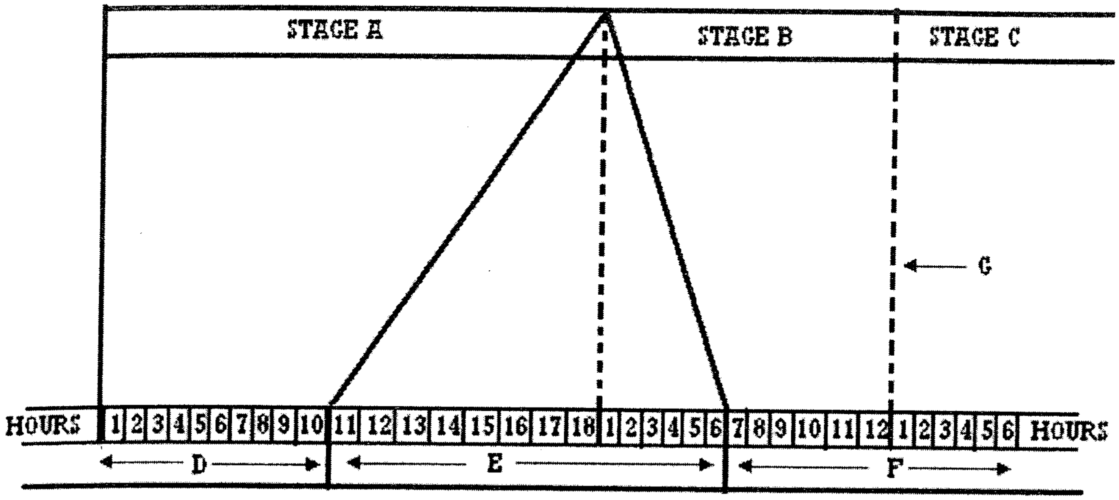
i. When should he allow the ewes to mate? Show your working.

ii. Explain one advantage of planning to have ewes lamb on 31st December 20(

2. a. Mention any four signs of calving in a cow.

b Explain any two causes of difficulties experienced by a cow during parturition.

3. The figure below is a timing chart for servicing cows. Use it to answer the questions that follow.



a. Identify stages A, B and C.

b. What process occurs at G.

c. What name is given to an activity that encourages the process that occurs at G.

d. State any three signs of the onset of stage A.

e. Explain the effect of inseminating the cow in the following stages.

i. Stage D.
ii. Stage E.
iii. Stage F.

4. State the importance of each of the following activities in livestock production:

a. Flushing.

b. Weaning.

5. Describe any two factors that limit the use of artificial insemination by a smallholder farmer.

6. With the aid of a well labeled diagram, describe the reproductive system of a hen.

7. With the aid of a well labeled diagram, describe the reproductive system of a bull.

3.5. Livestock improvement

1. Give the three methods of livestock improvement.

2. Mention any four examples of desirable characteristics that farmers look for when selecting livestock for breeding.

3. State any five aims of livestock improvement.

4. Explain how selection helps in improving livestock production.

5. In livestock production, describe the roles played by the following factors.

   a. Genes.

   b. The environment.

6. Differentiate between progeny testing and sib selection as used in animal breeding.

7. State any two advantages of cross breeding as a method of improving livestock.

8. Describe each of the following breeding systems in cattle.

   a. Inbreeding.

   b. Cross breeding.

9. A farmer would like to cross breed a bull with dominant genes for high milk production and a cow with recessive genes for low milk production.

   a. Given that "M" is a dominant gene for high milk production and "m" a recessive gene for low milk production, use a diagram to show how the cross can be conducted and the offspring that will be produced.

   b. From the diagram above, what can you say about the milk production in the offspring.

   c. Explain any one advantage of crossbreeding in cows.

   d. Give one limitation of cross bred animals.

Chapter Three

ANIMAL PRODUCTION

Answers

3.1 Livestock feeds and feeding

1. a. Roughages and concentrates.

   b. livestock feed nutrients and their sources.

Feed nutrients	Sources
Carbohydrates	Grains, grass, legumes
Lipids (fats and oils)	Legumes such as groundnuts, sunflower seeds, soya bean meals and animal products like fish meal
Proteins	Legumes, grass, fish meal, bone meal etc
Water	Rivers, lakes, streams, feedstuffs
Vitamins	Green grass, fish liver oil, fruits
Minerals	Mineral supplements, pastures, cereal grains

2. a. They provide animal cells with energy that is essential for various metabolic processes in the body.

   b. These are an important source of energy in the body.

   c. i. They are important for growth and the formation of new body tissues as well as repair of damaged tissues.

      ii. They are also essential in the production of meat, milk, eggs among many other livestock products.

   d. i. Helps in dissolving feed substances so that they can easily be absorbed into the blood stream where the whole body is able to use them.

      ii. It is also vital in maintaining the shape of cells and hence the whole body.

      iii. Helps in the regulation of body temperatures in that when temperatures are very low, water found in the blood helps to circulate the heat around the body, making the animals feel warm. When it is very hot, loss of water through the skin helps to cool the animal.

   e. These help to keep animals healthy and resist disease attacks.

   f. They are necessary for various body processes. For example, calcium and phosphorous are needed for the formation of strong bones and teeth as well as milk production. Iron is essential in the formation of haemoglobin, an important element in red blood cells while sodium helps in the functioning of body muscles.

3. a. For maintenance of essential bodily processes that keep the animal alive such as respiration, breathing, digestion.

   b. For production of such products like eggs, milk and meat.

4. a. Palatability: the feed that is formulated should be that which the animals can easily accept and consume. It is only through feed consumption where the purpose for feeding (maintenance and production) livestock is achieved.

   b. Digestibility of the feed: Nutrients found in livestock feeds are released into the blood stream and utilized by various body tissues only when the consumed feed has been successfully digested. Farmers should therefore ensure that when balancing feed rations, they are using feeds that are highly digestible.

   c. Cost of the feed: For farmers to produce a feed ration, a number of feed ingredients are required, some of which are expensive. As expected, farmers would love to minimize costs for producing livestock feeds. However, when doing this, the farmers should ensure that the feed ration produced is of higher standards.

5. a. i. It is a source of proteins that help in the repairing of damaged tissues.
   ii. Soya beans contain a high proportion of fats which supply layers with a lot of energy.
   ii. It also helps in the development of the deep yellow colour of egg yolks.

   b. A maintenance ration is the amount of feed that is given to a livestock to enable its body to maintain vital body processes that support life while a production ration is the amount of feed that is given to an animal in addition to the maintenance ration to enable it to produce and reproduce.

6. The first factor is the cost of the feed. Most of the livestock feeds are commercially produced making them very expensive to most small holder farmers. As such, when providing such feeds, farmers should ensure that only productive animals are kept as these help to minimize costs of raising the livestock.

   Farm animals are provided with different types of feeds depending on age. For instance, young ruminant animals primarily depend on milk from the mother while older animals require a lot of forage.

   Type of the animal is also an important consideration because different classes of farm animals require different types of feeds. While as ruminant animals require much roughage, non ruminant animals like pigs feed on more concentrate feeds. This is due to the differences in the digestive systems of the two classes of animals.

   Farm animals are kept for a number of reasons such as draught, milk and meat production. For example, draught animals require more energy giving feeds while milk producing ones need more protein concentrates. In respect of this, animals are provided with feeds that would enable them to perform the functions for which they are kept for.

   Feeds that are provided to livestock should be those that are digestible. Highly digestible feeds are easily assimilated into the body and used for a number of bodily processes.

Practical

7.

Maize meal 7%		19 parts maize meal
	21%	
Soya bean Meal 40%		14 parts soya bean meal
		<u>33 parts</u>

Work out

Maize meal $= \frac{19 \text{ parts}}{33 \text{ parts}} \times 990 \text{ kg}$

$= \underline{570\text{kg}}$

Soya Bean Meal $= \frac{14 \text{ parts}}{33 \text{ parts}} \times 990 \text{ kg}$

$= \underline{420\text{kg}}$

8. a. i. A blue black colour will appear on the spot where the iodine solution was placed.
   ii. A dark brown colour will be observed on the place where the iodine solution was placed.

   b. Maize flour.

   c. The blue black colour that appeared on the maize flour is an indication of the presence of starch which is a carbohydrate.

   d. The white paper on which the maize flour was rubbed on remained clear while the one on which the ground nut flour was rubbed on had a greasy spot (translucent spot).

   c. i. The flour can be packaged so that it is kept longer.
   ii. Extracting oil from the seed since the test that was conducted indicated that the flour contained oil.
   iii. It can be processed into peanut butter.
   iv. The seed can be roasted to improve its taste.

9. a. i. It provides animals with carbohydrates which are essential in providing energy to body cells.
   ii. It supplies large quantities of fibre in the alimentary canal that help to promote digestion.

   b. i. It can be crushed to extract oil and the residues used as groundnut cake which is a protein concentrate that supplies the animal with proteins.

   ii. The flour from the seed can be mixed with other feeds like maize meal to produce balanced feed Rations.

   c. i. Roughages.
   ii. Protein concentrates.

d. The first situation is fattening up in which case a young animal is stall fed for some time in order for it to gain weight quickly. The second situation is where an animal is finished up. This is where the animal is given a lot of protein concentrates so that they are able to reach the required weight for slaughtering.

e. Firstly, before oversowing is done, the pasture land on which specimen X is growing should be lightly grazed or burnt. This is done to reduce specimen X in the field to ensure easy establishment of stylo in the already existing pasture land. If the land is not light grazed or burnt, the young stylo plants may face stiff competition from the already established star grass for such resources like sunlight, nutrients, space and moisture which may lead to failure of the stylo to get established.

3.2 Livestock management and production systems

3.2.1 Sheep production

1. a. Karakul, Blackhead Persian, Merino, Dorper and Hampshire Down.

   b. i. Fast growth rate and maturity.
      ii. Good mothering ability.
      iii. Purpose for keeping the sheep, for example, mutton, wool or pelt (skin).
      iv. Resistance to parasites and diseases.

2. a. The house should be well ventilated to allow free circulation of air. The importance of this is that it helps to prevent the multiplication of disease causing organisms.

   b. The house should be constructed on a raised place to make sure that drainage is enhanced.

   c. It should be constructed using strong materials to offer protection from predators and thieves.

3. Sheep like to graze young grass and shoots at ground level. This is called nibbling.

4. a. A burdizzo resembles a pair of pliers and is used to crush the blood and nerve supply of the testes (above the testicles). Crushing of the cords cuts the blood supply of the testes causing them to degenerate. Sheep are castrated using this method when they are two to three months old.

   b. Involves making an incision on the scrotum through which the testes are removed by hands. Since it leaves an open wound on the scrotum, the farmer should take precautionary measures that would reduce pain, loss of more blood as well as prevent the wound becoming septic. The hands and materials used should be clean Once the testicles are removed, iodine or alcohol should be applied on the wound to promote quick recovery.

   c. This involves putting a strong rubber band on the upper part of the testes using an elastrator (an instrument used for stretching strong rubber ring). It aims at cutting blood supply to the testicles and scrotum. The stopping of blood supply to the testicles results into withering and falling off of the testes after a few weeks. The disadvantage of the method is that it causes pain to the animal over a long period

5. a. Docking involves cutting of the sheep's tail, a few centimeters away from the body Trimming is the cutting of overgrown hooves in order to reduce lameness.

   b. i. It prevents the build up dirt around the anal area which may be a breeding ground for parasites and diseases.

ii. Docking promote even distribution of fats throughout the whole body of the sheep. This increases the quality of the mutton.

iii. When a tail is present mating becomes difficult because the tail covers the genital area of the ewe. Cutting the tail therefore, makes mating in sheep to be easy.

3.2.2 Goat production

1. a. Malawi goat.
   b. East African goat.
   c. Toggenburg.
   d. British Saanen.
   e. British Alpine.
   f. Anglo- Nubian goat.

2. a. Warm houses help to reduce a pneumonia attack that is triggered by stress factors such as extreme cold weather.

   b. i. Protein concentrates help to induce heat and encourage ovulation in female goats. This therefore, helps the goat to conceive.

   ii. When provided to pregnant animals two months before parturition, they help to prepare the body of the female for kidding and full development of the unborn kid.

   c. i. Enable the goats to produce young ones with desirable traits. This causes the goats to be more productive.

   ii. It enable the goats to produce highly in relation to products that the farmer want to obtain from the goats such as milk, mohair and meat.

3. The first feeding habit is called browsing, which means eating leaves, shoots, flowers, twigs of shrubs and trees as well as barks of trees. The goats are able to do this because they are well adapted to browsing due to split upper lip, narrower mouthparts and longer legs for climbing. Secondly, goats also like grazing. In this case they eat grasses and other herbage that is close to the ground. It should also be noted that goats do not like feeding on pastures that have been trampled upon.

4. a. They are a source of income to farmers after selling products such as meat, milk and hides.

   b. Goats' meat and milk are a source of animal proteins in the diet of human beings.

   c. Dung and urine from goats are a valuable source of plant nutrients. When added in the garden, this manure promotes crop yields.

5. a. It should be well ventilated to enable the removal of excess moisture in the air which can cause the goats to become wet and so increase chances of pneumonic attacks.

   b. It should be warm so as to prevent goats from suffering from pneumonia.

   c. It should be built of strong building materials to prevent predators that can attack the goats.

   d. The Kraal should be roomy enough to provide room for exercises to the animals.

6. a. The selected breed should be a high yielding breed of such products like meat, milk or skins as this increases productivity on the farm.

   b. Farmers should select those breeds that are well adapted to prevailing climatic conditions, bearing in mind that different breeds are differently adapted to different climatic conditions.

   c. The breeds should be resistant to parasites and diseases. This is because parasites and diseases result into low productivity in goats.

   d. The goats should be those with a faster growth and maturity rate to allow a fast increase in the number of goats within a short period of time.

   e. Selected nannies should be those that have a good mothering ability to ensure that the kids are well looked after during infancy stages.

3.2.3 Beef production

1. Hereford, Afrikander, Brahman, Arberdeen Angus, Boran, Galloway, Charolais.

2. a. The animals put up weight very quickly because they do not spend a lot of energy searching for feed and water as these are provided to them in the Khola.

   b. The animals are protected from certain contagious diseases and parasites because they do not get into contact with other animals from different Kholas.

   c. The system helps to prevent trampling of young and tender pastures, which is common in other grazing systems.

3. a. Prevents undesirable males from breeding.
   b. Castrated animals are docile and strong.
   c. The meat from castrated bulls is tender and does not produce strong smells
   d. It completely removes libido from the males thereby calming them.

4. a. It helps to protect the dehorned animal from injuries that can be sustained when the animal is entangled in a bush.

   b. It helps to prevent injuries to people when handling the animal.

   c. It helps to prevent injuries that can be caused to other animals during fights.

   d. Removal of horns helps to create enough space for the animal in the stalls, feed bunks and when transporting the animals.

   e. It makes the animals docile and less aggressive, making them easy to handle.

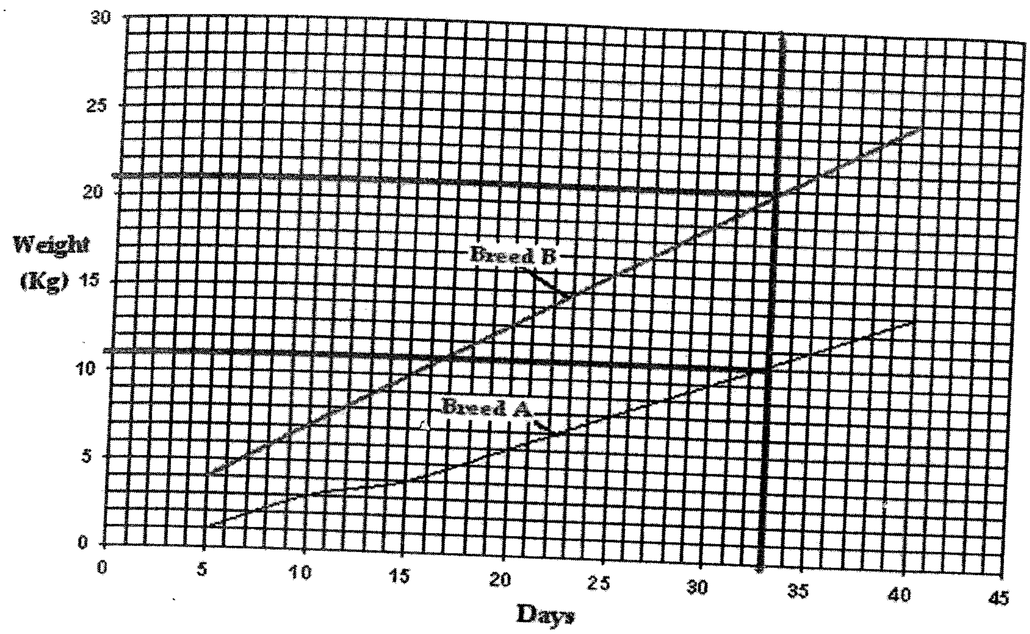
   f. It reduces financial losses from trimming damaged carcasses caused by horned animals during fights.

   g. Dehorned animals gain a price advantage at auction.

5. a. Beef animals require a lot of feeds that are rich in carbohydrates such as grass and maize meal to provide them with energy that is essential in supporting vital bodily processes such as respiration and movement.

   b. Beef animals also require feeds rich in proteins such as bone meal, bean meal and blood meal to enable them to build up more flesh around their bones.

6. a.



b. Difference = Live weight gain for Breed B - Live weight gain for Breed A
= 21Kgs - 11Kgs
=10 Kgs

c. i. Breed B has a higher live weight gain as compared to breed A. This makes breed B to produce more meat which is a source of income when sold.

iii.It is an efficient forager because it is able to put up more body mass using the same amount of feed as that provided to breed A.

d.

Income			K35,000
Less:	Variable costs		
	Feed	K8,000	
	Vert. Costs	K2,500	
	Casual labour	K6,500	
Total variable costs			K17,000
Gross margin			K18,000

7. a. i. Breed M = 340kg - 198kg/130days =1.09 kg/day

ii. Breed N = 395kg - 190kg/130 days = 1.54kg/day

b. Breed N

c. i.It had a very low initial weight (190kgs).
ii. Its finishing weight was higher (395kgs).
iii. It had a higher average live weight gain per day (1.54kgs/ day).
iv.It was able to gain more weight within 130 days under similar conditions like breed M.

8. a. It is a very cheap method of raising cattle as the cattle are left to wonder in sea for their own feed and water under minimal supervision of a herdsman.

b. Animals raised under extensive system are usually hardy and more resistant parasite and disease attack as they are conditioned to live under harsh condition

3.2.4 Dairy production

1. a. i. It provides nutritious feed nutrients to the growing foetus during its last stages development.
ii. It stimulates the alveolar cells, of the udder that helps to increase mil production.
iii.It helps in preparing the cows' body for parturition.

b. i. It is rich in antibodies that provide immunity to the calf.
ii. It is rich in food nutrients necessary for the development of the young animal.

2. a. The amount of milk produced increases as a cow is growing up until the fifth lactation.

b. Cows can be milked using either hands or machinery. Irrespective of the method chosen, the farmer should make sure that the cow is milked within 10 minutes as this coincides with the milk let down process.

3. a. Feeding
The cow should be provided with well balanced feed rations through out the gestation period in addition to plenty of fresh water. Additional feeds in form of concentrates should be provided during the last two months of the gestation period because it helps in developing the body of the cow for parturition and full development of the embryo.

b. Vaccination
The in calf cow should be frequently vaccinated, to increase the antibodies in the body that help to promote immunity to disease for the cow and the unborn calf.

4. a. i. It should have large udders that are tightly attached to the body to prevent injuries to the udder and teats as the cow is moving on pasture land

ii. The teats must be evenly spaced on the udder and be of sufficient size for easy milking.

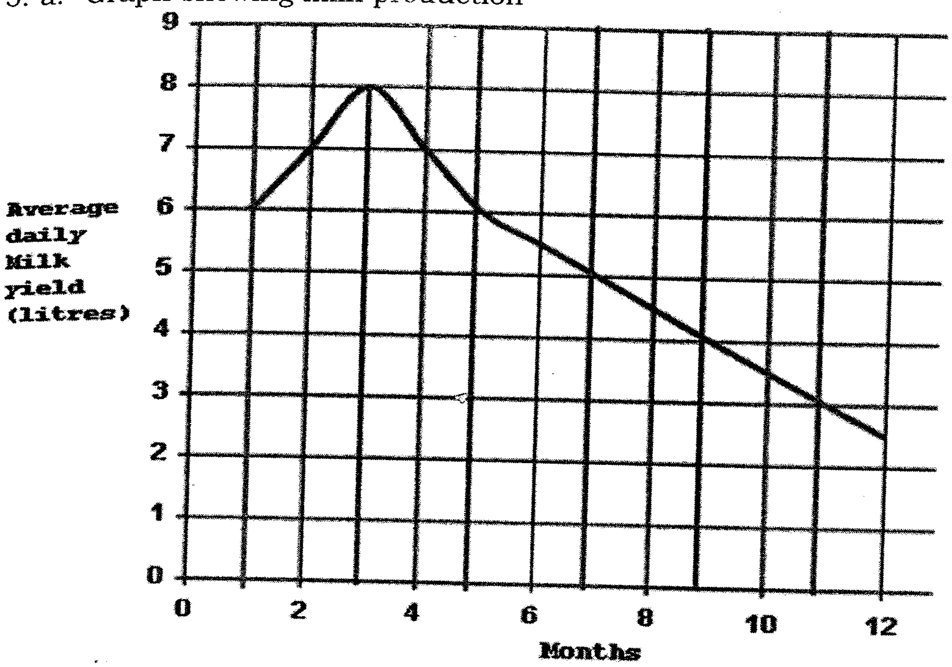
iii.It must have a large stomach which is an indication for capacity for more food and water intake.

iv.It must be docile with a mild temperament, to ensure more milk production. It should be noted that easily irritated animals are bad milkers.

b. i. Always ensure that the lactating cow is clean and healthy.
ii. Keeping all milking utensils clean.
iii.Using a milking shed that has a cemented floor and free from dust.
iv.All people handling the milk must be clean and free from diseases such as tuberculosis
v. The udder and teats should be thoroughly washed with a disinfectant.

Practical

5. a. Graph showing milk production



b. Month number 3.

c. Month number 10

d. i. The farmer is able to assess the productivity of the cow over a given period. This helps the farmer to ascertain profitability of the cow.

ii. Where productivity is declining, the farmer is able to plan for replacement stock from cows with good milk production traits.

iii. It helps to compare the productivity of the dairy enterprise with those of other farmers. This assists the farmer to make necessary improvements in weaker areas.

iv. The farmer is able to identify unproductive cows that can be replaced or culled so that costs are minimized.

e. i. Livestock numbers records: this shows the total number of dairy animals on the farm, number of animals purchased or sold. It also include births and deaths of animals.

ii. Breeding records which show details of total number of females that are served, date served, as well as number of males used to service the cows and breeds of the animals.

iii. Health record: these contain full details on dates when diseases or parasites were noticed, the symptoms that were noticed, drugs that were used and the cost for treating the animal. The record also show remarks on duration of the sickness, recoveries and deaths resulting from the disease.

iv. Feeding records: these have details on amount and types of feeds bought, used and remaining stock.

v. Birth records: contain details on number of calves born and the identification of the parents, their breed, date of birth, birth weight, sex of offspring and date of breeding.

6. a. It is Bettie.

   b. i. It has a very short calving interval (340 days) than the other cows. This makes it possible for the cow to calve each and every year.

   ii. It is an efficient feed converter because it is able to produce a higher amount of milk than Kambani and Jolly despite all the cows being fed on the same amount of feed per day.

   c. This is the period from one calving to the next.

   d. It is through servicing the cow within the first three months after lactation. The reason for this is to ensure that the cow conceive and calve within one year.

   e. i. Type of feed
   Animals that are fed on better quality feed produce good quality milk than those that are fed on poor quality feed.

   ii. Health of the cow
   Cows that are healthy produce good quality milk unlike those that are attacked by diseases such as mastitis and tuberculosis.

   iii. Breed of the animal
   Some breeds of cows produce high quality milk than others. For example the Guernsey produces milk with higher butter fat and protein content than the Holstein. (Guernsey milk with 4.68% Butterfat and 3.57% protein while Holstein produces 3.89% and 3.22% butterfat and protein content respectively).

7. a. Stage M

   b. The cow is not pregnant such that much of the feed that is consumed is used up into milk production.

   c. The cow is in calf and so, some of the feed nutrients are shared with the developing embryo.

   d. This is a period when a cow is not milked as it prepares for calving.

   e. i.It prepares the cow's body for the stress that accompanies the process of parturition.
   ii. It helps the cow to get sufficient feed nutrients for the development of the body in readiness for parturition.
   iii.It also helps to provide more feed nutrients to the unborn calf so that it becomes fully developed.

3.3. Livestock parasites and diseases

1. a. Foot rot.

   b. i. Teat the animals with antibiotics.

   ii. Provide a foot bath of copper Sulphate solution on a weekly basis.
   iii.Conducting regular trimming and examination.

2. a. i. The milk contains pus, blood or turns watery.
   ii. The udder and teats are swollen.
   iii.Serious cases may result into death of the cow.

b. i. Practicing strict hygiene practices and using disinfectants when milking to prevent infecting the cows with bacteria that causes mastitis.

ii. Using right milking techniques that do not cause injuries to the udder. Wounds on teats and udder are entry points for mastitis causing organisms.

iii. Treating the cow with antibiotics during dry off period to control bacteria.

iv. Providing clean and dry bedding materials to make sure that the cow's udder does not get in contact with dirt containing mastitis causing bacteria.

v. Providing well ventilated houses for cows. Such houses are important because they help in reducing humid conditions. It should be noted that humid conditions are ideal places for the multiplication of bacteria.

3. A i. The animal loses appetite.

ii. Difficulty in breathing and coughing.

iii. Mucous discharge from the nose.

iv. The animal becomes dull and sleepy.

v. High or low temperatures.

b. i. Avoiding stress causing factors like overcrowding, malnutrition, adverse weather conditions and sudden feed changes as these disturb the normal body defense mechanisms. (Take note that the microorganisms that cause pneumonia use the respiratory tract as their habitat. Hence, if the goats are stressed up, they can take advantage and cause the disease in the goats).

ii. Treating with antibiotics.

iii. Keeping goats in clean and dry Kraals that prevent the build up of disease causing organism.

4. a. Foot and mouth disease.

b. i. Slaughtering of all affected animals.
ii. Restricting movement of animals (quarantine) when there is an outbreak of the disease.
iii. Vaccinating the animals every six months.

5. a. i. Abortion that is followed by a brownish vaginal discharge.
ii. Inflammation of genital organs.

iii. Swollen testicles in bulls.

b. i. Slaughtering all affected animals.
ii. Do not touch the aborted foetus with bare hands.

6. a. The protozoa that causes trypanosomiasis interferes with red blood cells causing the animal to become anaemic.

b. It may cause abortion in cows.

7. a. It involves injecting livestock with vaccines that help to stimulate production of antibodies. These antibodies increase the body's immunity to diseases.

   b. This involves practicing hygiene on the farm with an intention of getting rid of a conducive environment for breeding of parasites and pathogens that can transmit or cause diseases in the animals.

8. a. A liver fluke.

   b. i. The animal loses weight and becomes weak because of loss of appetite.

      ii. Digestion is upset because the liver fluke attacks the bile duct causing it to become hard and fail to produce bile.

      iii. The animal becomes pale due to anaemia that results from damage of the liver by the liver fluke.

      iv. Serious attack of the digestive system results into swelling of the stomach due to the accumulation of body fluids in the abdomen. As a result, the animal becomes dull and depressed because of pain.

   c. i. Applying copper (II) sulphate on river banks, streams and marshy areas to control snails which are intermediate hosts of liver flukes.

      ii. Clearing grass and any other vegetation that can act as a habitat for snails.

      iii. Burning of swampy areas and other snail infested areas to destroy snails.

      iv. Drenching animals with tetrachloride every 6 months.

      v. Draining of waterlogged areas that are used for grazing animals.

      vi. Avoiding grazing animals in waterlogged areas.

      vii. Fencing of grazing lands to restrict movement of animals.

9. a i. The animal loses weight because the flukes interfere with the digestive system.

      ii. They destroy the liver and cause internal bleeding leading to anaemia.

      iii. If the animal is seriously attacked by the parasite, death may occur.

   b. Cattle, goats and sheep.

10. a. i The livestock loses appetite.

      ii. The parasite sucks digested feed in the intestines.

      iii. It can cause damage to tissues like muscles, brain and liver if the larva of the parasite enters the blood stream.

   h. i. Cooking meat thoroughly in order to kill the bladder worm that can infect human beings.

      ii. Using recommended drugs to drench the livestock.

iii. Proper disposal of human faeces so as to prevent infecting pasture lands with eggs of the parasite.

iv.Practicing rotational grazing in order to break the life cycle of the parasite.

v. Close inspection of meat before selling to consumers to ensure that infected meat is not sold. This helps to prevent reinfection of the consumers.

11. a. They cause a loss in the productivity of the animal (reduced milk production, weight loss, poor quality meat and lowered conception rate).

b. The parasites suck digested feeds from the alimentary canal of livestock resulting into malnutrition diseases.

c. Some internal parasites suck blood from the alimentary canal leading to increased cases of anaemia which is fatal.

d. Other internal parasites like tapeworms cause damage to tissues like muscles, brain and liver if the larva of the parasites enter the blood stream.

12 a.The adult tick lays eggs into the ground which later hatch into a larva. The larva climbs on to the first host (cattle) where it feeds on blood and becomes swollen up. This larva then moults and emerges into a nymph which after feeding on the blood on the first host drops on to the ground where it moults. This results into an adult tick which climbs onto the second host and mate.

b. i. It helps in breaking the life cycle of the tick by preventing the host animals from entering a tick infested area. This helps in starving the ticks which eventually die.

ii. It is also important because the farmer is able to control the ticks in a stage when they can not cause serious damages to the livestock.

3.4 Anatomy and physiology of reproductive systems of livestock

1. a. It is a Sow.

b. Long heat periods are important because a farmer is able to notice and make necessary arrangements like communicating with officials at an artificial insemination unit for the female to be served with a male.

c. i. - Frequent bellowing.
- The cow loses appetite.
- The vulva becomes swollen.
- Frequent urination.
- The cow mounts other cows.
- The cow stands still when mounted by other animals.
- it produces mucous discharge from the vulva.

ii. Assuming that the cow was on heat in the morning of 27th May, the heat period will last on 28th May.

length of oestrus cycle = 21 days

Next heat = 28th May + 21 days
= 17th June 2008

d. i. Gestation period for ewes = 150 days (5 months)
= 31st December - 5 months (31 + 30 + 31 + 30 + 28)
= 2nd August, 2009

ii. The ewe and lamb are able to get enough feed since lambing takes place during the rainy season when there is abundant pastures.

2. a. i. The vulva becomes red and enlarged.

ii. Discharge of mucous from the vulva.

iii. The cow becomes restless.

b. i. Servicing a heifer at an early stage before maturity. This brings complexities when calving because the body is not fully developed for the stress that is associated with calving.

ii. High birth weight of the calf that cause the reproductive tract to become smaller for normal calving.

iii. Abnormal orientation of the calf in the uterus. For example, if the calf is lying upside down or presented in a backward position, the cow may experience problems because the reproductive tract may become smaller. (The normal presentation of the calf is where the front legs appear first with the head resting between them).

3. a. i. Stage A is standing heat period.
ii. Stage B is out of heat period.
iii. Stage C is the fertile period.

b. Ovulation.

c. Steaming up.

d. i. The cow mounts other cows.
ii. The cow stands still when mounted by other animals.
iii. It produces mucous discharge from the vulva.

e. i. Stage D: This is too early for serving the cow. The cow may fail to conceive becaus the sperms may become weaker at the time when ovulation occurs.

ii. Stage E: This is the recommended time for serving the cow because the sperm have a high chance of meeting with the ova when they are still strong. It ma result into conception.

iii. Stage F: conception can not take place because the cow is out of heat such th it may not accept a male for mating.

4. a. i. It helps to induce heat in the female animal.
ii. It encourages ovulation which prepares the body of the female animal f conception.

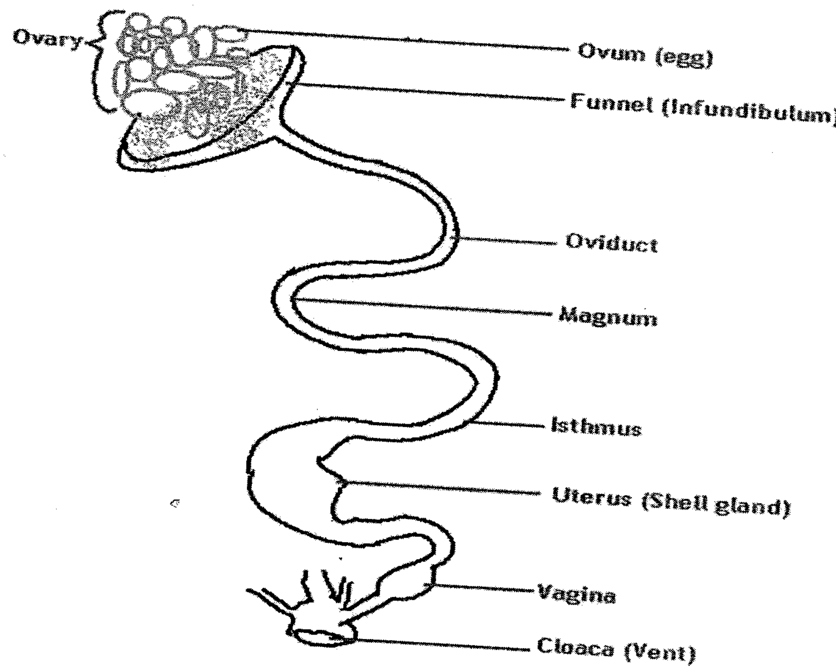
b. It helps in introducing a young animal to solid foods thereby allowing the fema animal to produce more milk.

5. a. Communication problems. This arises when farmers live in very far away place from the artificial insemination unit. In this case, the farmer may fail to inform th officials in good time for inseminating the animal.

   b. High costs: setting up an artificial insemination unit requires a lot of money fo training personnel and managing the whole system.

   c. Lack of knowledge: If the farmer is not knowledgeable on the heat period and oestrus cycle of the animal involved, there may be poor timing on when to conduct the insemination.

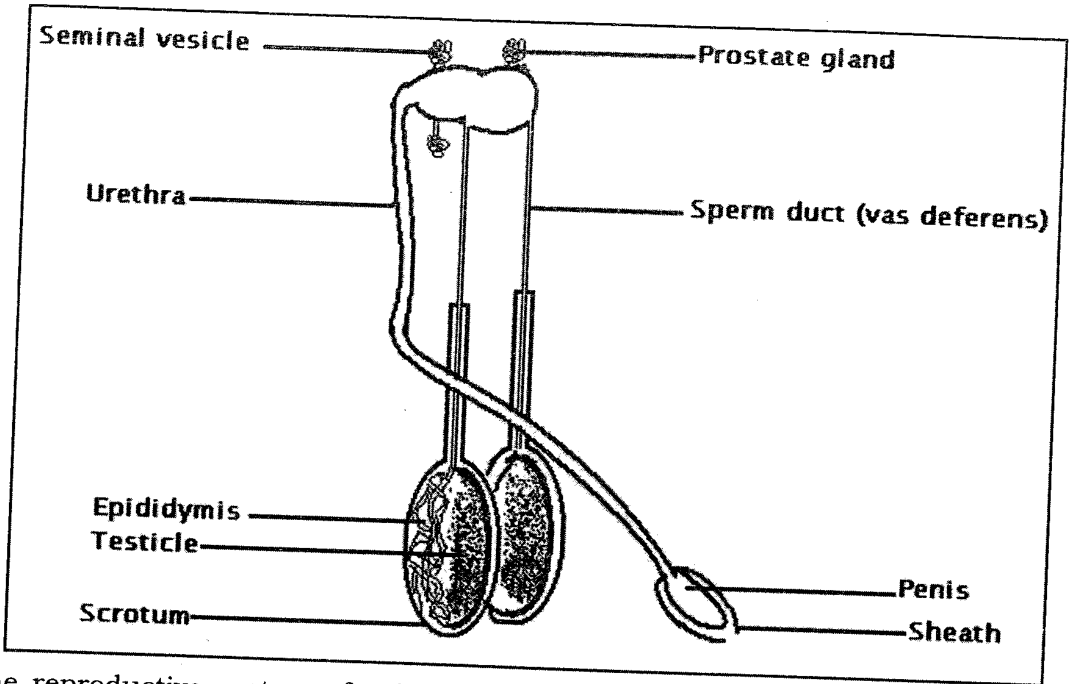
6. Diagram showing the reproductive system of a hen



The reproductive tract of a hen comprise of several parts, one of which is the ovary. The left ovary is the only functional one in a mature hen. This is responsible for releasing an egg that already contains a yolk into the oviduct. The oviduct is divided into five different parts, each of which has a specific function.

The first section of the ovary is the funnel or infundibulum. It is responsible for receiving an egg from the ovary. This section stores sperms and also acts as a site for the fertilization of the egg. The chalaza, which is a membrane which keeps the yolk in position, is also produced in this part. The second section is the magnum. This is a section where the albumen is added on to the egg. The third section is the isthmus. It is responsible for adding mineral salts, water and outer albumen membranes on the egg. Addition of the shell as well as the colour of the egg is done in the fourth section. This is called the shell gland or the uterus. Lastly, the egg goes into the vagina. It is responsible for sealing the pores on the shell as well as secreting fluids that help to reduce friction as the egg is being laid out through the cloaca or vent.

7. The reproductive system of a bull



The reproductive system of a bull consists of testicles, epididymis, sperm ducts, accessory glands and a penis. The testicles are responsible for the production of millions of sperms which are the male gametes. The sperms are stored in the epididymis, which is a coiled tube that surrounds each testicle. The testicles are enclosed in a bag like skin called the scrotum. Its function is to protect the testicles. The epididymis extends out from each testicle to form the sperm duct (vas deferens). This is responsible for transporting the sperms into the urethra. The function of the urethra is to conduct both urine from the urinary bladder and sperms out through the male copulatory organ, the penis. Midway in the male reproductive system, particularly where the urinary and the reproductive systems join, there are a number of accessory glands which include the prostate gland and the seminal vesicles. These glands produce sticky fluids that neutralize the acidity caused by the urine. They also provide nourishment to the sperms as well as acting as a mode for transporting the sperms. The mixture of sperms and the fluids from accessory gland produces semen which is ejaculated through the penis into the vulva of a cow. When resting, the head of the penis is protected by an extension of the skin called a sheath.

3.5 Livestock improvement

1. a. Introduction.
   b. Selection.
   c. Breeding.

2. a. Fast growth rate.
   b. Good mothering ability.
   c. Resistance to parasites and diseases.
   d. High productivity.

3. a. To increase yield of products that are obtained from livestock such as eggs, milk, meat.
   b. To improve the quality of livestock products.
   c. To increase disease resistance.
   d. To improve growth rate.
   e. To promote adaptability to climatic conditions.

4. Selection aims at identifying animals that have superior characteristics for breedin. Among the considerations that are made when selecting livestock for breeding ar fertile animals with large size and faster growth rate as well as those whose progen (offspring) have good characteristics. On the other hand, animals that have inferio traits such as high susceptibility to disease attack, slow growth rate and small bod sizes are prevented from breeding by slaughtering or by castration (for males). B allowing animals with desirable characteristics to mate and reproduce, the farmer is assured of obtaining animals that have improved qualities.

5. a. These are the basic units of inheritance that are found on chromosomes inside the nucleus of a cell. The genes are passed on from parents to offsprings during fertilization. Genes are responsible for determining the characteristics of farm animals. To a greater extent, genes may be regarded as the major limiting factor to livestock productivity. For example, animals whose genes are for low productivity cannot be expected to produce highly by good management practices.

   b. This has a function of determining the extent to which an organism is going to realize its potential. In livestock production, environment includes the physical set up in which the livestock is raised as well as the husbandry practices such as feeding, housing, parasite and disease control. Where the environmental conditions are very good, animals are able to produce high yields. But where environmental conditions are poor, the livestock may fail to produce to the fullest even if it inherited good characteristics from the parents.

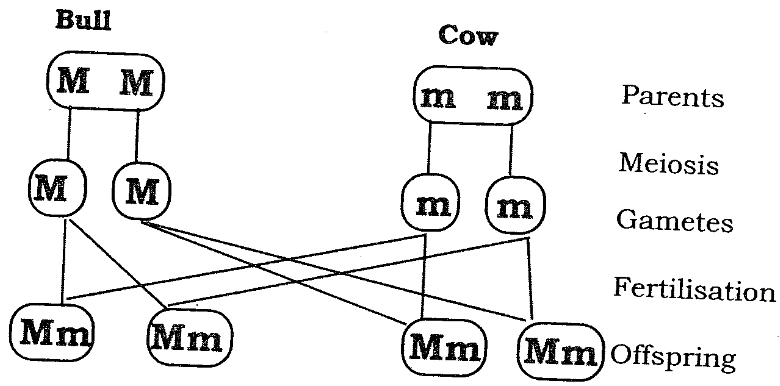
6. Progeny testing is assessing the performance of a male animal by mating it with a number of females with proven performance. The offspring that result are used as a basis for judging the performance of the male animal. Sib selection on the other hand is the selection of animals based on the performance of their mother.

7. a. It introduces new genes into the herd.
   b. It results into increased vigour in the offspring.

8. a. This is the production of offspring from closely related parents. If in breeding is continued for several generations, it helps in fixing certain desirable traits in the herd. The major set back is, however, that it results into in breeding depression, which is a reduction in the performance of the offspring.

   b. This involves mating of two animals from different breeds. For example, the Malawi Zebu cow with a Friesian bull. This type of mating is important because it helps to introduce new genes in the herd apart from increasing vigour in the offspring. The off spring from cross breeding is a higher performer in various aspects such as milk and meat production in cows and egg production poultry.

9. a. Diagram for cross breeding



b. All the offspring will be high milk producers.

c. It increases hybrid vigour in the offspring because of the genes that are inherited from both parents. As a result, cross bred offspring are high performers in a number of respects such as milk and meat production.

d. When compared with local breeds, cross bred animals require careful management practices for them to perform better.

Chapter Four

AGRICULTURAL MARKETING

Questions

4.1 Marketing forces

1. a. Explain the differences between price elasticity of demand and price elasticity of supply.

   b. Demand for maize is said to be inelastic while that of Irish potatoes is elastic. Explain one reason for this.

   c. With the aid of sketch graphs, describe the two types of demand.

2. Table below shows the supply and demand for fresh chambo fish at different prices in the market. Study it carefully and answer the questions that follow:

Table 2: supply and demand for fresh chambo fish.

Average price for each fish in Kwacha	Quantity demanded (Kg)	Quantity supplied (Kg)
80	10	88
70	21	79
60	30	70
50	40	61
40	50	50
30	63	37
20	77	23
10	92	10

a. Draw a graph to show the relationship between price and quantity of fish demanded and supplied on the market.

b. How much Chambo fish would be demanded when the price for each fish is K55?

c. What is the equilibrium price of chambo fish at the market?

d. Explain any one way in which the equilibrium price is important to a fish farmer.

e. i. What happens to the quantity demanded when the price of fish increases from K40 to K50?

ii. Calculate the price elasticity of demand.

iii. What does the answer above tell you about the price elasticity of demand for fish at the market?

f. Describe the effects of changing the price of fish at the market.

3. Describe the market situation in each of the following conditions of demand for an agricultural commodity.

   a. Elastic demand.

   b. Unitary demand.

4. a Work out the price elasticity of supply for beef if a price change from K500/kg to K450/kg resulted into an increase in the quantity of beef supplied from 900kg to 750kg.

   b. From the answer above, what price elasticity of supply does the beef have?

   c. Sketch a graphical presentation to show the price elasticity of supply for the beef.

   d. Comment on the significance of the price elasticity of supply of the beef?

4.2 Marketing channels and agencies

1. What is the difference between marketing channels and agencies?

2. List down any five functions that are played by middlemen.

3. Explain one way in which each of the following marketing agents contributes to the marketing of agricultural products:

   a. Producers.

   b. Wholesalers.

   c. Processing companies.

4. Mr Bamusi bought Irish potatoes from Mr Chitsulo's farm and sold them all to Ndaterapano Trading Company. The Irish potatoes were later bought by vendors, who in turn, sold them to different households in the city.

   a. i.Produce a marketing channel for the Irish potatoes. In the marketing channel include.
      ii. Names of individuals involved.
      iii. One function for each marketing agency involved.

4.3 Marketing costs and margins

1. a. Explain the relationship between marketing costs and margins.

   b List down any two sources of marketing costs.

2. Explain two ways in which trading of farm produce at community level is important.

3. Explain any one way in which each of the following activities carried out on a farm can reduce marketing margins and increase profits:

   a. A farmer performing some of the marketing functions.
   b. Farmer selling farm produce through agricultural cooperatives.
   c. Eliminating some marketing functions.

4. Explain any two causes of an increase in marketing margin for a produce.

4.4 Population distribution and marketing

1. Describe any five effects of population distribution on the marketing of agricultural products. (essay).

4.5 Trading of agricultural commodities

1. Define marketing.

2. State one way in which resources are used in each of the following agricultural activities:

   a. Trading.

   b. Marketing.

3. Suggest ways of promoting trading at:

   a. Community level.
   b. National level.

Essay

4. Explain any five barriers to trading at the international level.

5. Describe any five ways in which international trade in agricultural commodities is important.

Chapter Four

AGRICULTURAL MARKETING

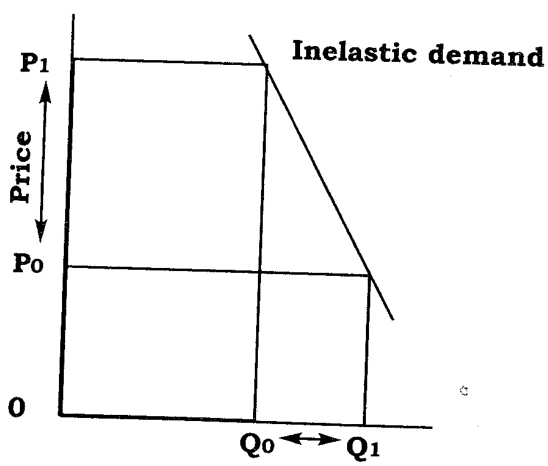
Answers

4.1 Marketing forces

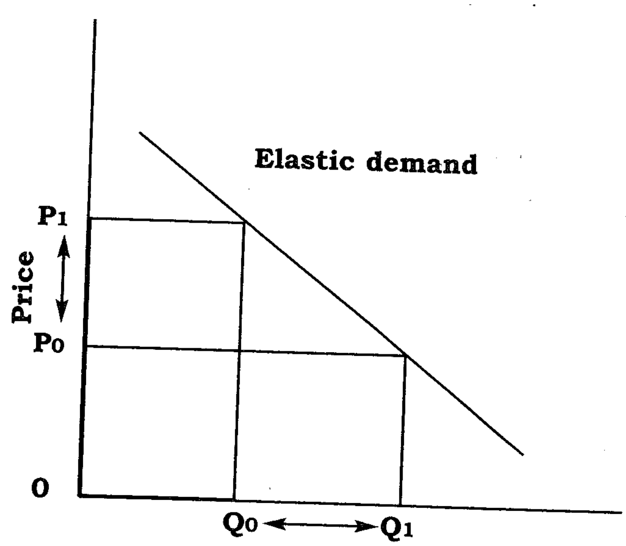
1. a. Price elasticity of demand is the responsiveness of the quantity of goods demanded to changes in the price of a commodity. On the other hand, price elasticity of supply is the responsiveness of the quantity of goods supplied to changes in the price of the commodity.

   b. The demand for maize is inelastic because it is a staple food (basic necessity) for most people in Malawi. Since most people regard maize as a basic need, they can not live without eating food prepared from maize, and so they keep on buying the commodities regardless of the price. On the other hand, the demand for Irish potatoes is elastic because it is regarded as a luxury. As such, most people can live without them. For this reason, a change in the price of the Irish potatoes results into a very big change in the quantity demanded.

   c. Sketch graphs for Inelastic and elastic demand



Demand for maize

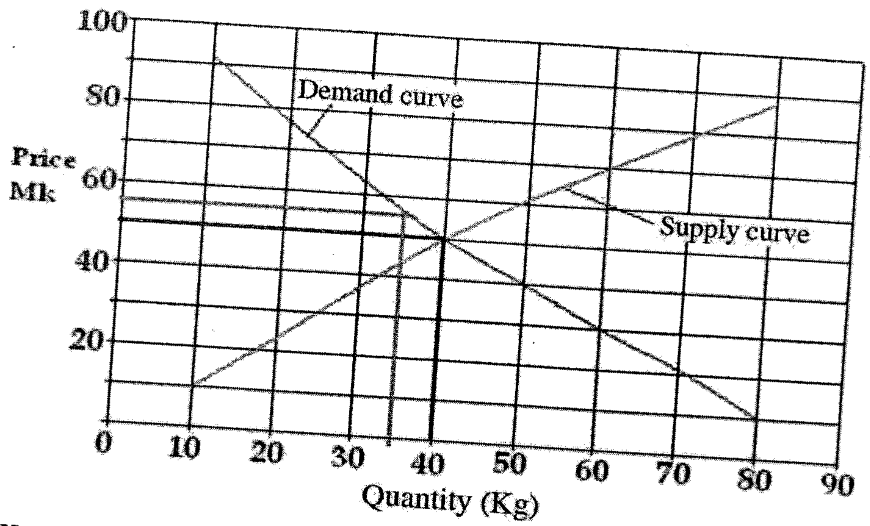


Demand for irish potatoes

In the figure above, the demand for maize is said to be inelastic. This is so because when the price of maize was very low (P_o) the quantity that was demanded was equal to Q_1. But when the price was greatly increased to P_1, the quantity demanded changed very little to Q_o. This small change in the quantity of goods demanded causes the demand curve to be very steep.

For the Irish potatoes, the demand is said to be elastic. The reason for this is that a very small change in the price of Irish potatoes (from P_o to P_1) resulted into a very big change in the quantity of Irish potatoes that were bought (i.e. from Q_1 to Q_o). The result of this very big change in the quantity demanded causes the demand curve to be gentle.

2. a. Demand and supply curves



b. = 35 Kg

c. = K50.

d. Since the equilibrium price represents the minimum price for the fish, the farmer will be able to forecast on how much money to expect from the sales.

e. i. The quantity demanded declines from 50 to 40 kgs.

ii.

$$\text{Price Elasticity of Demand} = \frac{\%\text{ change in quantity demanded}}{\%\text{ change in price}}$$

$$= \left(\frac{\text{Old quantity} - \text{New quantity}}{\text{Old quantity}} \times 100\right) \div \left(\frac{\text{Old Price} - \text{New Price}}{\text{Old Price}} \times 100\right)$$

$$= \left(\frac{50\text{kg} - 40\text{kg}}{50\text{kg}} \times 100\right) \div \left(\frac{K40 - K50}{K40} \times 100\right)$$

$$= \left(\frac{10\text{kg}}{50\text{kg}} \times 100\right) \div \left(\frac{-K10}{K40} \times 100\right)$$

$$= 20 \div 25$$

$$= \underline{0.8}$$

iii. The demand for fish is inelastic.

f. i. When a price is high, the demand of the fish will decline as more people will not be willing to buy the fish. However, lowering of the price would result into more people buying the fish.

ii. The quantity of the fish supplied on the market will also change in respect to any changes in the price. At higher prices, more fish will be supplied than at lower prices. The rationale behind this is that at higher prices, the producers will be able to make more profits.

2. a. The income realized from the sale of the produce raises the living standards of the people.
   b. It helps people working in other fields to get food.
   c. It promotes division of labour and specialization which enable people to become more skilled and more productive.

3. a. When a given set of marketing function is performed, there are charges that are made by the agents as payment for the services rendered. If farmers perform some of these marketing functions, the selling price of their produce is likely to increase so that it reflects the cost of the marketing functions performed. By selling their produce at higher prices farmers are therefore, able to make more profits.
   b. Farmers in cooperatives are better placed to bargain for better prices when selling their produce to traders than if each farmer is dealing directly with such traders. These high selling prices also lead to more profits on each individual farmer.
   c. If some marketing functions are completely eliminated, the costs for handling the goods are reduced. If the costs are minimized, farmers are able to make more profits.

4. a. Involvement of many marketing agents that tend to charge costs on activities that they perform on the products. The result is an increased difference between the selling price and the farm gate prices, hence increased marketing margins.
   b. Lower commodity prices at farm level that attract a lot of middlemen to buy and resell the commodities.
   c. Failure of farmer to perform some marketing functions which attract other marketing agents to set in and perform those functions. The marketing agents in turn charge on the services they perform, leading to increased selling prices.

4.4 Population distribution and marketing

Essay

Population distribution affects the direction in which commodities flow to. Generally, more goods are transported to areas that have more people than in areas with very few people. The reason for this is that it is usually anticipated that more people will demand more goods than fewer ones.

The second effect is that population distribution affects the length of the marketing channels. In places with very few people, direct marketing channels are more common. This is because the producer can easily reach out to all people in the area. But where there are more people, a lot of intermediaries are involved in order to ease congestion of consumers at the point of production. In this case, indirect marketing channels are more common.

The other effect is on the quantity of goods and services demanded and supplied. Normally, places with a lot of people have a high demand for goods and services than those with very few people. As a result of this increased demand, producers are attracted into those areas where they can supply more of the goods and services.

The way people are distributed in an area impacts a lot on the mode of transporting goods. In areas where there are few people, goods are generally transported using very simple modes of transport such as bicycles, head loads and wheelbarrows. These modes of transportation are able to meet the demand of the few people. However, in areas with more people, goods have to be transported very quickly so as to meet the ever increasing demand for the goods. In such cases, the use of Lorries, trains and even aeroplanes is very common.

Understanding of the way population of a given area is distributed also helps marketers to effectively segment the market so that only specific group is targeted. Once the targeted group is identified, the producers can then put in place necessary promotional activities that would help to bring about more sales.

4.5 Trading of agricultural commodities

1. Marketing is a process that starts with identification of consumer needs through market research and seeking out means of satisfying those needs through performing a number of marketing functions.

2. a. Trading: In trading, resources are directed towards bringing about more sales.

   b. Marketing: Resources are directed towards producing commodities that satisfy the needs of the consumers.

3. a. i. Promoting effective use of resources that would ensure production of surplus yields that can be sold out.

   ii. Promoting development of rural growth centres which provide markets for surplus agricultural produce.

   iii. Improving rural communication network for easy transportation of commodities.

   b. i. Improving communication network for easy transportation of the commodities

   ii. Promoting peaceful coexistence that provide a conducive environment for trading

   iii. Removing Value Added Tax (VAT) on agricultural commodities to ensure that consumers get these commodities at reasonable prices

Essay

4. *The first barrier is subsidies on commodities. A subsidy is the amount of money that is paid by a government on inputs that are used in production. These subsidies are aimed at reducing the costs that are incurred when producing commodities. The effect of these subsidies is that they make international trade less competitive in that they lead to lower prices for locally produced commodities than foreign ones.*

The second barrier is import quotas. These are restrictions on the quantity of goods that are allowed into a country. They limit the amount of goods that people in a country can obtain from other countries. The aim for these quotas is to protect home industries from foreign competition. The disadvantage of this restriction, however, is that domestic consumers are presented with a limited choice of goods.

Thirdly, there are tariffs or duty on imported and exported goods. Tariffs are charges that governments make on imported or exported goods. If the charges are very exorbitant, traders may not be willing to import from or export to other countries. The tariffs therefore limit the volume of international trade.

Embargoes are another important trade barrier. This is where a government makes a ban on trade with another country. When this ban is imposed on a particular country, there is a very big likelihood of a similar action from the other country. This therefore prevents countries from freely trading with each other.

Lastly there are foreign exchange controls. In this case the government sets out a limit on the amount of money that is paid by importers to foreign suppliers. Once the limit is reached, the importers are prevented from importing more goods regardless of the demand. This is a disadvantage to both consumers in the local economy and the foreign suppliers.

5. *Firstly, international trade is important because it enables countries to earn foreign currencies through the sale of agricultural commodities. This foreign currency is important to countries because they use it to obtain goods from other countries that are not locally produced.*

   International trade is also important to countries because it enables them to get revenue through taxation. When exporting or importing agricultural commodities, traders are required to pay a certain amount to the government as tax on the commodities. This amount of money collected is very important to countries as it is used for various development projects.

   Another point is that international trade helps to widen the market base for agricultural products where farmers are able to sell to. This encourages the farmers to increase their production by efficiently utilizing the resources at their disposal.

   The third point is that international trading offers consumers a wide range of products from all over the world. The significance of this is that consumers are able to get these products at reasonable prices than where a limited range of goods is available.

   Lastly, international trade is important because it provides employment opportunities to a wide range of people who are directly or indirectly engaged in agriculture. These job opportunities enable the people earn money for uplifting their lives.

Chapter Five

FARM BUSINESS MANAGEMENT

Questions

5.1 Farm records

1. A maize farmer had 1 hectare of land and recorded the following transactions:

- Bought fertilizer for K16,000.
- Paid K15,000 for casual labour.
- Paid K 20,000 for permanent labour.
- Bought seed for K7,500.
- Selling price of maize was K2,800 per bag.
- Sold 48 bags of maize and kept the rest for home consumption.
- Bought 6 hoes at K5,000 each.
- Harvested 60 bags of maize.

a. Produce a financial record.

b. Calculate the farmer's Gross Margin.

c. Calculate the farmer's profit.

2. A farmer kept the following record. Use it to answer the question that follows:

-Harvested 52 bags of maize.
-Sold 40 bags of maize at K2,500/ bag.
-The family consumed 12 bags of maize.
-Bought 6 bags of fertilizer at K5,500/bag
-Sold maize stalks for K4,000.
-Paid casual labour for K36,000.
-Bought 20kg maize seed at K150/kg.

Calculate the farmer's profit or loss.

3. a. Give any two examples of production records on a sheep farm.
   b. Give any two reasons for keeping production records on a sheep farm.

4. Table below shows a type of production record for groundnuts kept by Mrs Zabwera. Use it to answer the questions that follow.

Date	Work done	Number of persons	Number of days worked
07/01/08	Planting	3 women 2 children	1
20/01/08	1st weeding	5 men 3 women	2
06/02/08	2nd weeding	3 children 4 men 1 child	1

a. Name the type of production record.

b. Calculate the total number of mandays for the first weeding.

c. Assuming the farmer had a constant supply of 9 mandays every day, calculate the labour surplus or shortfall for the two days of the 1st weeding.

d. Explain two ways in which the record is important.

5. The table below shows a health record on cattle production for the 2008/2009 farming season. Use it to answer the questions that follow.

Date	Symptoms of disease	Drugs used	Cost of treatment
07/05/09	-High fever -Excessive salivation -Lameness	-	-
15/06/09	-Loss of appetite -Coughing -Difficult breathing -Nasal discharge	-Antibiotic	- K500
20/08/09	-Swollen udder -Clots in milk	-Tetracycline	- K700

a. Name the disease that attacked the cattle on 07/05/09. (1 mark)

b. What was the cause of the disease named in a? (1 mk)

c. Give one reason for not administering drugs to the diseased cattle on 07/05/09? (1 m.)

d. Calculate the cost of treatment incurred during the 2008/2009 farming year. (2 m)

e. Give three [two] ways of preventing the disease shown by the symptoms on 20/08/09. (2 m.)

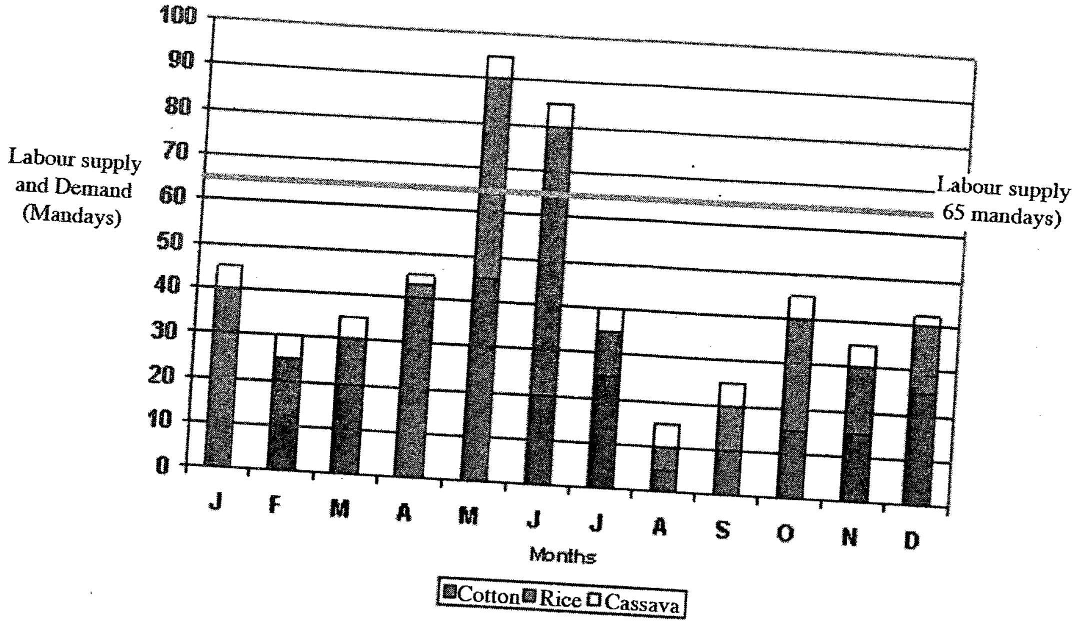
f. Name any other two farm animals that can be attacked by the disease shown the symptoms on 07/05/09.

6. A farmer bought an ox cart at K25,000. The oxcart had an expected life span of eight years and an estimated salvage value of K6, 500.

a. Explain two factors that would make the oxcart reach salvage value earlier than the expected life span.

b. Using a straight line method, calculate annual depreciation for the ox cart.

7. The figure below is a bar graph showing labour profile for cotton, rice and cassava. Us it to answer questions that follow.



a. During which months was labour insufficient?

b. In which month was labour demand lowest on the farm?

c. Calculate the total amount of labour needed to meet demand in the months mentioned in a.

d. Explain any two ways in which a farmer could solve the problem in a.

e. Name one crop which had the highest labour demand and another which had the least during the year.

f. Explain any two ways in which knowledge of labour demand for different enterprises is important.

g. Give one point to explain what would happen to labour demand if all the land on which cotton was grown was used for growing rice.

h. Other than labour, state any two farming resources which are used in production.

8. Table 3 shows transactions that were recorded on a farm in the year ending 31st December, 2009. Use it to answer the questions that follow.

Opening valuation	K8500
Closing valuation	K10000
Depreciation	K1000
Fertilizer application	K7000
Casual labour	K2000
Tobacco sales	K24000

18500

2000

a. Prepare a profit and loss account for this farmer.

b. What is the farmer's profit or loss.

c. Describe any three uses of this type of record on a farm.

d. Explain the benefit of calculating depreciation on a farm.

9. A farmer had the following assets at the beginning of September, 2006:
   - 10 hectares of land valued at K50,000.
   - 100 bags of maize valued at K500,000.
   - 200 broilers valued at K100,000.
   - 20 goats valued at K16,000.
   - 20 bags of feed valued at K60,000.

Prepare an inventory record for the farmer.

10. A vegetable farmer kept the following records.
    - Land tax K3,000.
    - 2 bags of CAN fertilizer at K1,200 per bag.
    - Wages of permanent labour K18,000.
    - Insecticides K1,300.
    - Cost of casual labour K6,000.
    - Packets of vegetable seed K3,000.
    - Depreciation cost of irrigation equipment K3,000.

    a. Classify the records into fixed cost and variable costs.

    b. What was the total variable cost?

    c. What was the total fixed cost?

    d. If the farmer sold 100 bunches daily on average for a period of 90 days at a price of K10 per bunch, calculate:

       i. The amount obtained from the vegetable garden.

       ii. The gross margin of the vegetables enterprise.

       iii. The profit/loss of the vegetable enterprise.

    e. What is the main difference between variable costs and fixed costs?

    f. Explain any one way in which farmers prefer calculating gross margins to profits.

11. A family has a total supply of labour of 40 mandays per month. Below is a table showing labour demand for one hectare of maize in mandays per month.

Month	**Mandays**
January	55
February	50
March	70
April	26
May	22
June	12
July	10
August	5
September	15
October	27
November	46
December	48

a. Construct a labour profile in a Bar graph for the maize enterprise.

b. Mention any four months during which the family can manage the amount of work available.

c. Which four months does the enterprise experience peak periods?

d. Mention the activities on the maize enterprise that lead to peak period?

e. Explain two ways in which you would manage the peak periods.

5.2 Decision making

1. a. State any characteristics of land that would influence choice of a crop enterprise.

   b Describe the following types of enterprise combination.

   i. Supplementary enterprises.

   ii. Complimentary enterprises.

   iii. Competitive enterprises.

2. You are provided with the following information.
   - Farmer A had only enough money to buy either a bag of fertilizer or a bag of livestock feed.
   - Farmer B had only enough land on which to grow either pasture or mangoes.
   - Farmer A decided to buy a bag of livestock feed while farmer B decided to grow pasture.

   a. State the opportunity cost for farmer A if a bag of fertilizer would have given him/her the highest profit.

   b. State the opportunity cost for farmer B if pasture had the lowest value.

   c. Name the type of enterprise combination if a farmer raises both beef and dairy cattle at the same time.

   d. Explain your choice of the type of enterprise combination in a.

3. Describe how a farmer's managerial skills would influence choice of enterprise combination.

4. Explain one way in which each of the following agricultural economics principles help to increase crop production.

   a. Substitution of inputs.
   b. Comparative advantage.
   c. Opportunity cost.

5. Explain one way in which each of the following enterprises can benefit from fish farming:

a. Crop production.
b. Beef production.

6. Farmers in a certain village have the following resources at their disposal:
   - Small farm holdings with clay soils.
   - A big river.
   - Livestock.

Explain how the farmers can use the available resources to maximize production.

7. A farmer has five hectares of land with the following enterprises: maize, beans, poultry and a fish pond.

   Describe how the farmer can combine the enterprises in order to reduce costs of production and maximize profits.

8. In agricultural production, farmers have to face many variables which may be outside their control and such variables are known as risks and uncertainties.

   a. Describe any risks and uncertainties that farmers may face in Malawi.

   b. Explain any five ways in which farmers in Malawi may adjust to risks and uncertainties.

9. State any two factors to consider when selecting enterprise combination.

5.3 Farm budgeting

1. A soil test of Mrs. Phiri's garden showed that it needs 42 kilogram's of phosphorous per hectare for optimum maize production. She intends to use 23:21:0+ 4S as a source of phosphorous.

   a. If there are 5 hectares of land, calculate the amount of the fertilizer required in kilograms.

   b. If a 50 kilogramme bag of 23:21:0+ 4S costs K 6, 500, calculate the amount of money required to buy the fertilizer.

2. A farmer has 2 hectares of land on which he/she grows MH 18. He/she however plans to make the following changes:

   - To apply 4 bags of urea instead of 6 bags per hectare at K1,300 per bag.
   - To sell 40 bags of maize at K1000 per bag to Chibuku products instead of K850 per bag to ADMARC.
   - To store maize in 50 sacks at K30 each instead of storing it in the nkhokwe.
   - To spend K500 instead of K300 on actellic.
   - To spend K600/hectare instead of K300/hectare on casual labour.

   a. Prepare a partial budget for this farmer.

   b Should this farmer go ahead with his/her plan? Explain your answer.

   c Explain two main uses of partial budget prepared in question 2a to this farmer.

   d Explain the major weakness of the partial budget prepared in question 2a.

3. A farmer has 10 hectares and has a choice of growing either more maize or groundnuts. The following information is available for use in decision making.

	Maize	Groundnuts	Beans
Yield (kilogrammes/hectare)	6000	2000	2500
Price (Kwacha/kilogramme)	30	50	40
Cost of seed/hectare	K500	K1200	K1000
Cost of fertilizer/hectare	K2400	-	K2000
Depreciation of Nkhokwe/year	K500	K500	K500

a. Calculate variable costs per hectare for maize, groundnuts and beans.

b. Which crop between maize and groundnuts would the farmer be encouraged to increase hectarage? Show your calculations.

c. If the farmer wanted to substitute beans for groundnuts, what advice would you provide? Show your calculations.

d. Explain one reason for considering depreciation when calculating profit.

e. What two important variable costs may have been forgotten in the data?

5.4 Agricultural cooperatives

1. State any three principles under which agricultural cooperatives are formed.
2. Explain any five benefits of agricultural cooperatives to small holder farmers.
3. Explain any five factors that contribute to the success of agricultural cooperatives.

6,420
115860

Chapter Five

FARM BUSINESS MANAGEMENT

Answers

5.1Farm records

1. a. A receipts and expenditure account

Sales and receipts | **Purchases and Expenses**

Details	**Amount (K)**	**Details**	**Amount (K)**
Maize Sales :48 bags @ K2,800 each	134,400	Fertilizer	16000
		Casual labour	15000
		Permanent labour	20000
		Seeds	7500
		6 hoes@ K5000 each	30000
		Cost of consumed maize: 12 bags @ K2,800 each	33600
		Balance Carried Down	12300
	134,400		**134400**

b.

Total output		60bags
Price / bag		K2,800
Gross Income		K168,000
Variable costs		
Fertilizer	K16,000	
Casual labour	K15,000	
Seeds	K7,500	
Total Variable costs		K38,500
Gross Margin		K129,000

c. Profit = Gross margin - Fixed costs (Permanent labour)

= K129, 500 - K20, 000

= K109, 500

2. A financial record

Sales and Receipts | **Purchases and Expenditure**

Details	**Amount (K)**	**Details**	**Amount (K)**
Sales: 40 bags @ K2,500 each	100, 000	Purchases: 6 bags fertilizer @ K5, 500 each	33, 000
Sales: maize stalks	4, 000		
		Casual labour	36, 000
		Purchases: 20kg maize seed @ K150/kg	3, 000
		Cost of consumed maize: 12 bags @ K2, 500	30, 000
		profit	2, 000
	104, 000		**104, 000**

3. a. i. Sheep numbers record.
   ii. Sheep breeding record.
   iii. Sheep health record.
   iv. Sheep feeding record.
   v. Birth record.

   b. i. For assessing the profitability of the sheep enterprise.
   ii. For assessing the method of sheep management that is currently under use.
   iii. They assist the farmer in planning for breeding of sheep.
   iv. Farmers are able to cull unproductive animals.

4. a. It is a labour record.

   b. Average man days per month are: Men 25 man days (1 x 25), Women=17.5 man days (0.7x 25), Children = 7.5 man days (0.3 x 25)

   2 days labour supply = (2 days x 5 men) + (0.7 x 3 women x 2 days) + (0.3 x 3 Children x 2 days)
   = 10 + 4.2 + 1.8 man days
   = 16 man days

   c. 18 man days - 16 man days
   = 2 man day surplus.

   d. i. It helps the farmer to establish the labour requirement on the farm so that peak and slack periods are assessed.

   ii. It helps the farmer to calculate the labour cost in relation to the output of the groundnuts so that the farmer is able to come up with an accurate budget.

   iii. It helps the farmer to assess the efficiency of the type of labour used on the farm.

5. a. Foot and mouth disease.

   b. Virus.

   c. The disease has no treatment

   d. K500 + K700 = K1,200.

   e. i. Treating with antibiotics such as tetracycline.

   ii. Using right milking techniques.

   iii. Practicing strict cleanliness and using disinfectants when milking.

   f. Pigs, goats, sheep.

6. a. i. Physical deterioration that is caused by rusting, tear and wear or rotting due to usage of the asset.

   ii. Obsolescence which results into an asset being put out of use due to advancements in technology. This causes the assets to become outdated and useless even though it is still capable of being used.

b.

Annual Depreciation $= \frac{\text{Cost - Salvage Value}}{\text{No. of years}}$

$= \frac{K(25{,}000 - 6{,}500)}{8 \text{ years}}$

$= \underline{K2{,}312.50}$

7. a. May and June.

b. August.

c. = 30 mandays (May) + 20 mandays (June)
= 50 mandays.

d. i. By hiring additional casual labour to supplement the already available labour.

ii. By growing crops that do not have similar labour requirements to ensure even distribution of labour through out the year.

e. Cotton had the highest labour demand while cassava had the least labour demand.

f. i. The farmer is able to establish the labour peak periods such that measures are put in place at the right time to prevent labour shortages during peak periods.

ii. The farmer is able to estimate the costs for labour in each enterprise which assist in ascertaining profitability of various enterprises.

iii. Helps the farmer to assess the efficiency of the type of labour used on the farm.

g. The demand for labour would decrease because farm operations will be decreased.

h. Capital, managerial skills and land.

8. a.

A Profit and loss Account for year ending 31st Dec, 2009

	Amount (K)		**Amount (K)**
Opening valuation	8,500	Sales: Tobacco	24,000
Depreciation	1,000	Closing Valuation	10,000
Fertilizer	7,000		
Casual Labour	2,000		
Profit	15,500		
	34,000		**34,000**

b. The profit is K15, 500

c. i. It helps in the calculation of profit (or loss) in a given period, and so, helps the farmer to check financial progress on the farm.

ii. It is useful in valuation of assets at the start and end of farming year thereby enabling the farmer to assess the performance of business in the course of the year.

iii. It helps the farmer to calculate the amount of tax to be remitted to the government in a given farming year basing on the profit (or loss) that has been made.

d. Depreciation is a cost that reduces the farm's profitability. Calculation of depreciation enables the farmer to have a true picture of the financial position of the farm business.

9. a. Farm inventory as beginning of September, 2006

ITEM	QUANTITY	ESTIMATED VALUE (MK)
Land	10 hectares	50, 000
Maize	100 bags	500, 000
Broilers	200	100, 000
Goats	20	16, 000
Feeds	Bags	60, 000
Total value		**726, 000**

10. a. fixed and variable costs

Fixed costs	Variable costs
Land tax	Fertilizer
Permanent labour	Insecticides
Depreciation	Casual Labour
	Vegetable seeds

b. MK11,500

c. MK24,000

d.i. = 100 bunches x K10/ bunch x 90 days
= MK90,000

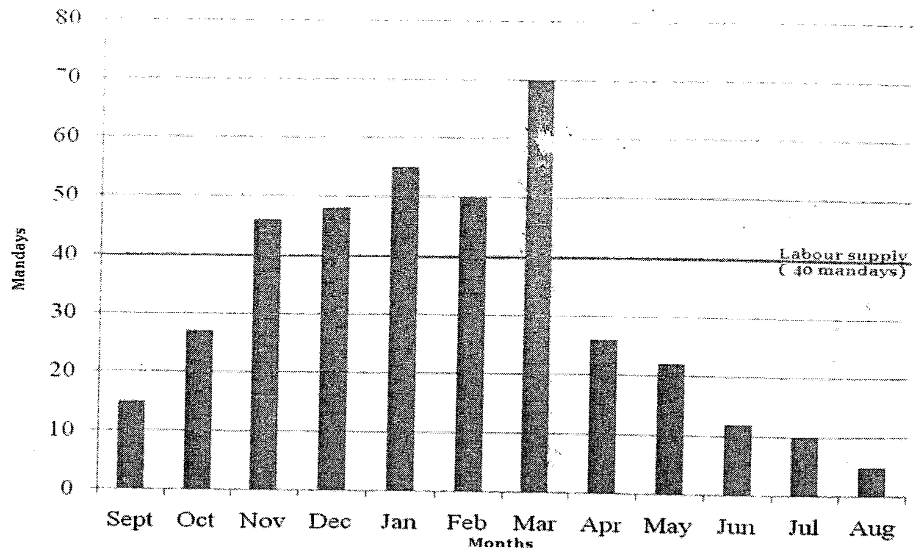
ii. Gross margin = value of output - variable costs
= MK 90,000 - MK 11,500
= MK 78,500

e. Profit/loss = Gross margin - fixed costs
= MK 78,500 - MK 24,000
= MK 54,500 (profit)

f. Variable costs change with the size of an enterprise while fixed costs do not change very much in relation to size of enterprise, but still farmers have to pay for them whether used or not.

g. Gross margins enable the farmer to assess the profitability of each and every farm enterprise.

11. a. **A Bar graph showing a labour profile for a maize enterprise**



b. September, October, April, May, June, July, August.

c. November, December, January, February, March.

d. Planting, weeding, fertilizer application and harvesting.

e. i. By hiring casual labour which helps in supplementing the already existing labour supply.
ii. Using farm machinery which helps farm operations to be completed in time.

5.2 Decision making

1. a. i. Size of the land,
ii. Quality of the land e.g. soil fertility, soil pH, drainage, water retention,
iii. Location of the land,

b. i. Supplementary enterprises are those that do not compete for resources with other enterprises. Rather, they make use of superfluous or excess resources that may be available on the farm. Examples of these resources may be farm buildings that are staying idle and spare labour.

ii. Complimentary enterprises are those in which a shortfall in one enterprise is overcome by another. The increase in one enterprise may result into an increase in another enterprise. For example, fish farming and vegetable production.

iii. Competitive enterprises are enterprises that compete or require similar resources in order to produce. The problem with this type of enterprise combination is that an increase in the productivity of one enterprise is followed by a decrease in the output in the other enterprise.

2. a. The revenue that is not realized for not producing crops using the fertilizer.

b. The lost revenue for not producing mangoes.

c. Competing enterprise.

d. Both dairy and beef animals require same resources such as pastures, labour and capital investment.

3. Generally farmers would tend to choose those enterprises that they can best produce using their skills and experience.

4. a. This principle requires that farmers should combine inputs in a way that minimizes costs while ensuring increasing productivity. This is important in that it helps farmers to use inputs economically by replacing expensive inputs with cheap ones that enable the farmers to maintain productivity because inputs that are saved in one enterprise may be useful in increasing the productivity in another enterprise.

   b. This economic principle suggests that farmers should concentrate in producing in enterprises that they can produce better using resources at their disposal such as land, capital, climate and labour.

   c. This is a principle that helps farmers to choose enterprises that will ensure that they are able to produce high yields by efficiently utilizing the limited available resources. it is very useful when farmers are faced with a challenge of choosing the most rewarding enterprise from a pool of other best alternative enterprises.

5. a. The water in the fish pond can be used for irrigation of crops, thereby allowing farmers to produce more crops.

   b. The fish can be used for the making of fish meal which is a good source of proteins and other feed nutrients to the beef animals.

6. Firstly the farmers may use part of their small farm holdings for the cultivation of crops. In order to increase crop productivity, the farmers may make use of the water in the river for irrigation of the crops. With this type of farming, the farmers may be able to harvest several times in a year, leading to increased productivity.

   Secondly, the farmers may also practice livestock production. Bearing in mind that their farm holdings are very small, the farmers may be encouraged to raise their livestock under the intensive system. The advantage of this system is that a large number of animals can be kept in a small area. In order to promote productivity, the farmers may use crop residues from irrigation farming to feed the livestock. Additionally, part of the land may be allocated to pasture production that can help to supplement livestock feeds.

7. In the first place, the farmer may engage in crop production on part of the land. In order to increase productivity, the farmer may practice a crop rotation system in which beans are alternated with maize. The advantage of this type of farming is that it helps in minimizing the costs for buying nitrogenous fertilizers as the beans will be able to fix nitrates that are greatly needed by the maize. This results into higher yields for the maize.

   The second way of minimizing costs of production would be where the farmer is engaged in producing feed rations for the poultry using beans, maize and fish. This will help to ensure that the cost for buying expensive commercially produced feeds is reduced. Furthermore, this farmer may also reduce costs for buying fertilizers as the manure from poultry may be used to increase fertility in a garden where beans and maize are grown.

   Another beneficial combination of enterprises is where the farmer is practising irrigation farming using water from the fish pond. This kind of farming may allow the farmer to harvest several times in a year, leading to increased productivity in the crop enterprises. The fish farming may also greatly benefit because of the feeds that are produced using the beans and maize.

8. a. i. Price fluctuations: This becomes a serious problem when the price of agricultural produce becomes very low due to such factors as over supply that may be caused by good weather conditions. Lower prices for agricultural produce reduces the profitability of the farm.

ii. Incidences of droughts can cause some crops to fail and animals to die. This results into losses being made in the affected enterprises.

iii. Heavy rainfall may cause damage to crops, livestock and farm infrastructure. Such damage also results into great losses in the farm's revenue.

iv. Pests and diseases are also another uncertainty that can cause great loss to the farmer as these may reduce the quantity and quality of crop yields.

v. Arson is also another problem that farmers may face. This is the intentional setting of fire on other people's property. Such fires may damage farm property leading to huge loses on the farm

vi. Theft of property also results into major loses on the farm.

b. i. By practising crop diversification in which the farmer grows different crops in the same field. The advantage of this is that it reduces the risk of total crop failure in cases of droughts, heavy rainfall, pests and diseases.

ii. Farmers may engage in more reliable enterprises in which the risk of failure is very low.

iii. Practising input substitution in which case the farmer strives to use less expensive inputs in production. For example, the farmer may decide to use organic manures instead of inorganic fertilizers that are very expensive. This helps to lower the cost of production.

iv. By insuring crops, livestock and farm infrastructure against such risks like droughts and arson. If these goods are insured, any damage caused to them may be replaced by the insurance company.

v. By storing food items so that they are made available in times of food scarcity.

vi. Practising input rationing in which case fewer than the required inputs are used in production. This help in reserving some inputs so that in terms of failure, the farmer may still have some inputs for use in further production.

9. a. Comparative advantage.
b. Risks and uncertainties.
c. Availability of resources such as land, labour, capital and managerial skills.
d. The farmer's family food requirement.

5. 3 Farm budgeting

1. a.

Ratio of the fertilizer	= 23:21:0+4s
Part of Phosphorus/ha	= 21
Amount of Phosphorus/ha	= 42kgs ie 2times the parts of phosphorus in the bag
Amount of fertilizer/ha	= {(23x2) + (21x2)+ (0x2) +(4x2)}kg
	= 96kg
Amount of fert. for whole plot	= 96kg x5ha
	= 480kgs

b. $\frac{480kg}{50kg} \times \frac{K6,500}{1}$

K62,400

2. a. partial budget.

COST OF CHANGE INCOME FROM CHANGE

Extra costs	(K)	(K)	Saved costs	(K)
◆50 Sacks @K30 each	1500		◆4 bags (urea @K1,300 each)	5,200
◆Actellic	200		Extra returns	
◆Casual labour (K300/ha)	600		◆40 bags sold @K1000 each	40,000
Total extra costs		2300		
◆Sacrificed income (40 bags @K850 each)		34,000		
Profit from change		8,900		
		45,200		**45,200**

b. The farmer would be advised to go ahead with the change because it results into saving an amount equal to K8,900.

c. i. It is used in estimating the effects that changes in the enterprise may have on the future profitability of the farm enterprise.

ii. It is used as a tool for deciding the most profitable of possible alternative enterprises or methods of utilizing farm resources.

d. It has only considered items that are going to be affected by the proposed change, thereby leaving out other equally important items that affect the profitability of the whole farm enterprise.

3. a.

Variable costs	Maize (K)	Groundnuts (K)	beans (K)
Seeds	500	1200	1000
Fertilizer	2400	-	2000
Total variable costs	**2900**	**1200**	**3000**

b.

Gross Margins

	Maize	Groundnuts
Output (Kg/ha	6000	2000
Price (K/kg	30	50
Gross Income	180,000	100,000
Total variable costs	2,900	1,200
Gross Margin (K)	177,100	98,800

The farmer would be encouraged to increase hectarage for maize because it has a very high gross margin.

c.

	Gross Margins	
	Groundnuts	Beans
Output (Kg/ha)	2,000	2,500
Price (K/kg)	50	40
Gross income	100,000	100,000
Total variable costs	1,200	3,000
Gross Margin (K)	98,800	97,000

The farmer would be advised to go ahead with the plan because groundnuts provide the highest gross margin and has lower variable costs per hectare than beans.

d. Depreciation is a cost that reduces the profits for the farm, as such if included when calculating profits, it helps to reflect the true picture of the farm's finances.

e. Cost for casual labour and pesticides.

5.4 Agricultural cooperatives

1. a. It should be legally recognized.
   b. It is run on democratic principles.
   c. Participation is voluntary.
   d. It is formed by people with common interests.
   e. Membership is open to all members of the community.

Essay

2. *Firstly, farmers are able to obtain farming inputs at very low prices. This is because when farmers are in a cooperative, they are able to buy commodities in bulk at wholesale prices which are very low. Through cooperatives, farmers are therefore able to minimize production costs leading to increased net profits.*

   Cooperative societies also enable farmers to mechanize some of their operations. This is possible because the farmers as a group can contribute towards the hiring of certain expensive farm machinery. When hired, each farmer is able to make use of the machinery than where each farmer is operating solely.

   Farmers in a cooperative are also able to access credit facilities more easily from lending institutions than on individual basis. This is because cooperatives are able to provide security for the loans. Such credit helps the farmers to expand their business enterprises.

   Cooperatives are also important in that they enable farmers to sell their commodities at better prices. The reason for this is that as a group, farmers in cooperatives have a greater bargaining power than where each farmer is operating individually.

   The other advantage of joining cooperatives for farmers is that farmers are able to make more profits, especially where the cooperatives are performing some marketing functions. Through performing such marketing functions like grading, they help in adding value to the commodities. This helps the farmers to sell their commodities at higher prices, hence the increased profits.

3. *Firstly, the cooperatives should have sufficient funds that are well managed. These funds are used in the building of important infrastructure like offices as well as for the daily running of the cooperative. It is also advisable to ensure that the accounts of the cooperative are periodically audited so as to prevent misuse of the funds.*

Secondly, cooperatives should have clearly known goals and objectives. Such objectives ensure that each member knows their expectations and work towards achieving the set goals.

Thirdly, cooperatives need to ensure that people that are employed to oversee the daily affairs of the cooperatives have the necessary expertise. This ensures that the activities of the cooperative are run smoothly.

Another important factor is availability of training opportunities for members. Where ever necessary, the members of the cooperative should be provided with relevant education such as on business management and record keeping that can ensure improved productivity.

Lastly, the success of a cooperative requires job security on the part of the management staff. It is only when these people are certain about the future of their jobs that they work hard.

Chapter Six

AGRICULTURAL TECHNOLOGY

Questions

6.1 Farm energy

1. State the use of the following sources of energy on the farm:

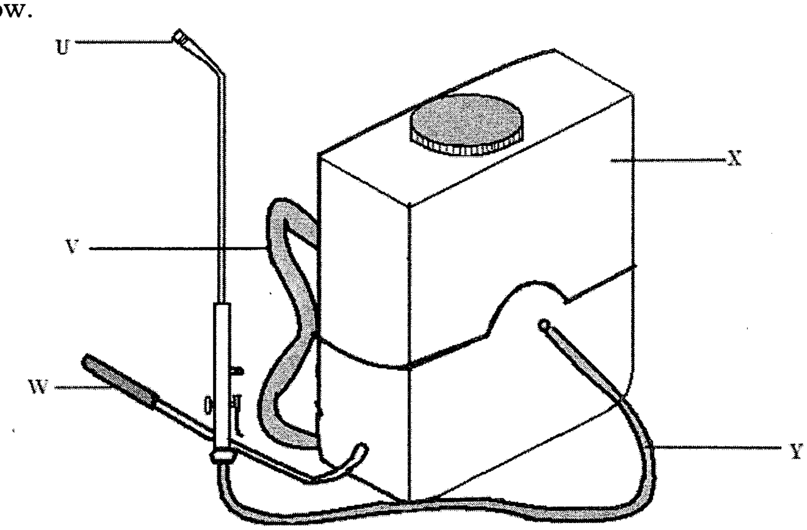
   a. Fuel energy.
   b. Solar energy.
   c. Mechanical power.
   d. Electrical energy.
   e. Hydro power.
   f. Wind power.
   g. State one advantage and disadvantage of the following sources of farm energy. (i) Solar energy (ii) Wind energy (iii) Fuel energy.

2. a. Name any two tillage implements which use human power.

   b. Differentiate between mechanical and solar energy.

3. List down any five safety measures when using farm energy.

4. Figure below is a diagram of a farm implement. Use it to answer the questions that follow.



   a. Name the farm implement.

   b. Name the parts labeled U, V, W, X, and Y.

   c. State one function of the part labeled U.

   d. State any one form of power that can be used to operate the farm implement.

   e. Explain one way in which use of the farm implement can improve beef production.

   f. Explain any one way of maintaining the farm implement.

5. a. Give any two sources of fuel energy on smallholder farms.

   b. State two environmental problems that may be caused by the sources of fuel energy given above.

   c. Explain one effect of each of the environmental problems stated above on agricultural production.

6.2 Irrigation systems and land drainage

1. State any two points to consider when selecting a water source for an irrigation system.

2. Explain two environmental problems that flood irrigation may cause.

3. Explain one difference between drip and sprinkler irrigation systems in terms:

   a. Water requirement.
   b. Efficiency in use of water.
   c. Facilitation of soil erosion.

4. a. Define soil salinity.

   b Explain one way in which irrigation can cause soil salinity.

5. Explain one way in which each of the following factors can influence the amount of water applied to a crop:

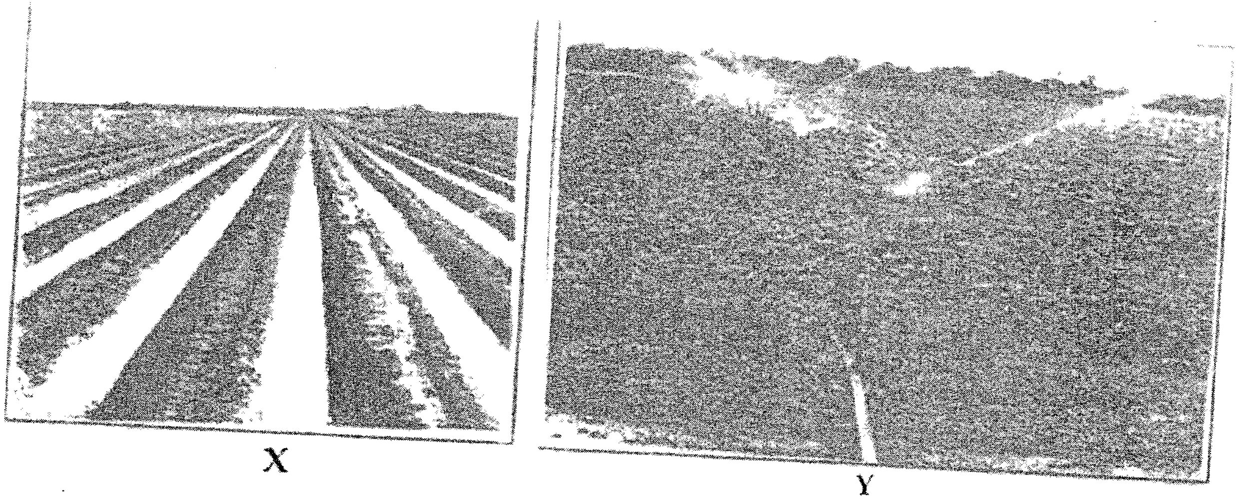
   a. Crop water requirement.

   b. Soil texture.

6. Low income farmers live in an area with the following characteristics:
   - Moderate sloping land with clay soils.
   - A dam with plenty of water situated on the upper part of the farms.
   - Plenty of unskilled labour.

If the area experiences a long dry spell every year, describe how a suitable irrigation system for maize production would be established and managed.

7. **Figure 2 shows diagrams of irrigation systems labeled X and Y. Use it to answer questions that follow:**



X Y

a. Identify the types of irrigation systems labeled X and Y.

b. State any three advantages of irrigation systems shown in Y.

c. State any three disadvantages of irrigation systems shown in Y.

d. Explain one advantage of planting trees in the windward side of a garden irrigated by the system in Y.

e. What is the name of the structures filled with water in X.

f. Explain any two problems caused by the structures named in e.

g. Briefly explain how system X of irrigation can alter the farming calendar.

h. List any three factors to consider when establishing the irrigation system in X.

8. Explain any two problems and solutions to the problems that are associated with surface irrigation.

9. a. What is land drainage?

b Give any three reasons for draining land.

6. Describe the two main methods of land drainage.

7. Explain any two effects of water logged soils on growth of arable crops.

10.a. Name the type of land drainage ideal for a farm where an ox-drawn plough is used.

b Explain your answer above.

6.3 Farm mechanization

1. Name any two examples of ox drawn implements.
2. Explain any five limitations of mechanization
3. a. State any two examples of gender biases on agricultural technology.

b. Describe any two implications of gender biased application of agricultural technology.

Essay

4. *Describe any five factors that should be considered when mechanizing a farm.*

5. *Explain any five advantages of mechanization.*

Chapter Six

AGRICULTURAL TECHNOLOGY

Answers

6.1 Farm energy answers

1. a. i. It provides energy for driving farm machineries.
   ii. For lighting.
   iii. For heating.

   b. i. For drying of crops.
   ii. For photosynthesis.
   iii. For producing electricity.

   c. For driving machineries.

   d. For driving motors that are used in operating various farm appliances.

   e. For generation of hydro electricity.

   f. i. For driving windmills that produce electricity.
   ii. For pumping water.

   g. Advantages and disadvantages of different energy sources.

Energy source	advantage	Disadvantage
Solar energy	It is cheap	Affected by weather conditions
Wind energy	Requires less servicing once installed	Irregular and uncertain power supply
Fuel energy	Very efficient	It is expensive

2. a. A plough and a hoe.

   b. Mechanical energy is power that is directly or indirectly transmitted through an engine through the use of electricity or fuels. Solar energy on the other hand is heat and light energy that comes from the sun.

3. a. Avoid exposure of fuels to an open flame.
   b. Switching off running engines when refueling at pump stations
   c. Do not wear loose clothing when working with machines
   d. Wearing sunglasses when working on the sun
   e. Electric wires should always be treated as live at all times

4. a. Knapsack sprayer.

   b. U: Nozzle.
   V: Shoulder strap.
   W: Pumping/priming lever.
   X: Chemical container/reservoir/tank.
   Y: Hose pipe.

c. To let chemicals out.
d. Man power.
e. It helps to control external parasites such as ticks which can transmit diseases that affect the health of the beef cattle.

f. i. Washing out the container after using to remove any remaining chemicals.

ii. Cleaning the nozzle in order to remove any foreign materials that can block the outflow of chemicals.

iii. Cleaning and applying oil to the trigger valve for easy handling.

5. a. Wood, Charcoal, Coal, Oil.

b. Deforestation and environmental pollution.

c. i. Deforestation leaves the land bare thereby exposing the top soil to agents of erosion.

ii. Air pollution due to the burning of fuels releases green house gases into the atmosphere which causes climate change. This can result into droughts and erratic rainfall patterns.
- Oil spill in water bodies can lead to death of aquatic forms of life.

6.2 Irrigation systems and Land drainage

1. a. The type of irrigation system.
b. Proximity to the water source.
c. Quality of the water.
d. Topography of the land.

2. a. Flooding of the land causes soil erosion leading to reduced soil fertility.
b. It leads to the spread of waterborne diseases that may cause harm to people and livestock.

3. a. Drip irrigation requires less amount of water because the water drips slowly from the pipes while sprinkler irrigation requires more water because it is applied all over the field.

b. Drip irrigation uses water more efficiently as the water only gets to the root base where it is needed. On the other hand, sprinkler irrigation wastes a lot of water as some of it may fall on leaves where it may evaporate. Additionally, some water may be blown away if the weather is windy.

c. Drip irrigation does not facilitate soil erosion since the water trickles very slowly below the ground. In contrast, sprinkler irrigation can facilitate soil erosion when large drops from the nozzles directly hit on bare soils. This may cause splash erosion.

4. a. Is a condition when the soils have a higher concentration of soluble salts.

b.i. If the source of water for irrigation has got a lot of salts, these salts can accumulate in the area under irrigation.

ii. In areas where rainfall is seasonal or very low, salts can accumulate on the surface. During the rainy season, the salts are dissolved and cause damage to the plants.

5. a. Some plants like rice require more water than others. In this case, plants with a lot of water requirement would need more water for irrigation than those that require less water.

   b. Areas with a lot of course textured soils (sandy soils) require more water for irrigation than those with an abundance of fine textured soils (clay soil). This is because course textured soils, due to their large particles, have large pore spaces which allow more water to pass through, thereby making them to keep less water. In contrast, fine textured soils have very small spaces that do not allow a lot of water to pass through, making them to keep more moisture for a very long time.

6. In the first place the recommended system for irrigating the area would be furrow irrigation. This choice of an irrigation system is ideal for the area because water in the furrows can be brought into the fields from the dam under the influence of gravity. Furthermore, the clay soils that dominate the area would ensure that more water is reaching the crops since ground seepage is very minimal in clay soils due to their tightly packed particles.

   In order to establish this type of irrigation the farmers could utilize the plenty of unskilled labour available in the area for the digging of the irrigation channels.

   After successfully establishment of the irrigation system, the farmers may also make use of the unskilled labour for managing the irrigation system. The activities that would be involved in managing the irrigation would include the removal of silt and trash that accumulate in the channels. In addition, the unskilled labourers may also be of great need for the maintenance and repairing of the channels.

7. a. i. X is furrow or channel irrigation.
      ii. Y is sprinkler irrigation.

   b. i. Water is evenly distributed to all parts of the field.
      ii. It does not require leveling of the land.
      iii. It minimizes the rate of soil erosion.

   c. i. It is expensive to buy the sprinklers and pipes.
      ii. The top soil may be hardened as a result of being hit continuously with water drops.
      iii. During a windy day, water distribution is uneven.
      iv. It requires a lot of labour to move the sprinklers all over the field.
      v. Water that falls on leaves may end up being evaporated.

   d. The trees act as a wind break thereby making it possible for water to be evenly distributed in the field.

   e. furrows.

   f. i. It may lead to salt accumulation especially if the water used has a high content of salts.
      ii. Water in the furrow erodes soils causing the furrows to enlarge.
      iii. If the furrows are not properly laid down they may cause difficulties when working with farm machineries.

   g. Use of irrigation enables farmers to grow crops through out the year because water is always available to crops. This is contrasted to where crop production is dependent on rainfall for supply of moisture to crops.

h. i. Topography.
   ii. Source of water.
   iii. Profitability of the crop enterprise.
   iv. Soil type.
   v. Crop water requirement.

8. a. If the water for irrigation has a high salt content, the salts may accumulate in the furrows and cause damage to crops. This problem may be overcome by growing of salt tolerant crops that can reduce the salt content of the soil. Additionally, the salts may be flushed out by flooding the land with fresh water.

   b. If the furrows are constructed poorly the water moving in them can erode and enlarge the furrows. This problem can be solved by making sure that the furrows are not constructed at a steep angle.

9. a. This is the removal of excess water from a waterlogged area.
   b. i. To reclaim land on which agricultural production can done.
      ii. To improve soil aeration.
      iii. To increase soil temperature so as to enhance crop production and work of microorganisms.

   c. i. Open ditch method
      This is where ditches are constructed on the land so that water is carried down the slope by gravitational force. The ditches help in reducing the water table, thereby encouraging the growing of crops. The problem with the method is that the ditches reduce the size of arable land apart from hindering the use of machinery on the farm.

      ii. Underground drainage
      This is where pipes or ceramic tiles are laid underground in order to remove excess water that result from seepage. The advantage of the method is that working with machinery is easy because it leaves no open ditches on the land. However, the tiles and pipes are very expensive to purchase and install.

   d. i. Waterlogged soils are usually very cold. This reduces the rate of chemical reactions in plant roots, resulting into poor plant growth.

      ii. Waterlogged soils provide a conducive environment for denitrifying bacteria which causes nitrates to be changed into nitrogen gas that can easily be released into the atmosphere.

      iii. Waterlogged soils are poorly aerated because the pore spaces are filled with water.

10. a. Sub surface drainage.

   b. The system does not leave the land with open ditches, like in surface drainage. As such, there are no obstructions that are faced with the system when working with machineries.

6.3 Farm mechanization

1. a. Ox-cart.
   b. Plough.
   c. Ridger.
   d. Harrows.
   e. Cultivators.

2. a. It requires skilled labour to operate the machines in order to do a perfect job.

   b. It leads to unemployment of several unskilled labourers because the machines perform jobs that can be done by several people.

   c. Mechanisation requires a lot of capital investment for purchasing, maintaining and repairing the machineries.

   d. It is not profitable on small scale farms because of large overhead costs that are involved in running some machinery.

   e. Lack of adequate maintenance and skill of operating the machinery can cause them to break down, thereby causing some farm operations to be delayed or not performed at all. This may lead to serious losses on the farm business.

3. a. i. Men driving tractors.
      ii. Use of a plough by men only.
      iii. Food processing by women.

   b. i. Gender biases result into low productivity because men and women do not have equal access to agricultural technology. As a result, there is less skilled labour to operate the machinery, hence low productivity.

      ii. It results into low agricultural development because there is less effort that is made towards increasing production as people, especially women feel sidelined on issues of agricultural technology.

Essay

4. *The first factor is capital availability. This is one of the limiting factors to farm mechanization because without adequate amount of money farmers may fail to purchase machinery. Even when the machinery is bought, there is a greater need for maintainance by buying spare parts, fuel and general serving of the equipment. All these activities require a greater financial backing.*

   In cases where farmers are not able to raise enough capital, they can try soliciting funds from other money lending institutions. If farmers are able to access these credit facilities, farm mechanisation could be a possible undertaking.

   Size of the land is another important factor for mechanizing farms, especially in crop production. Mechanization is more practicable on larger pieces of land than small ones. The reason for this is that on large pieces of land farmers are able to cultivate more crops that result into large profits after selling the produce. These large profits in turn, cover up the overhead costs for maintaining the machinery.

   Most of the machinery used on farms are quite complex such that they require people with adequate knowledge to operate them. In this case, where the farmer lack expertise, mechanization of the farm may be very difficult.

   As already noted above, mechanization goes along with huge overhead costs. In order to prevent making losses, the farmer should critically consider the profitability of the enterprise to be mechanized. Where it is clear that the enterprise is likely to end up in making losses, it would be very wise for the farmer to consider other means of production other than mechanisation.

5. *Firstly, mechanisation is important in that it makes farm operations to be completed in time. This helps the farmer to save time and labour for other equally important activities on the farm.*

Timeliness in farm operations has an advantage of helping the farmer in producing high yields. The sale of the increased output therefore results into more profits being realized on the farm.

In crop production, use of various farm machineries enables the farmer to cultivate very large pieces of land. In so doing, the farmer is able to benefit from economies of scale because inputs that are used on such farms are bought at lower wholesale prices. As a result, costs of production are minimized leading to increased profitability.

Farm mechanisation is also beneficial in that it makes work to be enjoyable and easy to do unlike when manual power is used. The result of this is that farm work is done more effectively and efficiently.

Where mechanisation is effectively organized, farmers are able to overcome the problem of labour shortage during peak periods. This is because farm mechanisation is labour saving, meaning that very few people are required to operate a given machine at one time. In this case, any additional labour on the farm can be assigned other operations.

50
0
160

50
0
160

Chapter Seven

AGRICULTURAL EXPERIMENTATION

Questions

7.1 Basic principles of agricultural experimentation and Report writing

1. a. What does evaluation of results in a Fertilizer Trial on maize involve?

   b Explain any one way in which a Fertilizer Trial on maize would contribute to food security.

2. A form four class would like to carry out an experiment to determine the effectiveness of farm yard manure and compost manure over green manure on maize.

   a. What would be the control for this experiment?

   b. Design the experiment in three blocks.

3. a. Explain one way in which each of the following activities in agricultural experimentation is important:

   i. Observing and measuring plants in the net plot.

   ii. Analyzing data.

   b Explain one way in which variety trials in agriculture could ensure food security.

4. A form four class intends to carry out an experiment to find out the appropriate rate of Urea on maize production. The class teacher advises them to use the following information:

Treatments	Rate of Urea/ha
1	0kg
2	40kg
3	50kg
4	120kg
5	160kg

Random Number Table

3	1	8	5	9	4	2	6	7

a. In the experiment, state:

   i. the variable.

   ii. The control treatment.

   iii. Using the random Number Table, design the experiment in three blocks.

   iv. Why should the students use more than one block?

b Describe each of the following steps when carrying out an experiment on Variety Trial in groundnuts:

i. Setting up an experiment.

ii. Clarifying aims.

iii. Drawing conclusions.

5. Form four students conducted an agricultural experiment to find out the effect of different rates of fertilizers on maize yield.

a. What was the variable in this experiment?

b. What was the control of the experiment?

c. Explain any one observation that could have been recorded during this experiment.

6. Table 2 show growth rate of bean seedling. Use it to answer questions that follow.

	DAYS	1	2	3	4	5	6	7	8
Growth rate (mm)	Seedlings from plump seeds	17	25	30	36	50	64	82	96
	Seedlings from shrunken seeds	14	16	22	25	30	33	49	56

a. Draw a line graph with two curves; one on seedlings which germinated from plump seeds and another on those from shrunken seed.

b From the graph

i. Give the mean shoot length of each set of seedlings after 5½ days of germination.

ii. Describe the growth rate of seedlings from plump and shrunken seeds within the first day of germination.

iii. Explain why there is a difference in the curves on the graph after day 4.

c State any three conditions which would have been held constant during the experiment.

d State one conclusion you would make from the results to show why it is necessary to select seeds before planting.

7. Form four students would like to find out the effect of four different spacings (30 cm, 60 cm, 90 cm and 120 cm) on the yield of maize in their school garden.

a. Design an experiment in three blocks using randomized block design.

b. Describe how the experiment would be conducted using the following guide lines:

i. Carrying out the experiment.

ii. Collecting data.

iii. Recording of data.

iv. Analyzing of data.

c Give one reason for repeating the experiment.

d In your opinion, which of the above spacing would you say is the control? Give a reason for your answer.

8. Table 2 shows results of an experiment on maize. Use it to answer questions that follow.

Plots	1	2	3	4	5
Plot size (m2)	50	50	50	50	50
Rate of DAP (kg/ha)	0	100	200	300	400
Rate of Urea (kg/ha)	0	100	200	300	400
Yield of maize (kg)	2.5	40	30	10	5

a. What was the aim of the experiment?

b. What name is commonly given to the treatment in plot 1?

c. Why it is necessary for an experiment to have a treatment named in question b.

d. Which fertilizer rate gave the highest maize yield?

e. Explain one reason for the answer to question c.

f. Explain one reason for harvesting 2.5 kg of maize in plot 1 despite the absence of fertilizer.

g. What was the yield of maize obtained in plot 4?

h. Convert yield of maize obtained from plot 4 to yield in kilogrammes per hectare.

i Explain any two ways in which the nutrient status of the soil in plots should have been analyzed before applying fertilizers.

j Explain any one reason for analyzing the soil in plots before applying the fertilizers.

9. Table 2 shows a wrong randomized block design of an experiment on maize variety trial by students at a certain secondary school. Use it to answer the questions that follow.

	Block 1	**Block 2**	**Block 3**
Plot 1	NSCM 41	MH 16	NSCM 41
Plot 2	Local Maize	MH 12	Local Maize
Plot 3	MH 16	Local Maize	MH 12
Plot 4	MH 12	MH 16	NSCM 41

a. Mention four mistakes in the experimental design.

b. Using the information in table 2, lay out the correct field plan for randomized block design.

c Apart from randomization, explain two other ways in which the students could have made the results of the experiment more reliable.

d Explain two main methods which the students could have used to collect data in the experiment.

e Explain any two ways in which the results from this experiment could have been analyzed.

10. State any two ways in which agricultural experimentation can help to improve agricultural production.

11. Explain any one role of each of the following principles in agricultural experimentation:

a. Replication of experiments.

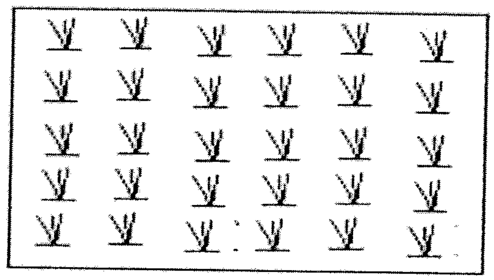
b. Use of a control.

12. Explain the importance of writing reports for agricultural experimentation.

13. Describe the following aspects of a report for an agricultural experimentation.

a. Title.
b. Introduction.
c. Materials and methods.

14. The figure below shows one of the experimental plots to find out the effect of irrigation on the yield of sorghum. Use it to answer the questions that follow.



a. Indicate on the figure by circling all the sorghum plants that would be recommended for observation.
b. Give a reason for the recommendation in a.
c. Explain the need for identifying a sample of sorghum plants for observation from the recommendation made in a.
d. Assuming that the experiment was successful, explain any one way in which such an experiment would ensure sufficient food production.
e. Explain one way in which reporting on such experiment would increase sorghum production.

15. Farmers in a certain area have a serious problem of weeds in maize. Describe the "scientific" approach to experimentation that should be followed in order to overcome the problem.

Chapter Seven

AGRICULTURAL EXPERIMENTATION

Answers

7.1 Basic principles of agricultural experimentation and Report writing

1. a. It involves interpretation of data that has been collected and analysed in an experiment.

   b. The reports that are released after the experiments contain recommendations on best farming practices. If farmers are able to follow such recommendations on their farms, they can produce more products from their farms that help to ensure food security.

2. a. Farm yard manure.

   b. Experimental design.

	Block 1	Block 2	Block 3
Plot 1	Green Manure	Compost manure	Farm Yard Manure
Plot 2	Farm Yard Manure	Green Manure	Compost manure
Plot 3	Compost manure	Farm Yard Manure	Green Manure

3. a. i. -The net plot is not greatly affected by some external forces that may tend to influence the results of the experiment.

   -By concentrating on the sample plants in the net plot alone, the experimenter saves a lot of time for other equally important activities since the sample plants are representatives of the whole plant population being studied.

   ii. This enables the researcher to scrutinize the data collected for easy comparison and evaluation on the experimental treatments.

   b. They help in coming up with recommendations that can increase the production of food.

4. a. i. Rate of fertilizer.

   ii. 0kg of fertilizer (Urea).

   iii. Experimental design.

	Block 1	Block 2	Block 3
Plot 1	T3	T1	T5
Plot 2	T1	T5	T4
Plot 3	T5	T4	T2
Plot 4	T4	T2	T3
Plot 5	T2	T3	T1

   iv. It helps the students to get reliable results because environmental interferences are avoided.

b. i. This involves planning or designing the experimental set up. It includes choosing treatments or variables of the experiment.

ii. Aims of the experiments are the goals that the experiment seeks to achieve. The aims of the experiment defines the scope that the experiment is going to cover. For this reason, the aims should be made as clear as possible.

iii. This includes a brief summary on the major findings of the experiment. The conclusion should be based on the aims and objectives that were set at the beginning of the experiment.

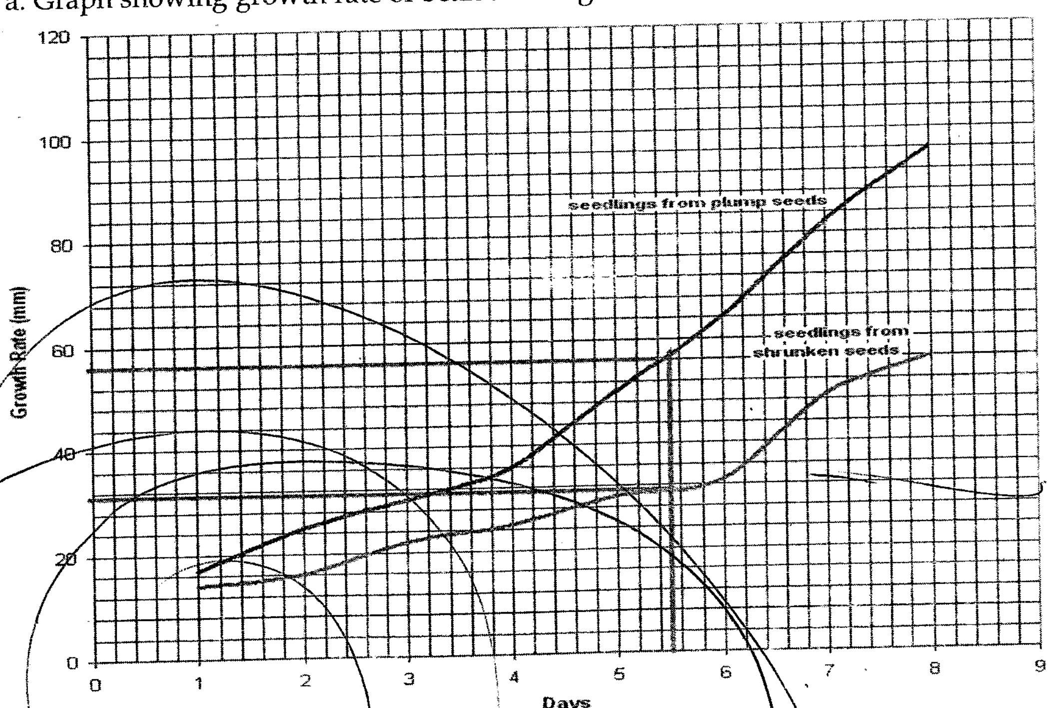
5. a. Fertilizer rates.

b. Maize plants without fertilizer.

c. i. The colour of the leaves is likely to differ depending on the amount of fertilizer applied. For example plants that are applied with the right amount of fertilizer will have a greener colour while those with no fertilizer will show a yellow colour, an indication of lack of nitrogen and other elements.

ii. Some plants are likely to be taller than others depending on the amount of fertilizer applied. For instance, plants without any fertilizer are likely to be short and stunted due to lack of essential plant nutrients that promote plant growth.

6. a. Graph showing growth rate of bean seedlings



b. i. - Seedlings from plump seeds: = 56 mm
-Seedlings from shrunken seeds = 30 mm

ii. The seedlings from plump seeds are growing steadily while those from the shrunken seedlings are growing at a slower rate.

iii. The seedlings from plump seedlings are able to grow rapidly because they are healthy and have sufficient food reserve to support plant growth. On the other hand, seedlings from the shrunken seedlings grow very slowly because they have insufficient food reserve to support growth.

c. i. Temperature.
ii. Moisture content.
iii. Soil fertility.

d. Seedlings from plump seeds grow faster than those from shrunken seedlings.

7. a. A randomized block design for an experiment.

	Block 1	Block 2	Block 3
Plot 1	30cm	120cm	90cm
Plot 2	60cm	30cm	120cm
Plot 3	90cm	60cm	30cm
Plot 4	120cm	90cm	60cm

b. i. During the experiment, there is need to handle all the treatments in a similar manner, except for the variable that is under investigation. In the experiment above, the experimenter should make sure that all necessary crop husbandry practices like planting, weeding, application of fertilizer and pesticides are all done on the treatments. The only thing to be changed in this experiment would be the spacing. Care must be taken to make sure that when performing the husbandry practices bias is avoided by randomizing these husbandry practices.

ii. This involves collection of any information that may have a bearing on the final results of the experiment. The data may be collected by either using direct observation or measuring. For example, attributes like leaf colour, occurrence of pest and diseases may be directly observed while as plant height and area may be measured. When collecting the data, there is need to collect this within a sample population that is chosen randomly in a net plot. This helps to save time and labour since the sample plants act as representatives of the whole plant population.

iii. Once collected, the data should be recorded immediately in note books or journals. At a convenient time, the data is then transferred into tables or graphs for easy analysis.

iv. This is concerned with scrutinizing the collected data for easy comparison between the treatments. It involves calculation of percentages, averages and ranges for all numerical data.

c. It helps in overcoming the effects of external factors in the results of the experiment.

d. 30cm spacing because it is the mostly used spacing for growing maize.

8. a. To find out the effect of different fertilizer rates on maize yield.

b. Control treatment.

c. It acts as a basis for comparing and judging the performance of other treatments in the experiment.

d. 100kg/ha of DAP and 100kg/ha of Urea.

e. The fertilizer rate supplied the optimum level of nutrients to the crops unlike in plots 3, 4 and 5 where the rates were excessive. When fertilizer is applied in excess amounts it results into scorching of the plant roots causing yields to be low.

f. The plot had some soil nutrients which the plants were able to utilize.

g. 10 kg

h.

Area of field	$= 50m^2$
But 1 hactare	$=10{,}000m^2$
∴ Area of the field in ha	$= \frac{50m^2}{10{,}000m^2}$
	$= 0.005$ ha
Yield on 0.005 ha	$= 10kg$
Yield on 1 ha	$=x$
	$\frac{10kg}{0.005 \text{ ha}}$
	$= \underline{2000kg/ha}$

i. By randomly collecting a few soil samples from different areas inside the experimental plots. The soil can then be analyzed in laboratories for availability of plant nutrients.

j. It enables the researchers to know the amount and type of fertilizer to apply in the plots.

9. a. i. NSCM 41 has been repeated in plot 1.
ii. Local maize has been repeated in plot 2.
iii. Block 2 has MH16 repeated.
iv. NSCM 41 is repeated in block 3.

b. A Randomized block design

	Block 1	Block 2	Block 3
Plot 1	NSCM 41	MH 12	MH 16
Plot 2	Local Maize	NSCM 41	MH 12
Plot 3	MH 16	Local Maize	NSCM 41
Plot 4	MH 12	MH 16	Local Maize

c. i. Replication where the experiment is done several times in order to come up with a fair conclusion.

ii. Using one of the treatments as a control treatment on which the performance of the other treatments are compared and judged.

d. i. By direct observation of qualitative attributes such as leaf colour, incidence of pest and disease attack.

ii. By measuring quantitative attributes such as length of the plants and leaf area.

e. i. Calculating averages for example by adding the total yield obtained for each treatment and dividing the results by the number of plots.

ii. Calculation of range in which the highest value is compared with the lowest value so as to determine the extent to which the variables differ in a given attribute, for example height, quantity.

10. a. They come up with recommendations on best agricultural practices.

b. They enable researchers to refine findings of previously conducted experiments by redoing the experiment.

11.a. Repetition of an experiment several times helps to overcome external influences that may affect the results of the experiment. It therefore makes the results of an experiment to be more valid.

b. A control treatment enables the researcher to easily compare and judge the performance of other treatments in the experiment

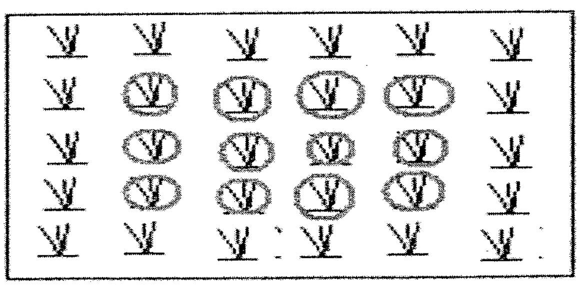
12. Reports are a basis for communicating the findings of an agricultural experimentation to various interested people. For example, farmers may wish to know more about the results of the experiment in order for them to put the recommendations into practice. Furthermore, fellow researchers may use the report for them to investigate further on the finding of the experiment so that fair conclusion is arrived at.

13. a. This is the heading of the experiment. It must be as brief as possible while at the same time providing more information about the nature of the experiment.

b. This gives an overview of the whole experiment by providing a number of statements on a number of issues in the experiment. For example, it contains a statement of the subject in which the title of the experiment, related theories and a brief historical background are given. The introduction also contains a statement of purpose under which a description of what the experiment achieved is given. There is also a statement of scope in which there is an explanation on the areas that were covered in the experiment.

c. This part highlights all the materials that were used in the experiment, their quantities and what they were used for. Additionally it also describes how the treatments were chosen, type of experimental design and the husbandry practices that were conducted. For each material and method used, there should be a justification for it.

14. a. Diagram showing sample plant for observation in an experiment



b. The circled plants are not affected by border effects.

c. They are the true representatives of the treatment in the plot because they are not greatly affected by external factors like alluvial deposits from neighboring plots.

d. Farmers would be able to grow sorghum several times in a year, thereby increasing the yield obtained.

e. The report acts as a way of communicating recommendations of the experiment to farmers. If farmers adopt irrigation farming for sorghum, the yield for sorghum is likely to increase.

15. The first stage is problem identification. In case of the weeds, the farmers should try to identify the type of weeds that are infesting the maize gardens in the area. These weeds may be identified using methods of weed classification such as seed type, leaf shape, growth habit and feeding habits among many others.

The next stage is clarification of aims and objectives. This is important because it elaborates on what the experiment seeks to achieve. In this case, the farmers may seek to find out the best methods ~~that can~~ effectively overcome the weeds.

Once the aims and objectives have been clarified, the farmers may then design the experiment. This stage involves making a plan on how the experiment can be conducted. In doing this, the farmers need to consider such factors like the number of treatments or plots to be used, number of times the experiment will be conducted (replication) and the type of randomization to be used. With respect to the problem of weeds that the area is facing, farmers may chose to have five plots for the experiment in which different methods of weed control at farm level may be studied. These would include chemical, mechanical, physical, biological and cultural control.

Upon successfully designing the experiment, farmers can then start collecting relevant data that can have a bearing in the final results of the experiment. This may include the number of times the weeds are controlled in each plot, the cost of controlling the weed as well as the yield that is obtained from each plot among many others. The collected data is properly recorded in tables or graphs (charts).

The other important stage is analysis of the recorded data. This involves categorizing the collected data into meaningful units for easy interpretation. This can be done by calculating averages, ranges or percentages. In this experiment, for instance, the farmers may calculate the average yield that is obtained from the five plots as well as calculating the yield percentage for each plot.

Analysis of the data is followed by evaluation of results. This part is concerned with discussing the results of the experiment and giving reasons for each result obtained.

The final stage in the experimentation would be to draw a conclusion on the findings of the experiment. This part summarizes the major findings of the experiment in relation to the aims and objectives of the experiment. In addition, the farmers would come up with a recommendation on the best method of weed control.

Chapter Eight

CHALLENGES IN AGRICULTURAL DEVELOPMENT

Questions

8.1 Population growth and the environment

1. Rapid population growth results into soil erosion.
   a. Draw a five step cause effect problem chain for soil erosion.
   b. Describe how rapid population growth accelerates soil erosion.
   c. Explain how soil erosion may cause silting and flooding.
   d. List down any two effects of soil erosion on the physical environment.
2. a. Explain how rapid population growth may contribute to the occurrence of droughts.
   b. Explain how rapid population growth contributes to depletion of water resources.
   c. State any five ways of conserving water.
3. Explain one effect of each of the following factors on agricultural production.
   a. Land degradation.
   b. Pollution of rivers.
   c. Food insecurity in a family.

8.2 Population growth and food security

1. State any four scientific and technological innovations, in maize production, from planting to harvest which ensure food security.
2. Explain any two ways in which fish farming may be used to ensure food security for a growing population.
3. Explain one way in which each of the following agricultural services ensures food security for the growing population:
   a. Seed technology.
   b. Provision of farm inputs.
   c. Crop protection.
   d. Irrigation.
   e. Agricultural credit.

4. Examine the role of estates in ensuring food security in Malawi.

5. List down any four qualities of a good storage facility for maize.

6. Explain any one way in which each of the following technologies would ensure high food production:

   a. Crop diversification.

   b. Use of herbicides.

   c. Mechanization.

   d. Food storage and preservation technology.

   e. Fertilizer production.

7. Explain briefly how food security is affected by each of the following:

   a. Storage facilities.

   b. Planting early maturing varieties.

   c. Food processing technologies.

8. a. Name any two examples of drought tolerant crops.

   b Explain any two ways in which drought tolerant crops may ensure food security.

9. Examine how mixed cropping and mixed farming can help in ensuring food security in a community.

10. A farmer keeps local breeds of cattle and goats. He has the following resources and services at his disposal: large arable land, dambo land, a research station and a veterinary station. Despite having these resources and services, he faces perpetual poverty and food insecurity.

    Explain how the farmer can use the resources and services available to overcome the problems.

8.3 Agro based Industries

1. a. Name any two agro based industries that sell fertilizers to farmers.

   b. Explain one role of agro based industries in supporting the growing population.

2. Explain any five ways in which ADMARC contributes towards food security in Malawi.

8.4 Agricultural Development Services

1. List down examples of the following.

   a. Agricultural development agencies.

   b. Agricultural development services.

2. Describe the services that are offered by the following agricultural development agencies.

    a. Agricultural Research stations.

    b. Marketing agencies.

    c. Extension and training.

    d. Processing companies.

    e. Agricultural Communication Branch.

3. Discuss the importance of agricultural development services in ensuring food security in a country.

8.5 Gender and Agricultural Development

1. What is the meaning of gender bias?

2. Give any three examples of gender biases in agriculture.

3. Discuss the implication of gender bias in agricultural development.

4. Explain any two ways in which participation of female farmers in agricultural development can be improved.

8.6 Population and Land Policies in Agricultural Development

1. Define the term land tenure.

2. State any two ways in which population policies in Malawi promote agricultural development.

3. Describe the three land tenure systems in Malawi and assess how they affect agricultural development:

    a. Customary tenure.
    b. Public tenure.
    c. Private tenure.

4. Explain two ways in which equitable land distribution is important in Malawi.

5. Explain how each of the following policies on agriculture can assist in national development:
    a. Population policy.
    b. Equitable land distribution policy.

8.7 HIV/AIDS and Agricultural Development

1. Explain any one way in which HIV and AIDS can limit contribution of women to agricultural production.

2. Describe the impact of HIV/AIDS on the following:

   a. Labour supply on the farm.

   b. Productivity of the farm.

   c. Farm resources.

1a Distinguish between esp & edp. (2mks)

b) i) Calculate esp at Chanlikana.

MK80.00 → 95.00/kg

6000kg to ((4mks)

ii) degree (1mk)

iii) Sketch (2mks)

iv) advice. (1m)

2. Pros

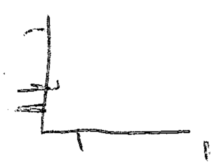
$6000 \div \frac{200,000}{1} \div \frac{75}{100} \div \frac{50}{100}$ 18 $\frac{15}{50} \times 100 = 18.75$

b.

$60,000 \times \frac{1}{200,000} \times \frac{100}{75} \times \frac{100}{50}$ $\frac{1500}{6000} = 25\%$

$edp = \frac{25}{18.75} = 1.3$

3. Elastic



Seed rate = expt plant pop/ha ÷ seed size ÷ purity% ÷ germ%

Seed rate = $\frac{\text{expt plant pop/ha}}{\text{seed size} \times \text{purity\%} \times \text{germ\%}}$

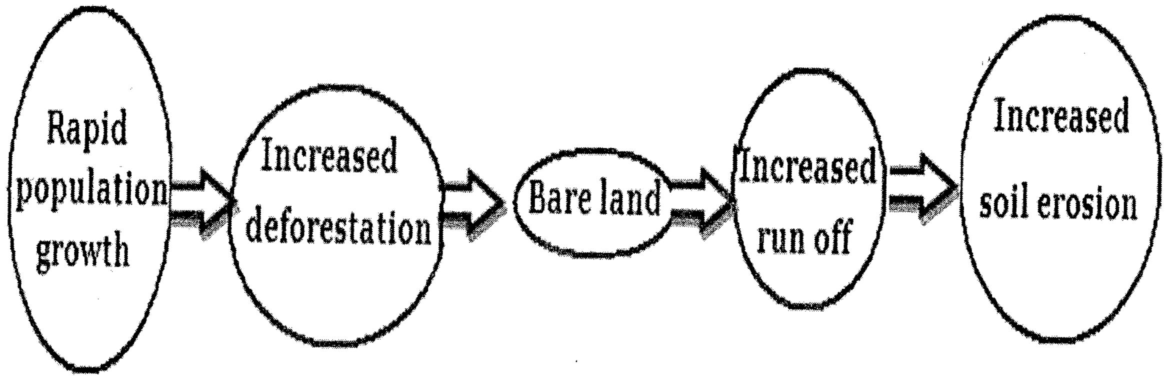
Chapter Eight

CHALLENGES IN AGRICULTURAL DEVELOPMENT

Answers

8.1Population growth and the environment

1. a. A cause effect problem chain for soil erosion.



b. As the population is rapidly increasing, a lot of forest areas are cleared in order to provide land for settlement and cultivation. This leaves the land bare, thereby exposing it to agents of soil erosion such as running water and wind.

c. When soils are eroded by water they are deposited into water bodies. This siltation reduces the volume of the water bodies such that any excess water that is emptied into the water body can easily spill over as flood water.

d. i. Loss of fertile top soil.
ii. Siltation of water bodies.
ii. Pollution of water bodies.

2. a. Growth in population results into more demand for forestry resources leading to rapid deforestation. When more trees are cut, rainfall becomes scarce because the rate of evapotranspiration is reduced due to increased run off.

b. Rapid population growth leads to cutting down of trees which leaves the ground bare. When trees are cut, the rate of run off is increased while the rate of infiltration is reduced. The reduced infiltration results into low ground water table level.

c. i. Construction of dams.
ii. Making contour bunds.
iii. Construction of box ridges.
iv. Planting vegetative cover.
v. Mulching.

3. a. If land is degraded, crop production is affected in that such type of land does not adequately support the growing of crops. Crop harvests are usually very low on degraded land.

b. i. Pollution of rivers and other water bodies with chemical fertilizers, sewage and animal excreta lead to eutrophication. This is the overgrowing of aquatic plants due to addition of dissolved nutrients from the above stated sources. The problem with this is reduced levels of oxygen in the rivers which cause deaths in fish and other important creatures in water.

ii. Some of the pollutants are toxic such that they poison aquatic forms of life such as fish and plants. Serious poisoning levels may lead to death of these living things.

c. i. If a family is facing perpetual food shortages, the active family members are always preoccupied with searching for food to eat. This makes such families to have less time for attending to farm operations. In so doing production on the farm is very low.

ii. A family that does not have adequate food is usually weak and less productive in agriculture.

8.2 Population growth and food security

1. a. Planting of improved seed varieties.
   b. Use of herbicides and pesticides.
   c. Use of chemical fertilizers.
   d. Use of farm machinery.

2. a. The fish when harvested can be used as a source of food. Additionally, excess fish can be sold out and the proceeds used as a source of income for buying food.

   b. Fish farming can provide water for irrigating crops. If such water is utilized, a farmer can produce a lot of food that can be utilized at home or sold out to other people in a country.

3. a. This has resulted into production of improved seed varieties that are able to grow and mature fast to resist disease and pest attack and that are high yielding. These improvements ensure that food is available to people through out the year.

   b. Provision of such inputs like seeds and fertilizers to very poor small holder farmers empowers them to produce sufficient yields. These yields are used by the farming families while the surplus is sold out to other people who are not engaged in agriculture. This ensures food security to the entire population.

   c. Protection of crops from weeds, pest and disease attack through the use of chemicals help to reduce incidences of crop damage. This results into high yields that can ably support the growing population.

   d. Irrigation farming helps to alter the farming calendar in that it reduces dependence on rainfall which is very unreliable. Through irrigation, farmers are able to produce and harvest several times in a year leading to increased food availability for the growing population.

   e. Easy access to credit facilities enable farmers to increase their capital base that helps them to easily access modern farming inputs that lead to increased yields.

4. Estates in the country own very large tracts of land. Despite this, they concentrate much on producing cash crops instead of food crops. As such, they play a very minor role in ensuring food security in the country. However, if most of the estates owners can change attitude and start producing a lot of food crops, the country can easily become food secure. Estates may also help in ensuring food security by providing farmers with training in improved farming technologies as well as offering soft loans to their tenants for them to get farm inputs.

5. a. They must be water proof to prevent mould from developing.

   b. They must be cool to prevent overheating of the grains.

   c. They should be air tight to keep out pathogens and pests.

   d. They must be clean to reduce multiplication of pests and diseases.

6. a. This involves growing different crops on the farm. It has an advantage that failure of one crop to do well will be covered by the other crops.

   b. Herbicides help in the control of weeds that cause a reduction of crop yield. The herbicides therefore enable farmers to harvest more yields.

   c. This ensures cultivation of large pieces of land which result into the production of more food.

   d. This helps to minimize wastage that is caused due to pest attack thereby ensuring that food items are kept for very long time without getting bad.

   e. The fertilizers help to increase soil fertility which help to increase crop yields.

7. a. Good storage facilities help to prevent loss of harvest through damage that can be caused by pests, diseases and moisture.

   b.i. It ensures early maturity of crops that help farmers to get food in times of food shortage.

   ii. Where rainfall is unreliable, early maturing varieties can still do well giving a chance to the farmer to get some harvest.

   c. These technologies enable food to be kept for long time without getting damaged. This food can be used in times of food shortages.

8. a. Cassava, millet, sorghum

   b.i. They are adapted to survive in low moisture conditions thereby enabling farmers to obtain yields in times of droughts.

   ii. They mature early thereby overcoming shortage of food that comes due to short rainfall season.

   iii. Most of the drought tolerant crops like cassava and sorghum are easy to store making them to be available at all times.

9. When different crops are grown on the same farm farmers are able to reduce the risk of total crop failure in times of droughts. This is because some crops can withstand drought conditions thereby enabling the farmer to still get yields in times of drought. On the other hand, mixed farming help farmers to minimize costs of production while maintaining high yields. For example, manure from livestock is used to enrich soil fertility leading to high crop yields. In the same vein, crop residues can be used as livestock feeds resulting into increased productivity in the animals.

10. Firstly, the farmer may use the arable land for the growing of arable crops. In order to increase productivity the farmer may obtain improved crop varieties and advice on best farming practices from the near by research station.

Secondly, the dambo land may be dedicated to the growing of pastures which can be used to feed the cattle and goats on the farm. The research station may also be of greater use as it may provide the farmer with improved pasture species that are of high quality and more yielding. These quality pastures would ensure high productivity in the animals.

The research station could be of greater significance in livestock production. This may be used for obtaining improved breeds of goats and cattle that can help to boost production. The farmer may also be advised on best husbandry practices for managing the livestock. This facility may provide farmers with artificial insemination services that help in upgrading local breeds of livestock.

The veterinary station may also be of greater use to the farmer, particularly in livestock production. The farmer may use this facility for dipping the animals to control external parasites and drugs for the control of diseases. If the goats and cattle are protected from parasites and diseases, production is greatly enhanced.

8.3 Agro based Industries answers

1. a. Kulima Gold, Agora, OPTIMECH, ATC

   b. i. They process agricultural commodities into forms that can be kept for very long time, thereby ensuring availability of these items in times of scarcity.

   ii. They provide employment to both skilled and unskilled labour, there by enabling a lot of people to earn incomes that help in uplifting their lives.

2. In the first place, ADMARC buys various food produce from farmers and sell them to other people who are not engaged in the production of the food. Through this, people through out the country are able to easily access the food.

   Secondly, ADMARC has good storage facilities for various foods. This ensures that food items are not damaged due to poor storage methods. In this way, food availability is ensured through out the year.

   In cases where some areas have faced droughts and failed to produce enough food, ADMARC plays a role of distributing the foods to those areas so that people are able to buy them. This is made possible because the organization has a lot of depots that are strategically located in areas through out the country.

   ADMARC also plays a crucial role in stocking and selling modern farm inputs to farmers. These farm inputs help farmers to produce more yields that ensure food security in the country.

   Lastly, ADMARC is involved in setting up minimum prices for agricultural commodities. This ensures that farmers are not exploited by traders. By setting up good prices for the products, ADMARC help to motivate farmers to produce more food crops that help to promote food security in the country.

8.4 Agricultural Development Services

1. a. i. Research stations.
   ii. Rab Processors.
   iii. Bakhresa Grain and Milling Company.
   iv. Kulima Gold.

b. i. Conducting research.
   ii. Irrigation.
   iii. Seed technology.
   iv. Provision of farm inputs.
   v. Soil testing.
   vi. Agricultural credit.

2. a. These are concerned with conducting of research on crop and livestock production with the aim of developing new technologies that help improve agricultural productivity. The results of the experiments are then communicated to farmers for them to apply certain recommended farming practices.

   b. Various marketing agents are involved in the process of marketing by performing a number of marketing functions such as assembling, grading, standardizing, transporting, and processing among others. Marketing of agricultural produce is very important because it enables a wider population that is not directly involved in agriculture to get access to the produce.

   c. This is concerned with providing technical support as well as advice to farmers on modern farming practices that help to increase production. The main roles of this agency is to train farmers, presenting farmers' problems to researchers for further research and orienting extension workers in their fields. It also promote health and population messages to communities to ensure sustainable use of the environment.

   d. This is mainly concerned with transforming of raw materials from agriculture into finished products. This is an important service for it helps to add value to farm produce apart from helping in promoting easy storage of the produce.

   e. This is responsible for informing farmers on new technologies on improved crop varieties, livestock breeds and other farm inputs. This is done through radios, magazines, films and video shows.

3. The first one is land husbandry services. This is an important service because it helps to bring about awareness on vulnerability of land and how the land can be conserved. Through various land conservation measures, soil fertility is maintained which helps to promote crop yields that ensure food security.

   Secondly, provision of irrigation services to farmers also help in increasing yields. Through use of irrigation facilities provided by the department of irrigation and water development, farmers are able to obtain high yields because they are able to harvest more than once in a year.

   Thirdly there is provision of farm inputs to farmers. By enabling farmers to easily access farm inputs, they are able to increase productivity on the farm which helps in ensuring food security.

   Soil testing is also another important service that is provided by research stations in the country. It involves analyzing soil samples in order to make recommendations on the type of fertilizer to be used. The advantage of this is that crop production is increased because farmers are able to apply the right fertilizer that can increase crop yield.

   Lastly, crop protection is another significant service that is offered by agricultural development agents such as Department of Agricultural Research. This service ensures that crop losses due to weeds, pest and disease attack are minimized by producing resistant varieties and recommending suitable chemicals for control. The result of this is increase in crop yields.

8.5 Gender and Agricultural Development

1. a. Gender bias is favoritism or discrimination of one sex as a result of stereotypes that comes in due to the socially prescribed roles.

   b. i. Ownership of land by men only.
   ii. Men using machinery while women use very simple forms of technologies.
   iii. Accessing of agricultural loans and inputs by men only.
   iv. Growing of cash crops by men while women grow food crops.

2. a. Gender biases result into low productivity because men and women do not have equal access to agricultural technology. As a result, there is less skilled labour to operate the machinery, hence low productivity.

   b. It results into low agricultural development because there is less effort that is made towards increasing production as people, especially women feel sidelined on issues of agricultural technology.

3. a. Increasing access to education to women and girls so that they become educated and become self reliant.

   b. Ensuring easy access to credit facilities to women to enable them raise enough capital for promoting agricultural production.

   c. Empowering more women to take up decision making roles at various levels of the society. This is important because they may easily influence formulation of policies that help to raise the status of women in society.

   d. Discarding cultural and traditional values that force women to play a subordinate role to men. This would ensure that women have powers to make important decisions in agriculture without relying much on men.

8.6 Population and land Policies in Agricultural Development

1. This is the ownership of land and any other conditions or regulations relating to such ownership.

2. a. It provides for traditional leaders to distribute customary land to the landless community members.

   b. It makes a provision for enterprising farmers to lease land for commercial farming from government.

   c. It provides for government to buy back idle land and redistribute it to needy farmers.

3. a. This is a system of land ownership in which every member of the community has a right of possession on the land. Usually the allocation of land to individuals is in the hands of community leaders such as chiefs. There are a number of problems that are associated with this tenure system. For instance, the land can not be used as security for accessing loans from banks. Secondly, there is little investment on the land since any such investment benefits the whole community. Following this, customary land tenure leads to low agricultural development as it does not encourage the land users to make positive contribution towards land conservation.

b. This is the system of land ownership in which all the rights of ownership are in the hands of the government. People may be allowed to use the land, but the problem is that they can be removed from such lands whenever the government wants it. Due to this, people may not be very willing to invest on such pieces of land as such; agricultural development is also very low.

c. This is where the rights of land ownership are vested in an individual. It is regarded as the most effective form of land ownership because the owners can use the land as collateral for obtaining funds from banks. Such funds can be used for developing the land in many ways. This system of land ownership helps to promote agricultural development because the owners put a lot of effort to conserve and efficiently utilize the land since they have absolute rights to the land.

4. a. If people have got easy access to land, they can use such land for agricultural practices that help to improve people's well being.

b. It helps to uplift the economic status of people because they can use the land as collateral for securing loans from banks. The loans can be used to increase capital that helps in increasing agricultural productivity.

5. a. It helps to reduce population growth rates to manageable levels thereby reducing pressure on environmental resources.

b. It ensures that farmers have easy access to land for the growing of both food and cash crops. This helps to improve food security as well as raising the living standards of the people.

8.7 HIV/AIDS and Agricultural Development

1. Culturally it is expected that when a family member has HIV or suffering from AIDS, a woman is supposed to attend to the sick. This practice reduces the labour supply on the farm. In the instance that a woman is suffering from AIDS, the normal labour contribution by her is also reduced very much. This reduced labour supply results into low levels of production.

2. a. Firstly, if the farmer or any family member is suffering from various diseases because of HIV, there is reduced contribution of labour on the farm. This is caused by the body weakness that results from the illness. Secondly, if death occurs due to HIV/AIDS, there is a permanent loss of the labour supply on the farm.

b. Productivity on the farm is reduced because of the reduced labour supply due to ill health that comes as a result of HIV/AIDS.

c. HIV/AIDS results into depletion of important farm resources like capital and human resources. As people are suffering from HIV/AIDS, a lot of money is spent on the patients for drugs, food, transport and other items. Attending to the sick also affects the managerial input of the farm such that production is low due low input levels.

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